From Ghost SICs to SIC-POVMs: A Galois Reduction of Zauner's Conjecture

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Symmetric informationally complete positive operator-valued measures (SIC-POVMs, or SICs) are optimal minimal measurements for quantum state tomography. Problem 1 of Horodecki, Rudnicki and Życzkowski's open problems in quantum information asks for SICs in infinitely many dimensions. By work of Appleby–Flammia–Kopp and Radchenko–Wheeler, ghost SICs, which satisfy the SIC equations with Hermiticity up to parity, exist unconditionally for $d \geq 4$. We study the passage from ghosts to SICs. We compute all isolated points of the complexified SIC variety for $d = 4, 5, 6$; exactly half are SICs. We prove that no symmetry of the SIC equations maps a ghost to a SIC, so the step is purely arithmetic, and that a SIC exists in dimension d whenever the ghost overlap units satisfy a place law: each Galois conjugate is real or unimodular according to the infinite place of the real quadratic field F below it. If the unit field is abelian over F, the place law is the Stark relation $c_- = \kappa$; equivalently, a Galois conjugate of an associated near-group fusion category is unitary exactly when it is pseudo-unitary. For d a power of a prime $p \equiv 5 \pmod 6$ or of 3, an infinite family, we prove that under the algebraic part of Stark's conjecture $c_- = \kappa$ is equivalent to the triviality of κ on the Hilbert-class-field part of the unit field. That condition involves no complex conjugation, and it follows from a sign-free multiplet hypothesis. We state an explicit GL_2 reciprocity law for the squared ghost overlaps that implies $c_- = \kappa$, and we prove it for the computed algebraic overlaps in $d = 5, 7, 11, 15$ by Artin reciprocity and a check in every ray class. Zauner's conjecture remains open: the algebraic part of Stark's conjecture and the multiplet hypothesis are not proved here.

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I. INTRODUCTION

A. Background and motivation

A symmetric informationally complete positive operator-valued measure (SIC-POVM, or simply SIC) in $\mathbb{C}^d$ is a set of d^2 unit vectors ψ_j satisfying $|\langle\psi_j, \psi_k\rangle|^2 = 1/(d+1)$ for $j \neq k$[1,2]. SICs are of interest well beyond this definition. They are minimal informationally complete measurements, and they are optimal among these for linear quantum state tomography[2,3]. They give quantum key distribution protocols[4], they serve as the reference measurement in the quantum-Bayesian (QBism) formulation of quantum theory as a probability calculus[5], and they are maximal sets of d^2 equiangular lines in $\mathbb{C}^d$, that is, equiangular tight frames, in frame theory and signal processing. Ref. 6 surveys these applications. Zauner conjectured that SICs exist in every dimension and may be chosen covariant under the Weyl–Heisenberg (WH) group. Problem 1 of Ref. 7 asks for a construction in an infinite sequence of dimensions. Since every WH-covariant SIC is a SIC, it suffices to construct WH-covariant SICs, i.e., WH orbits of a single fiducial vector. This is also the form taken by essentially all known exact constructions of SICs[8–10]. A conditional construction for all $d > 3$ was given by Appleby, Flammia, and Kopp (AFK)[11]. It depends on the Stark conjectures and on a twisted convolution identity (TCI). Radchenko and Wheeler (RW)[12] subsequently proved algebraicity of the relevant special values of the modular quantum dilogarithm and established the rank-one TCI. AFK[13] then extended the TCI to all admissible tuples. As a consequence, ghost SICs are now known to exist unconditionally in every dimension $d \geq 4$. Ghost SICs satisfy the SIC equations with Hermiticity replaced by Hermiticity up to parity. The remaining step is to pass from a ghost SIC to a genuine SIC. Ref. 13 formulates this step in terms of a Galois automorphism with a prescribed interaction with complex conjugation. The present paper studies this problem from three related perspectives:

(a) the relationship between the numbers of SICs and ghosts;
(b) the existence of modular solutions;
(c) the implication from ghost SICs to SICs.

Using the results of Refs. 12 and 13, we show that these questions are equivalent to Zauner's conjecture and reduce to the same arithmetic obstruction.

B. Main results

The principal contributions of the paper are as follows.

1. **Complete solution counts** (computational; Sec. III). The complexified SIC varieties V_d for $d = 4, 5, 6$ have 512, 4000, and 6912 isolated points, respectively. Exactly half are SICs, and for $d = 4, 5$ all remaining points are ghosts.
2. **A no-go result for Clifford symmetries** (Lemma IV.1). No element of the extended Clifford group maps a ghost to a SIC.
3. **Galois and finite-field criteria** (Proposition IV.3, Theorem IV.6). A Galois orbit contains a SIC if and only if transposition acts on that orbit as a conjugate of complex conjugation.
4. **Galois inversion** (Proposition V.3). Galois realization of transposition is equivalent to the existence of a field automorphism inverting all ghost overlap units.
5. **Structure theorems** (Theorems VI.6, VI.7, and VI.8). In the RW formulation, any failure of the Galois realization problem is confined to a specific class of pseudo-unitary solutions and is subject to a parity constraint.
6. **The place law** (Theorems VI.22 and VI.23). The place law (PL) determines the type of each Galois conjugate from the underlying infinite place of $F = \mathbb{Q}(\sqrt{N(N+4)})$. If the place law holds, then a WH-covariant SIC exists.
7. **Single-unit and 2-adic reductions** (Theorem VI.24, Corollary VI.27, and Proposition VI.28). The archimedean input may be reduced to a single generating unit, while one parity of the resulting obstruction is determined by finite-prime data.
8. **A class-field-theoretic formulation** (Proposition VI.29). If the overlap-unit field M/F is abelian, then the place law is equivalent to the identity $c_- = \kappa$, and hence to a single Frobenius congruence. If M/F is only assumed Galois but $\mathrm{Gal}(M/F)$ has a unique element of order 2, the archimedean input disappears altogether (Proposition VI.30).
9. **A unitarity formulation** (Theorem VI.53, Corollary VI.54). The Stark solution defines a near-group fusion category $\mathcal{C}_{E_*}$ that is not pseudo-unitary, while exactly half of its Galois conjugates are. For a pseudo-unitary conjugate, Izumi's unitarity equations reduce to a single Hermiticity condition, which is exactly the SIC-type reality condition. A SIC exists as soon as one pseudo-unitary conjugate is unitary, and

the place law says that the unitary conjugates are precisely the pseudo-unitary ones. Unitarity is also equivalent to the minimality of $\sum_{u\neq 0}|\sigma(E_*(u))|^2$ (Proposition VI.56).

10. **An explicit reciprocity law** (Conjecture VI.40, Proposition VI.42, Table VII). We state a reciprocity law of Arakawa type for the squared ghost overlaps: the Artin symbol of a principal ideal (ϖ) acts on the labels through the matrix $\phi(\varpi)\in GL_2(\mathbb{Z}_d)$ of ϖ in the order of the Zauner symmetry, with ϖ normalized to be positive at ∞_-. At the generator $\varpi\equiv -1$ of the class of c_- the matrix is $-I$, and hence the law implies $c_-=\kappa$ and Z(d). We verify the law exactly, prime by prime, for $d=5$ and $d=9$; for $d=9$ the class group of F acts on a multiplet of two ghosts. For the computed algebraic overlaps in $d=5,7,11,15$, where F has class number 1, we prove the law at every prime by Artin reciprocity and a check in each ray class (Lemma VI.43, Table VII).
11. **Prime-power dimensions** (Theorem VI.46, Corollaries VI.47 and VI.49). If $d=p^k$ with $p\equiv 5 \pmod 6$ prime, or $d=3^k$ with $k\geq 2$, then the subgroup of the Galois group that contains every complex conjugation has a unique involution. Hence, under the algebraic part of the Stark picture, $c_-=\kappa$ is equivalent to the triviality of κ on the part of the unit field contained in the Hilbert class field of F. That condition involves no complex conjugation, holds automatically when the class number of F is odd, and follows from a sign-free multiplet hypothesis (Conjecture VI.48). The family is infinite and contains 167 dimensions $d\leq 2000$.

a. The main reduction. For each $d\geq 4$, write Z(d) for the existence of a WH-covariant SIC in dimension d, and M for the field generated by the overlap units of the Stark solution. Then

$$\boxed{\text{Z}(d) \impliedby \text{PL} \overset{M/F \text{ abelian}}{\iff} c_-=\kappa \impliedby \text{archimedean rank-one Stark}}$$

Here c_- is complex conjugation at the ramified infinite place ∞_- of F, and κ is inversion of the overlap units. The first implication is Theorem VI.23, and the equivalence is Proposition VI.29. The identity $c_-=\kappa$ is the classical Stark relation $\varepsilon^{1+c_-}=1$ written in these coordinates, and abelianness of M/F is the algebraic part of Stark's conjecture. If $\mathrm{Gal}(M/F)$ has a unique element of order 2, abelianness is not needed: M/F Galois, $\kappa\in\mathrm{Gal}(M/F)$ and ramification at ∞_- already force $c_-=\kappa$, hence PL and Z(d) (Proposition VI.30). That hypothesis on the class group can be traded for total reality of the trace field (Conjecture VI.10), which then forces $c_-=\kappa$; under these algebraic hypotheses TR, HS and PL are

equivalent, so the trade does not weaken the archimedean input, and the ramification at ∞_- that it needs is unconditional on a set of dimensions of positive density (Proposition VI.32, Corollary VI.33, Theorems VI.37 and VI.38). In categorical terms, PL says that a Galois conjugate of the near-group category $\mathcal{C}_{E_*}$ is unitary exactly when it is pseudo-unitary, and one unitary pseudo-unitary conjugate already gives Z(d) (Theorem VI.53, Corollary VI.54). Finally, $c_- = \kappa$ is the instance at $\varpi \equiv -1$ of an explicit reciprocity law for the Artin symbols of principal ideals, which we verify exactly for $d = 5, 9$ (Conjecture VI.40, Proposition VI.42). For d a power of a prime $p \equiv 5 \pmod 6$, or of 3, $c_- = \kappa$ is equivalent, under abelianness, ramification at ∞_- and R2, to an unramified condition, the triviality of κ on $M \cap H_F$ for the Hilbert class field H_F (Theorem VI.46). Problem 1 follows if Z(d) holds for infinitely many d. Neither PL, nor abelianness, nor the reciprocity law, nor the multiplet hypothesis is proved here in any infinite family.

C. What this paper does, and does not, contribute

We state the scope of the paper plainly, because the reduction above could be misread as a near-solution of Problem 1.

a. What was already known. AFK[11] showed that SICs exist in every dimension $d > 3$ if the Stark conjectures and the TCI hold. The identity has since been proved[12,13]. Hence "Stark's conjecture implies Zauner's conjecture" already follows from existing work. Our implication Z(d) $\Leftarrow$ PL is not a new kind of conditional result, and it does not by itself produce SICs in any new dimension. In particular, *this paper does not prove Problem 1*, and the place law is not known for any infinite family of dimensions.

b. What is new. The paper contributes in six ways.

1. *A smaller hypothesis.* The place law is strictly weaker than the Stark conjectures used in Ref. 11. It is only the archimedean part, it concerns only the overlap units of one explicit solution, and it does not require knowing the field these units generate. The paper therefore isolates the minimal piece of Stark's conjecture on which SIC existence depends. It also gives equivalent forms of it, ranging from the total positivity of a single algebraic number to, under an algebraic hypothesis, a single Frobenius congruence, and it embeds the place law in an explicit reciprocity law for all principal ideals (Sec. VI G), the form in which comparable statements for $\zeta(0)$-invariants are proved. Under the

same algebraic hypotheses the place law is equivalent to total reality of the trace field, with no condition on the class group, and the ramification at ∞_- this uses is proved unconditionally on a set of dimensions of positive density (Proposition VI.32, Theorem VI.38, Sec. VI F).

2. *Limits on possible proofs.* Several natural strategies are shown to be insufficient:
   - no symmetry of the SIC equations maps a ghost to a SIC (Lemma IV.1);
   - existence arguments of degree or continuity type fail for dimensional reasons (Sec. III C);
   - information from norms and congruences detects only one bit of the place law, and that bit is fixed by the field alone (Proposition VI.28);
   - the overlap-unit field is not contained in any cyclotomic extension of F (Proposition VI.31), and hence no argument based on roots of unity or Gauss sums can supply the missing step.

   Together these show that any proof must construct non-cyclotomic abelian extensions of a real quadratic field ramified at exactly one real place. This is the hard core of Hilbert's twelfth problem for real quadratic fields.
3. *Removing the infinite places on an infinite family.* For d a power of a prime $p \equiv 5 \pmod 6$, or of 3, the subgroup of the Galois group containing the complex conjugations has a unique involution. Under the algebraic part of Stark, the archimedean identity $c_- = \kappa$ is then equivalent to an unramified condition, and it follows from a sign-free hypothesis on the Galois multiplet of ghosts (Sec. VI H).
4. *Structure theorems.* In the RW formulation, Theorems VI.6, VI.7 and VI.8 locate exactly where a failure of the Galois step would have to occur.
5. *A bridge to fusion categories.* The archimedean input is equivalent to a unitarity statement for Galois conjugates of an explicit near-group category (Sec. VI I). By Izumi's classification, the only condition that can fail for a pseudo-unitary conjugate is a Hermiticity equation. This connects the SIC problem with the Haagerup–Izumi unitarity question and with the tools of subfactor theory.
6. *New computations.* We give complete solution counts for $d = 4, 5, 6$ and exact minimal polynomials for 100 overlap-unit traces. We verify the place law in the SIC dimensions $d = 4, 5, 9, 13$, and give class-field certificates (for $d = 13$, a Frobenius test) showing that there the (sign-free) overlap-unit field is the full ray class field of conductor $d'\infty_-$,

as predicted by the AFK construction. For $d = 5$ and $d = 9$ we verify the explicit reciprocity law of Sec. VI G exactly at 1211 primes, and for the computed algebraic overlaps in $d = 5, 7, 11, 15$ we prove it at every prime (Lemma VI.43). We also verify, for every $4 \leq d \leq 20000$, that ∞_- ramifies in the ray class field of conductor $d'\infty_-$ (Table VI).

c. Relevance for quantum information. The measurements at stake are exactly the WH-covariant SICs used in the applications above, and the results bear on how they can be obtained.

- A construction of SICs in infinitely many dimensions along the AFK–RW route now needs only the place law. In dimensions where $\mathrm{Gal}(M/F)$ has a unique element of order 2, it needs only algebraic input (Proposition VI.30).
- Ghost solutions cannot be turned into SICs by any Clifford symmetry (Lemma IV.1), while in the dimensions computed SICs make up exactly half of V_d, and for $d = 4, 5$ the other half consists of ghosts. A numerical or algebraic search that lands on a ghost must therefore change the Galois conjugate, not the symmetry frame.
- The class-field results name the fields in which exact SIC fiducials should be sought: abelian extensions of F of conductor $d'\infty_-$. This agrees with the exact solutions of Refs. 8–10 and the AFK construction.

In short, the paper is best read as a precise map of the obstruction to Problem 1, with evidence for where it lies. It is not a step that constructs SICs in infinitely many dimensions.

d. Conjectures and their roles. The conjectures used in the paper play three different roles.

- *Central:* GR (transposition is Galois, Conjecture V.2), RU (real-or-unitary dichotomy, Conjecture VI.9) and PL (Conjecture VI.20). PL implies RU on the relevant orbit, and it implies GR. Its categorical counterpart, that pseudo-unitary Galois conjugates of $\mathcal{C}_{E_*}$ are unitary (Conjecture VI.55), is implied by PL and already gives $\mathrm{Z}(d)$. The explicit reciprocity law (Conjecture VI.40) implies the half of PL above ∞_-, and hence $\mathrm{Z}(d)$.
- *Supporting:* completeness of V_d (Conjecture III.1) and total reality (TR, Conjecture VI.10). Under the algebraic hypotheses of Proposition VI.32, TR is not merely supporting: it is equivalent to PL (Corollary VI.33).
- *Consequences and shadows of PL:* the half split (HS, Conjecture VI.12), the sign

statement, the sign–type correlation, and the parity criterion of Corollary VI.27. Section VII is a self-contained roadmap, and we recommend reading it after Sec. II. It contains a dependency diagram, the full chain of reductions, a status table marking every statement as proved, conjectural or false, and a precise list of what remains (Sec. VII C). Along the way we obtain computational facts that appear to be new: complete point counts for V_d, and the observation that V_d consists only of SICs and ghosts in the dimensions checked.

II. THE COMPLEXIFIED SIC VARIETY

The SIC equations involve complex conjugation, and hence they are not polynomial equations and Galois theory does not act on their solutions. We remove conjugation by complexifying: we replace the projector $\psi\psi^\dagger$ by a rank-one idempotent $xy^T/(y^Tx)$ with independent x and y. The result is a variety V_d defined over $\mathbb{Q}$. Both SICs and ghosts are points of V_d, distinguished only by how they behave under conjugation. This section sets up V_d, identifies these two loci, and records what the existence of ghosts already implies.

A. Definition and symmetries

Let $\omega = e^{2\pi i/d}$, let $Xe_j = e_{j+1}$, $Ze_j = \omega^j e_j$, and let $D_p = \tau^{p_1p_2}X^{p_1}Z^{p_2}$ for $p \in \mathbb{Z}_d^2$ and a fixed phase τ with $\tau^2 = \omega$. Let $U_Pe_j = e_{-j}$ be the parity operator, and let $\mathcal{P} = \{D_qU_P : q \in \mathbb{Z}_d^2\}$ be the set of displaced parities. Each element of $\mathcal{P}$ squares to a scalar.

Definition II.1. V_d is the set of rank-one idempotents $\Pi \in M_d(\mathbb{C})$ (that is, $\Pi^2 = \Pi$ and $\operatorname{tr}\Pi = 1$) such that

$$(d+1)\,\operatorname{tr}(\Pi D_p)\,\operatorname{tr}(\Pi D_p^\dagger) = 1 \qquad \text{for all } p \neq 0. \tag{1}$$

Writing $\Pi = xy^T/(y^Tx)$ identifies V_d with a subvariety of $\{(x,y) \in \mathbb{P}^{d-1} \times \mathbb{P}^{d-1} : y^Tx \neq 0\}$ cut out by bidegree-$(2,2)$ equations.

Lemma II.2. *1. The left side of (1) does not depend on the phase convention for D_p.*

2. The system (1) is stable under $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$, and hence V_d is defined over $\mathbb{Q}$ and $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $V_d(\overline{\mathbb{Q}})$ entrywise.

3. *Transposition* $T(\Pi) = \Pi^T$ *and conjugation* Ad_U *by any element* U *of the Clifford group preserve* V_d*. Transposition commutes with the Galois action.*
4. *The map* $\sigma(\Pi) = \Pi^\dagger$ *preserves* V_d*, and* $\sigma = c \circ T$*, where* c *is entrywise complex conjugation.*

Proof. For (1), a phase change $D_p \mapsto \lambda D_p$ multiplies $D_p^\dagger$ by $\bar{\lambda} = \lambda^{-1}$.

For (2), the automorphism $\omega \mapsto \omega^k$ sends D_p to a phase times $D_{(p_1,kp_2)}$, and sends $D_p^\dagger = D_p^{-1}$ to the inverse of the image. The products in (1) are therefore permuted.

For (3), D_p^T is proportional to $D_{(-p_1,p_2)}$, and Clifford elements permute the D_p up to phases. Both T and the Galois action are defined entrywise over $\mathbb{Q}$, and hence they commute.

For (4), $\mathrm{tr}(\Pi^\dagger D_p) = \overline{\mathrm{tr}(\Pi D_p^\dagger)}$, and hence (1) for $\Pi^\dagger$ is the complex conjugate of (1) for Π. □

Part (4) of Lemma II.2 is the starting point for everything that follows. The involution $\sigma = c \circ T$ is the only place where complex conjugation enters, and its fixed points are the SICs.

B. SIC and ghost loci

Twisting σ by a displaced parity gives a second family of real points, the ghosts.

Definition II.3. The *SIC locus* is $S_d = \mathrm{Fix}(\sigma) \cap V_d = \{\Pi \in V_d : \Pi = \Pi^\dagger\}$. Its points are exactly the WH-covariant SIC fiducial projectors, since a Hermitian rank-one idempotent is a projector and (1) then reads $|\,\mathrm{tr}(\Pi D_p)|^2 = 1/(d+1)$.

For $u \in \mathcal{P}$, the *ghost locus* is $\Gamma_u = \{\Pi \in V_d : \Pi^\dagger = u\Pi u^{-1}\}$. We write $\Gamma_d = \bigcup_{u\in\mathcal{P}} \Gamma_u \setminus S_d$. These are the parity-Hermitian rank-one ghosts of Refs. 11 and 13: their Weyl symbols $\mathrm{tr}(\Pi D_p)$ are real up to a fixed phase.

Thus SICs and ghosts are the fixed points of two closely related antiholomorphic involutions of the same variety, $c \circ T$ and $\mathrm{Ad}_u^{-1} \circ c \circ T$. The key difference between them is that the ghost locus is known to be nonempty in every dimension, by the recent results of AFK and RW.

Theorem II.4 ([12,13])**.** *For every* $d \geq 4$ *there is* $\tilde{\Pi} \in \Gamma_{U_P}$ *whose entries are algebraic. In particular* $V_d(\overline{\mathbb{Q}}) \neq \emptyset$*.*

This is Theorem 1.3 of Ref. 13 with $r = 1$. The admissibility condition $r < (d-1)/2$, $\frac{d^2-1}{r(d-r)} \in \mathbb{Z}$ holds for $r = 1$ and $d \geq 4$. Algebraicity of the entries follows from Ref. 12 (Thm. 1) through the dictionary of Ref. 13 (Prop. 2.1).

Corollary II.5 (modular solutions)**.** *For every $d \geq 4$ and all but finitely many primes ℓ, the variety V_d has points over $\overline{\mathbb{F}}_\ell$.*

Proof. Normalize $\tilde{\Pi}$ such that its entries lie in the ring of N-integers of a number field F, for some integer N. Take $\ell \nmid Nd(d+1)$ and reduce modulo a prime of F above ℓ. The reduced matrix is still a rank-one idempotent of trace one, and it satisfies the reductions of (1). □

Thus, item (b) of the introduction holds unconditionally for every dimension. It does not, however, distinguish ghosts from SICs: these modular points come from ghosts, and reduction modulo ℓ does not see complex conjugation. To see how the two loci compare, we next determine V_d completely in the smallest dimensions.

III. COMPLETE SOLUTION COUNTS

This section asks how far geometry and numerics alone can go. We compute V_d completely for $d = 4, 5, 6$ and find that exactly half of its points are SICs and, where tested, the other half are ghosts. We then check that relaxing the WH group to other abelian Heisenberg groups gives no new SICs, which confirms that V_d is the right object. Finally, we explain why no continuity or degree argument can replace these counts. The conclusion of all three is the same: the passage from ghosts to SICs is not geometric.

A. The varieties V_4, V_5 and V_6

To find all isolated points of V_d, we solved a random square subsystem of $2d-2$ linear combinations of (1), in bihomogeneous coordinates on $\mathbb{P}^{d-1} \times \mathbb{P}^{d-1}$, by polyhedral homotopy continuation (HomotopyContinuation.jl). We then kept the endpoints that satisfy the full system with residual below 10^{-6} and have $y^T x \neq 0$. Every isolated point of V_d is an isolated solution of the square subsystem, and hence this procedure finds all isolated points of V_d, up to the usual numerical caveats on path tracking. The resulting counts are listed in Table I.

TABLE I. Isolated points of the complexified SIC variety.

d	paths	singular endpoints	$\lvert V_d\rvert$	$\lvert S_d\rvert$	$\lvert V_d \setminus S_d\rvert$
4	560	48	512	256	256
5	14 614	9 718	4 000	2 000	2 000
6	218 343	78 192	6 912	3 456	3 456

Two features stand out. First, the values of $|S_d|$ agree with the known numbers of WH SIC fiducials in these dimensions, which confirms that the computation is complete on the SIC side. Second, in each dimension exactly half of the isolated points are SICs. To identify the other half, for $d = 4$ and $d = 5$ we tested each non-Hermitian point $\Pi = xy^T/(y^Tx)$ for the condition $[\bar{y}] = [ux]$ with $u \in \mathcal{P}$, which is the ghost condition $\Pi^\dagger = u\Pi u^{-1}$. Every non-Hermitian point passed this test (Table II). Thus, in these dimensions, V_d consists of SICs and ghosts only, in equal numbers. This suggests the following conjecture.

TABLE II. Every non-Hermitian point of V_4 and V_5 is a displaced-parity ghost.

d	non-Hermitian points	in Γ_{U_P}	in $\bigcup_{u\in\mathcal{P}} \Gamma_u$
4	256	64	256
5	2000	80	2000

Conjecture III.1 (completeness)**.** *For every $d \geq 4$, the variety V_d is finite, $V_d = S_d \sqcup \Gamma_d$, and $|S_d| = |\Gamma_d|$.*

Theorem II.4 gives $\Gamma_d \neq \emptyset$. Thus, if V_d is finite, Conjecture III.1 implies $S_d \neq \emptyset$ for all $d \geq 4$, which is Zauner's existence conjecture for WH SICs. The equality of counts is therefore at least as hard as Zauner's conjecture. The rest of this paper explains why.

B. Heisenberg groups of non-cyclic abelian groups, $16 \leq d \leq 36$

The variety V_d is built from the WH group of the cyclic group $\mathbb{Z}_d$. Problem 1 asks only for SICs, and hence one might hope to find them more easily as orbits of the Heisenberg group of a different abelian group of order d. We tested this numerically. For a finite abelian group A of order d, let X^a ($a \in A$) be translations and Z^b ($b \in \hat{A}$) be characters. Define

$$f(\psi) = \sum_{a,b} |\langle \psi, X^a Z^b \psi\rangle|^4 / \|\psi\|^8.$$

Then $f \geq 2d/(d+1)$, with equality exactly for A-Heisenberg-covariant SIC fiducials. We minimized f by L-BFGS with an analytic gradient (verified against finite differences to 10^{-9}), using 60 random starts per group.

The positive controls behaved as expected. Cyclic $\mathbb{Z}_d$ succeeded with $13, 6, 6, 1, 1$ hits out of 60 for $d = 16, 18, 20, 25, 27$. The Hoggar group $\mathbb{Z}_2^3$ succeeded in 30 of 30 runs. The non-cyclic groups $\mathbb{Z}_2^2$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, $\mathbb{Z}_3^2$ and $\mathbb{Z}_2 \times \mathbb{Z}_6$ failed, in agreement with Ref. 14, which covers $d < 14$.

TABLE III. No SIC was found for any non-cyclic abelian Heisenberg group (60 starts each).

d group A	best gap $\min f - \frac{2d}{d+1}$
16 $\mathbb{Z}_4^2$; $\mathbb{Z}_2 \times \mathbb{Z}_8$; $\mathbb{Z}_2^2 \times \mathbb{Z}_4$; $\mathbb{Z}_2^4$	0.026; 0.095; 0.055; 0.268
18 $\mathbb{Z}_3 \times \mathbb{Z}_6$	0.096
20 $\mathbb{Z}_2 \times \mathbb{Z}_{10}$	0.079
24 $\mathbb{Z}_2 \times \mathbb{Z}_{12}$; $\mathbb{Z}_2^2 \times \mathbb{Z}_6$	0.133; 0.161
25 $\mathbb{Z}_5^2$	0.037
27 $\mathbb{Z}_3 \times \mathbb{Z}_9$; $\mathbb{Z}_3^3$	0.103; 0.034
28 $\mathbb{Z}_2 \times \mathbb{Z}_{14}$	0.150
32 $\mathbb{Z}_2 \times \mathbb{Z}_{16}$; $\mathbb{Z}_4 \times \mathbb{Z}_8$; $\mathbb{Z}_2^2 \times \mathbb{Z}_8$; $\mathbb{Z}_2 \times \mathbb{Z}_4^2$; $\mathbb{Z}_2^3 \times \mathbb{Z}_4$; $\mathbb{Z}_2^5$	0.169; 0.088; 0.177; 0.029; 0.170; 0.610
36 $\mathbb{Z}_2 \times \mathbb{Z}_{18}$; $\mathbb{Z}_3 \times \mathbb{Z}_{12}$; $\mathbb{Z}_6^2$; $\mathbb{Z}_2^2 \times \mathbb{Z}_9$; $\mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_6$	0.199; 0.188; 0.191; 0.191; 0.079

The evidence is reasonably strong for $d \leq 20$, where the cyclic control succeeds repeatedly. For $d \geq 24$ it is weak, because the cyclic control also failed at 60 starts for $d = 24, 28, 32, 36$. In the range tested, no SIC was found outside the cyclic WH group, which supports working with V_d.

C. Why "soft" existence arguments fail

The counts above are computed, not proved. It is natural to ask whether a topological argument could prove existence directly, as is done for many existence problems in geometry. A dimension count shows that this cannot work. Restrict to fiducials invariant under a subgroup G of the Clifford group, with image $\bar{G} \subset SL_2(\mathbb{Z}_d)$. The moduli $|\operatorname{tr}(\Pi D_p)|$ are then constant on $\bar{G}$-orbits in $\mathbb{Z}_d^2 \setminus \{0\}$. One therefore needs about $(d^2-1)/|\bar{G}|$ real equations on a G-eigenspace whose typical complex dimension is about $d/|\bar{G}|$. The ratio of equations to unknowns stays near $d/2$ for every choice of G.

Intermediate-value, degree-theoretic or Morse-theoretic existence arguments on a sym-

metric slice need this ratio to be at most one, except in very small dimensions. The only "soft" argument insensitive to overdetermination is a count of points of a zero-dimensional variety, which is the approach of the previous sections. It leads back to the Galois question. (This section is a heuristic dimension count, not a theorem.)

Together, the counts, the non-cyclic search and the dimension count indicate that geometry alone does not explain the existence of SICs. The remaining sections treat the problem arithmetically.

IV. GALOIS STRUCTURE

This section shows that the passage from ghosts to SICs is arithmetic, and gives two criteria for it.

A. Real structures and the no-go lemma

We first describe SICs and ghosts as the real points of two real structures on V_d, and then show that no symmetry of the SIC equations exchanges them.

a. Two real structures. Both S_d and Γ_u are fixed-point sets of antiholomorphic involutions of V_d:

$$S_d = \mathrm{Fix}(c \circ T), \qquad \Gamma_u = \mathrm{Fix}\big(\mathrm{Ad}_u^{-1} \circ c \circ T\big).$$

A bijection $\Gamma_u \to S_d$ would follow from an automorphism h of V_d satisfying $h\,(\mathrm{Ad}_u^{-1}\, cT)\, h^{-1} = cT$. The natural candidates are the symmetries of the SIC equations.

b. The no-go lemma. The extended Clifford group cannot supply such an h.

Lemma IV.1 (no-go)**.** *Every element of the extended Clifford group, whether unitary $\Pi \mapsto U\Pi U^\dagger$ or antiunitary $\Pi \mapsto U\bar{\Pi}U^\dagger$, commutes with $\sigma(\Pi) = \Pi^\dagger$. Hence it maps S_d to S_d and Γ_d to Γ_d. In particular, no symmetry of the SIC equations maps a ghost to a SIC.*

Proof. We have $(U\Pi U^\dagger)^\dagger = U\Pi^\dagger U^\dagger$ and $(U\bar{\Pi}U^\dagger)^\dagger = U\overline{\Pi^\dagger}U^\dagger$. If $\Pi^\dagger = u\Pi u^{-1}$, then $(U\Pi U^\dagger)^\dagger = (UuU^\dagger)(U\Pi U^\dagger)(UuU^\dagger)^{-1}$. Here $UuU^\dagger \in \mathcal{P}$ up to a phase, because Clifford elements normalize the WH group and commute with U_P, which represents the central element $-I \in SL_2(\mathbb{Z}_d)$. The antiunitary case is the same, using $\bar{u} \in \mathcal{P}$. □

Thus, any passage from ghosts to SICs must come from arithmetic rather than from geometry.

B. Galois and finite-field criteria

Since geometry cannot connect ghosts to SICs, we now characterize the passage arithmetically, first through Galois groups and then through reduction modulo primes.

a. The Galois criterion. Fix an embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$, and let $c \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be the corresponding complex conjugation.

Lemma IV.2 (algebraicity of SICs)**.** *If a WH-covariant SIC exists in dimension d, then $S_d \cap V_d(\overline{\mathbb{Q}}) \neq \emptyset$.*

Proof. In real coordinates $(\mathrm{Re}\,\Pi, \mathrm{Im}\,\Pi)$, the conditions for a point of S_d form a system of real polynomial equations with coefficients in the field $K = \mathbb{Q}(\cos\frac{2\pi}{d}, \sin\frac{2\pi}{d}) \subset \overline{\mathbb{Q}} \cap \mathbb{R}$. By the Tarski–Seidenberg transfer principle, a first-order formula with coefficients in K that holds over $\mathbb{R}$ also holds over the real closure of K. That real closure is contained in the real closed field $\overline{\mathbb{Q}} \cap \mathbb{R}$, and hence the system has a solution with coordinates in $\overline{\mathbb{Q}} \cap \mathbb{R}$. □

Proposition IV.3 (Galois criterion)**.** *Let $O \subset V_d(\overline{\mathbb{Q}})$ be a $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-orbit. Then $O \cap S_d \neq \emptyset$ if and only if there exist $\Pi \in O$ and a conjugate $c' = g^{-1}cg$ of c such that*

$$c'(\Pi) = \Pi^T.$$

Proof. For $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the point $g\Pi$ lies in S_d if and only if $cT(g\Pi) = g\Pi$. Since T commutes with g, this is equivalent to $g^{-1}cg\,(T\Pi) = \Pi$, that is, $c'(T\Pi) = \Pi$. Because $c'^2 = \mathrm{id}$, applying c' to both sides shows

$$c'(T\Pi) = \Pi \iff T\Pi = c'(\Pi).$$

□

Corollary IV.4 (ghost criterion)**.** *Let $\tilde{\Pi} \in \Gamma_u$, as provided by Theorem II.4. Complex conjugation acts on $\tilde{\Pi}$ as $c(\tilde{\Pi}) = \mathrm{Ad}_{u^T}^{-1}(\tilde{\Pi}^T)$. A SIC exists in the Galois orbit of $\tilde{\Pi}$ if and only if, for some point Π of that orbit, transposition is realized by a conjugate of c rather than by c composed with a displaced parity.*

Equivalently: there is $\tau = c'c \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, a product of two complex conjugations, acting on the ghost $c(\Pi)$ as conjugation by the displaced parity $u^T \in \mathcal{P}$.

Proof. From $\tilde{\Pi}^\dagger = u\tilde{\Pi}u^{-1}$ we get $cT\tilde{\Pi} = \mathrm{Ad}_u\,\tilde{\Pi}$, and hence $T\tilde{\Pi} = \mathrm{Ad}_{u^T}(c\tilde{\Pi})$, using $T\,\mathrm{Ad}_v\,T = \mathrm{Ad}_{(v^T)^{-1}}$. Substituting into Proposition IV.3 gives $c'(\tilde{\Pi}) = \mathrm{Ad}_{u^T}(c\tilde{\Pi})$, that is, $\tau(c\tilde{\Pi}) = \mathrm{Ad}_{u^T}(c\tilde{\Pi})$. □

Corollary IV.4 is the intrinsic form of the "Galois automorphism sending $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ that intertwines with complex conjugation" that Ref. 13 (§1.2) identify as the missing ingredient. In the conjectural picture of Refs. 11 and 15, the field generated by V_d is a ray class field M of $K = \mathbb{Q}(\sqrt{\Delta})$ ramified at both infinite places ∞_1, ∞_2. The two complex conjugations $c_1, c_2 \in \mathrm{Gal}(M/K)$ at these places are conjugate under the nontrivial automorphism of K, and $\tau = c_2 c_1$ is predicted to act as parity. Abstract class field theory guarantees the existence of c_1, c_2. The *action* of τ on the specific Shintani–Faddeev values is the archimedean ($|u^\sigma| = 1$) part of the rank-one Stark conjecture, which is open.

Corollary IV.5 (equivalences)**.** *Assume V_d is finite for the dimensions considered. Then the following hold.*

1. *"Every ghost forces a SIC" holds in dimension d if and only if the criterion of Corollary IV.4 holds for the ghost of Theorem II.4. If it does, a SIC exists in dimension d.*
2. *"$|S_d| = |\Gamma_d|$ for all $d \geq 4$" implies Zauner's existence conjecture for WH SICs, by Theorem II.4.*
3. *In terms of the permutation action of $\mathcal{G} = \langle \mathrm{Gal},\, T,\, \mathit{Clifford} \rangle$ on V_d, we have $|S_d| = \#\,\mathrm{Fix}_{V_d}(cT)$ and $|\Gamma_u| = \#\,\mathrm{Fix}_{V_d}(\mathrm{Ad}_u^{-1}\,cT)$. These are equal whenever the two elements are conjugate in $\mathcal{G}$. By Lemma IV.1, any conjugating element must involve Galois.*

b. The finite-field criterion. At primes whose Frobenius is conjugate to complex conjugation, the count of SICs becomes a count of Frobenius-Hermitian points over finite fields.

Theorem IV.6 (finite-field criterion)**.** *Assume V_d is finite. Let M be a finite Galois extension of $\mathbb{Q}$ containing ω and the entries of all points of V_d. Let ℓ be a prime that is unramified in M, does not divide $d(d+1)$, and lies outside the finite set of primes at which distinct points of V_d become congruent. Suppose some Frobenius element $\mathrm{Frob}_{\mathfrak{L}}$ above ℓ is conjugate to c in $\mathrm{Gal}(M/\mathbb{Q})$. Then*

$$|S_d| = \#\big\{\bar{\Pi} \in V_d(\overline{\mathbb{F}}_\ell) :\ \bar{\Pi}^{(\ell)} = \bar{\Pi}^T\big\},$$

where $\bar{\Pi}^{(\ell)}$ is the entrywise ℓ-th power. By Chebotarev's density theorem, such primes have positive density. Hence a SIC exists in dimension d if and only if, for one such ℓ, the variety V_d has a Frobenius-Hermitian point over $\mathbb{F}_{\ell^2}$.

Proof. Reduction modulo $\mathfrak{L}$ gives a bijection $V_d(\overline{\mathbb{Q}}) \to V_d(\overline{\mathbb{F}}_\ell)$ that intertwines $\mathrm{Frob}_\mathfrak{L}$ with the ℓ-power map and commutes with T. Write $\mathrm{Frob}_\mathfrak{L} = g^{-1}cg$. Then the set $\{\Pi : \mathrm{Frob}_\mathfrak{L}(\Pi) = \Pi^T\}$ equals $g^{-1}S_d$, which has $|S_d|$ elements. □

Remark IV.7. Corollary II.5 supplies modular points at almost all primes. Theorem IV.6 shows that existence of SICs requires the stronger, Frobenius-Hermitian kind of point, and only at primes of a specific Frobenius class. Identifying that class needs knowledge of M, and hence modular arithmetic alone does not bypass the Galois problem.

Remark IV.8 (Brumer–Stark)**.** Dasgupta and Kakde proved the Brumer–Stark conjecture unconditionally[16]. It yields units u with $|u^\sigma| = 1$ at every complex place, but only for complex-multiplication (CM) extensions of totally real fields. The SIC field is not CM: its complex conjugations at ∞_1 and ∞_2 differ by τ, which acts as parity. The parity-fixed subfield is CM, but it only sees $\cos\theta_p$, where $\sqrt{d+1}\,\mathrm{tr}(\Pi D_p) = e^{i\theta_p}$. Thus, the modulus-one information needed for SICs is lost there, and the Brumer–Stark theorem does not transfer directly.

V. GHOST UNITS AND THE GALOIS REALIZATION PROBLEM

Section IV reduced the passage from ghosts to SICs to a Galois question on V_d: by Corollary IV.4, the Galois orbit of a ghost contains a SIC exactly when transposition acts on it through a conjugate of complex conjugation. In this section we express that question in terms of a finite family of algebraic numbers attached to a single ghost, its Weyl coefficients. They turn out to be real and to come in inverse pairs, and the Galois question becomes whether one field automorphism inverts all of them at once. This is the form in which the problem is passed to the finite quantum dilogarithms in Sec. VI.

A. Weyl coefficients of ghosts

Use the convention $D_{-p} = D_p^{-1} = D_p^\dagger$. Let $\tilde{\Pi} \in \Gamma_{U_P}$ be a ghost, as provided by Theorem II.4, and let $\nu_p = \mathrm{tr}(\tilde{\Pi} D_p)$ be its Weyl coefficients. Since the D_p form a basis of $M_d(\mathbb{C})$,

the ν_p determine $\tilde{\Pi}$. They have the following elementary properties.

Lemma V.1. *1. $\nu_p \in \mathbb{R}$ for all p.*

2. $\nu_{-p} = 1/((d+1)\nu_p)$ for $p \neq 0$.

3. $U_P \tilde{\Pi} U_P$ has Weyl coefficients ν_{-p}.

Proof. For (1), $\overline{\nu_p} = \mathrm{tr}(\tilde{\Pi}^\dagger D_p^\dagger) = \mathrm{tr}(U_P \tilde{\Pi} U_P D_{-p}) = \mathrm{tr}(\tilde{\Pi} D_p)$. Part (2) is (1). For (3), $U_P D_p U_P = D_{-p}$. □

Parts (2) and (3) together say that conjugation by the parity operator acts on the Weyl coefficients as inversion, up to the constant factor $d+1$. For a SIC the coefficients would instead satisfy $|\nu_p|^2 = 1/(d+1)$. The difference between the two cases is therefore whether inversion of the ν_p is realized by complex conjugation (a SIC) or only algebraically (a ghost).

B. Galois realization of transposition

Corollary IV.4 combines two different requirements:

(i) transposition acts on the orbit through *some* Galois element;

(ii) that element is conjugate to c.

Requirement (ii) is the archimedean part of the problem, and it is the subject of the rest of the paper. We first isolate requirement (i), which is purely algebraic.

Conjecture V.2 (GR)**.** *Let $O \subset V_d(\overline{\mathbb{Q}})$ be a Galois orbit containing a ghost. Then there is $h_O \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with $\Pi^T = h_O(\Pi)$ for some (equivalently, every) $\Pi \in O$.*

The two forms are equivalent because T commutes with Galois: if $T\Pi = h\Pi$, then $T(g\Pi) = (ghg^{-1})(g\Pi)$. By Proposition IV.3, the orbit O contains a SIC if and only if h_O can be chosen conjugate to c.

For the ghost $\tilde{\Pi}$, Lemma V.1 turns Conjecture V.2 into a statement about its Weyl coefficients alone.

Proposition V.3. *Let $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ fix τ. Then $U_P \tilde{\Pi} U_P = h(\tilde{\Pi})$ if and only if $h(\nu_p) = 1/((d+1)\nu_p)$ for all $p \neq 0$. In that case $T(\tilde{\Pi}) = (c\,h)(\tilde{\Pi})$, and hence Conjecture V.2 holds on the orbit of $\tilde{\Pi}$, with $h_O = ch$.*

Proof. Since h fixes every D_p, the matrix $h(\tilde{\Pi})$ has Weyl coefficients $h(\nu_p)$. Compare these with Lemma V.1(2),(3). Since $\{D_p\}_{p\in\mathbb{Z}_d^2}$ is a basis of $M_d(\mathbb{C})$, orthogonal for the Hilbert–Schmidt inner product, a matrix is uniquely determined by its Weyl coefficients $\mathrm{tr}(\,\cdot\, D_p)$. Finally, $T\tilde{\Pi} = c(\tilde{\Pi}^\dagger) = c(U_P\tilde{\Pi}U_P) = c\,h(\tilde{\Pi})$. □

To state the result in its final form, write $\nu_p = \varepsilon_p/\sqrt{d+1}$. We call the ε_p the *overlap units* of the ghost. By Lemma V.1 they are real and satisfy $\varepsilon_{-p} = \varepsilon_p^{-1}$. Proposition V.3 then says that Conjecture V.2 asks for a field automorphism that inverts the whole family (ε_p) simultaneously, up to sign. Zauner's conjecture asks in addition that this automorphism, composed with c, be a complex conjugation; equivalently, that $|\varepsilon_p| = 1$ for all p under some embedding of $\overline{\mathbb{Q}}$ into $\mathbb{C}$. Thus the existence of a SIC in dimension d is a question about one explicit finite family of algebraic numbers. In Sec. VI we identify these numbers with special values of finite quantum dilogarithms, where Stark arithmetic becomes available.

VI. FINITE QUANTUM DILOGARITHMS AND STARK ARITHMETIC

This section is the core of the paper. Subsection VI A transfers the problem to the RW special values. Subsections VI B and VI C give structure theorems and the reality and signature conditions. Subsections VI D–VI E develop the place law, which subsumes those conditions. Subsection VI I recasts the place law as a unitarity statement for Galois conjugates of the near-group category attached to the Stark solution.

A. The dictionary

The ghost overlaps are built from the special values $F_\gamma^\pm(u)$ of Ref. 12. Under the dictionary of Ref. 13 (Prop. 2.1, eqs. (8)–(10)), the map $p \mapsto -p$ corresponds to $u \mapsto -u$. We therefore study the RW variety directly.

Let $(G, \langle\cdot\rangle)$ be a metric group of order N, and let $Z = Z(G, \langle\cdot\rangle)$ be its finite set of finite quantum dilogarithms (Ref. 12, Def. 4, Thm. 8). Set $K_0 = \mathbb{Q}(\zeta_{4N})$, which contains $\sqrt{N}$ and the values of $\langle\cdot\rangle$. Every $E \in Z$ satisfies $E(0)^2 = \sqrt{N}E(0) + 1$ and $E(u)E(-u) = \langle u\rangle^{-1}$ for $u \neq 0$ (Ref. 12, Thm. 5). Let $Z_\pm = \{E : E(0) = (\sqrt{N} \pm \sqrt{N+4})/2\}$ and $L = K_0(\sqrt{N+4})$, and assume $\sqrt{N+4} \notin K_0$. The reflection $\rho E = E \circ (-1)$ maps Z to itself (Ref. 12, Prop. 6) and commutes with Galois. For the solution E_* introduced below, we write R2 (reflection

realized by Galois) for the statement that some $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ satisfies $h(E_*) = \rho E_*$. Under the dictionary, R2 is the form taken by GR (Conjecture V.2).

Following (Ref. 12, §6), call E *real* if $\overline{E(x)} = \langle x \rangle E(x)$ (the ghost type), and *unitary* if $\overline{E(x)} = \langle -x \rangle E(-x)$, which gives $|E| = 1$ (the SIC type). A solution that is both real and unitary is even, that is, $\rho E = E$. Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ satisfy $\sigma(\sqrt{N+4}) = -\sqrt{N+4}$.

a. From unitary solutions to SICs. The passage from a unitary-type Galois conjugate of E_* to a SIC does not require transferring the TCI or the order-$(2m+1)$ symmetry to the conjugate datum. Galois stability of V_d does the work, through the following results.

Lemma VI.1 (Hermiticity criterion)**.** *A point $\Pi \in V_d$ is Hermitian, that is, $\Pi \in S_d$, if and only if $(d+1)|\operatorname{tr}(\Pi D_p)|^2 = 1$ for all $p \neq 0$.*

Proof. Use the convention $D_p^\dagger = D_{-p}$, and put $a_p = \operatorname{tr}(\Pi D_p)$. Then $\operatorname{tr}(\Pi^\dagger D_p) = \overline{\operatorname{tr}(\Pi D_{-p})} = \overline{a_{-p}}$. Since $\{D_p\}$ is a basis of $M_d(\mathbb{C})$, Π is Hermitian if and only if $\overline{a_{-p}} = a_p$ for all p; for $p = 0$ both sides equal 1. By (1), $(d+1)a_p a_{-p} = 1$ for $p \neq 0$.

If $(d+1)|a_p|^2 = 1$ for all $p \neq 0$, then $\overline{a_{-p}} = 1/((d+1)a_{-p}) = a_p$, and hence Π is Hermitian. Conversely, if Π is Hermitian then $\overline{a_p} = a_{-p}$, and $(d+1)|a_p|^2 = (d+1)a_p a_{-p} = 1$. □

Corollary VI.2 (unit-circle criterion)**.** *Let $\tilde{\Pi} \in V_d(\overline{\mathbb{Q}})$ be a ghost with normalized overlaps $\varepsilon_p = \sqrt{d+1}\,\operatorname{tr}(\tilde{\Pi} D_p)$. If some $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ satisfies $|\sigma(\varepsilon_p)| = 1$ for all $p \neq 0$, then $\sigma(\tilde{\Pi})$ is a WH-covariant SIC fiducial projector in dimension d.*

Proof. By Lemma II.2(2), $\sigma(\tilde{\Pi}) \in V_d$. The automorphism σ maps each D_p to a root of unity times $D_{\pi(p)}$ for a permutation π of $\mathbb{Z}_d^2 \setminus \{0\}$. Hence the moduli $|\operatorname{tr}(\sigma(\tilde{\Pi}) D_q)|$ are the numbers $|\sigma(\operatorname{tr}(\tilde{\Pi} D_p))| = |\sigma(\varepsilon_p)|/\sqrt{d+1}$, using $\sigma(\sqrt{d+1}) = \pm\sqrt{d+1}$. All of these equal $1/\sqrt{d+1}$, and hence Lemma VI.1 applies. □

Corollary VI.2 is an unconditional form of the observation in Ref. 12 (Rem. 7). To connect it with the finite quantum dilogarithm statements we use the AFK dictionary in the following form.

Lemma VI.3 (dictionary)**.** *Let $d \geq 4$ and let $\tilde{\Pi}$ be the AFK ghost of dimension d (rank $r = 1$), with $\gamma = A_t = L^{2m+1}$, its attracting fixed point τ, and $E_* = F_\gamma^+$ on $G_d = \mathbb{Z}^2/\mathbb{Z}^2(\gamma - I)$. For every $p \not\equiv 0 \pmod d$ there are a root of unity ζ_p and an element $u_p \neq 0$ of G_d such that*

$$\varepsilon_p = \zeta_p \, E_*(u_p)^{-1},$$

where ε_p is the normalized overlap $\sqrt{d+1}\,\operatorname{tr}(\tilde{\Pi}D_{p'})$ for the index p' obtained from p by the twist matrix G of Ref. 11, Eq. (1.50). The roots of unity ζ_p and the map $p \mapsto u_p$ are given by formulas with rational coefficients, independent of the embedding.

Proof. By Ref. 11 (Def. 1.11, (1.12), (1.46), (1.47)) with $r = 1$, the ghost fiducial is

$$\tilde{\Pi} = \frac{1}{d} I + \frac{1}{d\sqrt{d+1}} \sum_{p \notin d\mathbb{Z}^2} \tilde{\nu}_{Gp} D_p,$$

with normalized ghost overlaps

$$\tilde{\nu}_p = \phi_p \, \mathbf{v}_{A_t}^{d^{-1}p}(\rho_t).$$

Here $\phi_p = (-1)^{s_d(p)} e^{-\pi i \Psi(A_t)/12} \xi_d^{-f_{jm} Q(p)/f}$ is a root of unity (Ref. 11, Def. 1.30 and Eq. (1.44)), $\mathbf{v}$ is the Shintani–Faddeev modular cocycle, and ρ_t is the fixed point of A_t. The cocycle is evaluated only at the non-integral arguments $d^{-1}p$, $p \not\equiv 0$.

By Ref. 13 (Prop. 2.1), for $u \notin (\gamma^\top - I)\mathbb{Z}^2$, that is $u \neq 0$ in G_d, one has

$$F_\gamma^{\pm}(u) = e^{\pi i \Psi(\gamma)/12} \, \mathbf{v}_\gamma^{\frac{1}{N}(I-\gamma^{-1})Su}(\tau)^{-1},$$

and the integral arguments correspond exactly to $u = 0$. The computation in Ref. 13 (§5), namely $\frac{1}{N}(I-\gamma^{-1})S(L-I)^\top = -d^{-1}L^{-m-1}S$, shows that every argument $d^{-1}p$ is of the form $\frac{1}{N}(I-\gamma^{-1})Su$ modulo $\mathbb{Z}^2$, for a unique class $u = u_p$, which is nonzero. Periodicity of $\mathbf{v}$ in its argument at real-multiplication points (Ref. 17, Prop. 4.35) completes the identification, and $F^+ = F^-$ away from $u = 0$. □

Remark VI.4. Lemma VI.3 is assembled from the statements of Refs. 11 and 13 as quoted above. The matching of conventions (orientation of τ, the matrix S, the twist G, and row versus column vectors) is the bookkeeping carried out in Ref. 13 (§2, §5). A reader should check it against the original. The essential content is simple: each overlap is a single cocycle value at a non-integral argument, and hence $E_*(0)$ never occurs.

The use of Lemma VI.3 in Theorem VI.14 is insensitive to these conventions. The proof needs only two structural facts:

(i) each normalized overlap is a root of unity times $E(u_p)^{\pm 1}$ for a single solution value with $u_p \neq 0$;

(ii) the solution E lies in the Galois orbit of E_*.

A convention mismatch of the usual kind does not affect either fact. Examples are $u_p \mapsto \pm u_p$, $G \mapsto G^{-1}$, $S \mapsto S^\top$, $\mathbf{v} \mapsto \mathbf{v}^{-1}$, or evaluation at the other fixed point of A_t, which replaces

E_* by a Galois conjugate. HS and RU are assumed for all Galois conjugates, and hence the conclusion is unchanged.

As a consequence, all arithmetic of the ghost overlaps is carried by the finite quantum dilogarithm values $E_*(u)$, $u \neq 0$.

B. Structure theorems

Lemma VI.5. *L is a CM field. If $E \in Z_+$ is real and all its values lie in L, then $\sigma E \in Z_-$ is real. If moreover E is not even, neither is σE.*

Proof. K_0 is CM and $\sqrt{N+4}$ is real, and hence complex conjugation is central in $\mathrm{Gal}(L/\mathbb{Q})$. Hence $c(\sigma E) = \sigma(cE) = \sigma(\langle\cdot\rangle E) = \langle\cdot\rangle \sigma E$, using that σ fixes $\langle\cdot\rangle$. Evenness is preserved since $\sigma\rho = \rho\sigma$. □

Theorem VI.6. *Suppose $Z_+ = \{E_*, \rho E_*\}$ with E_* real and not even. Then at least one of the following holds:*

(a) there is $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ with $h(E_*) = \rho E_*$;

(b) Z_- contains a real solution that is not even.

In particular, if Z_- contains no non-even real solution, then reflection is realized by Galois on E_*.

Proof. Every $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ fixes K_0, and hence g maps Z to itself. It also fixes $\sqrt{N+4}$, hence fixes $E(0)$, and hence g permutes Z_+. By hypothesis Z_+ consists of exactly the two elements E_* and ρE_*.

Suppose first that some $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ satisfies $g(E_*) \neq E_*$. Since $g(E_*) \in Z_+ = \{E_*, \rho E_*\}$, necessarily $g(E_*) = \rho E_*$, which is (a).

Otherwise every $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ fixes E_*, that is, fixes each value $E_*(u)$. By Galois theory every value of E_* then lies in L, and Lemma VI.5 shows that σE_* is a non-even real solution in Z_-, which is (b). □

Theorem VI.7. *Let $E_* \in Z_+$ be real. If σE_* is unitary, then $h := (\sigma^{-1} c \sigma)\, c$ lies in $\mathrm{Gal}(\overline{\mathbb{Q}}/L)$ and satisfies $h(E_*) = \rho E_*$.*

Proof. Put $c' = \sigma^{-1} c \sigma$. From $c(\sigma E_*) = \langle\cdot\rangle \rho\sigma E_* = \sigma(\langle\cdot\rangle \rho E_*)$ we get $c'(E_*) = \langle\cdot\rangle \rho E_*$, and by definition $c(E_*) = \langle\cdot\rangle E_*$. Both c and c' invert roots of unity and fix $\sqrt{N+4}$, and hence $h = c'c$ fixes L. Then $h(E_*) = c'(\langle\cdot\rangle E_*) = \langle\cdot\rangle^{-1} \langle\cdot\rangle \rho E_* = \rho E_*$. □

Theorem VI.7 is the finite-dilogarithm form of "Zauner ⇒ GR". Theorem VI.6 is genuinely weaker in its hypothesis. It does not ask that the solutions with $E(0) < 0$ be unitary, only that they not be real.

Theorem VI.6 assumes that Z_+ is a single reflection pair. The next result removes that hypothesis when the relevant orbit has odd size. Let c denote complex conjugation and put $\tilde{c}(E) = \langle\cdot\rangle^{-1}\, c(E)$. By Ref. 12 (Thm. 5(iv), Rem. 4), $\tilde{c}$ is an involution of Z. Real solutions are its fixed points, and unitary solutions are the fixed points of $\rho\tilde{c}$.

Theorem VI.8 (orbit version)**.** *Let $E_* \in Z_+$ be real, and let $O = \mathrm{Gal}(\overline{\mathbb{Q}}/L) \cdot E_*$. Suppose $\rho O \neq O$, that is, reflection is not realized on E_* by any element of $\mathrm{Gal}(\overline{\mathbb{Q}}/L)$. Then $\sigma(O) \subset Z_-$ contains no unitary solution and no even solution. Moreover, $\tilde{c}$ restricts to an involution of $\sigma(O)$ whose fixed points are exactly the real solutions in $\sigma(O)$. Consequently*

$$\#\{\textit{real solutions in } \sigma(O)\} \equiv |O| \pmod 2.$$

In particular, if $|O|$ is odd, then Z_- contains a non-even real solution.

Proof. Let $\Gamma = \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ and $G_L = \mathrm{Gal}(\overline{\mathbb{Q}}/L)$, which has index 2 in Γ. Complex conjugation normalizes both groups.

Take $g \in G_L$ and put $x = \sigma g$. Since E_* is real and $cxc \in \Gamma$ fixes $\langle\cdot\rangle$,

$$\tilde{c}(xE_*) = \langle\cdot\rangle^{-1}(cxc)(cE_*) = (cxc)E_* = x\,hE_*, \qquad h := x^{-1}cxc.$$

Both x and cxc flip $\sqrt{N+4}$, and hence $h \in G_L$. Hence $\tilde{c}$ maps $\sigma(O)$ to itself.

If xE_* were unitary, then $\tilde{c}(xE_*) = \rho xE_* = x\rho E_*$, and hence $hE_* = \rho E_*$. That would put ρE_* in O, and hence $\rho O = O$, a contradiction.

If xE_* were even, then $\rho E_* = E_*$, since ρ commutes with Galois. Again $\rho O = O$, a contradiction.

The parity statement holds because an involution of a finite set fixes a number of points congruent to the size of the set modulo 2. □

Theorem VI.6 is the case $|O| = 1$, since in that case $\rho O \neq O$ forces E_* to be L-rational. When $|O|$ is even, the conclusion is only that $\sigma(O)$ consists of non-unitary solutions, each real or of mixed type. Excluding that possibility requires essentially the full unitarity of Z_-. Thus, the cardinality hypothesis in Theorem VI.6 is doing real work, and an unconditional orbit-wise statement would need a new idea.

a. Complete solution sets in small cases. Theorem 5(iii) of Ref. 12 places E in an eigenspace of $W = \langle\cdot\rangle^{-1} \circ \mathcal{F}$, with eigenvalue λ satisfying $\lambda^3 = \int \langle x\rangle^{-1} dx$. This reduces the pentagon system to a few quadratic equations in eigenspace coordinates, which we solved completely by homotopy continuation.

TABLE IV. All finite quantum dilogarithms in small cases.

N	metric group	$\lvert Z\rvert$	Z_+	Z_-
5	γ of cycle (7)	5	4 real, two ρ-pairs	1 even (real and unitary)
6–9	cyclic, $\langle x\rangle = \zeta_{2N}^{x(x+N)}$	4	1 real ρ-pair	1 unitary non-even ρ-pair
16	$\gamma = \alpha^3$, $\alpha = \left(\begin{smallmatrix} 3 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (SIC, $d = 4$)	8	4 real, two ρ-pairs	4 unitary non-even

For $N = 6, 7, 8, 9$ the hypotheses of Theorem VI.6 hold. A PSLQ test at 300–400 digits shows that $E_*(1) \notin L$, while $E_*(1) + E_*(-1) \in L$ with relation height at most 4. Thus, $\{E_*, \rho E_*\}$ is a single $\mathrm{Gal}(\overline{\mathbb{Q}}/L)$-orbit.

For the $d = 4$ SIC group ($N = 16$), let θ be a generic rational combination of the coordinates. At 120 digits, $\theta(E) \notin L$ for every $E \in Z_+$, and $\theta(E) + \theta(\rho E) \in L$. The sums over the non-reflection pairs are not in L, while the sum over all four solutions is. Hence the orbit of E_* contains ρE_*.

In all these cases, then, reflection is realized by Galois.

b. A larger scan. We applied the same solver to all 94 conjugacy classes of hyperbolic $\gamma \in SL_2(\mathbb{Z})$ with $5 \le N \le 30$. In every class, each solution is either real or unitary. Every solution in Z_+ is real, and every solution in Z_- is unitary. No non-even real solution with $E(0) < 0$ occurs; the only real solution with $E(0) < 0$ is even ($N = 5$). The size $\lvert Z_+\rvert$ ranges over $2, 4, \ldots, 12$. The set Z_- is empty for some classes with $N = 12, 14, 21, 28$. In every such class $\sqrt{N+4} \in K_0$, as it must be: otherwise $\sigma E_* \in Z_-$. The converse fails; for example, $N = 12$ with cycle $(4, 4)$ has $Z_- \neq \emptyset$. When $\sqrt{N+4} \in K_0$, any automorphism sending $d_- \mapsto d_+$ must move the cyclotomic coefficients. The pseudo-unitary companion predicted by Thornton's theorem (Ref. 18, Thm. IV.3.6) then lies in the solution set of a Galois-conjugate metric datum, not in Z_- itself.

In Izumi's construction the dimension of ρ is $-\sqrt{N}/E(0)$ (Ref. 12, §5.4). Hence $E(0) < 0$ corresponds to positive (Frobenius–Perron) dimension. Unitary-type solutions with $E(0) < 0$ are exactly Izumi's input data for unitary near-group categories. The exceptional case of Theorem VI.6 is therefore a pseudo-unitary solution of real type.

C. Reality and signature conditions

Conjecture VI.9 (real-or-unitary dichotomy)**.** *For the metric groups arising from hyperbolic $\gamma \in SL_2(\mathbb{Z})$, every finite quantum dilogarithm is of real type or of unitary type. Equivalently, the involution $\tilde{c}$ of Z satisfies $\tilde{c}(E) \in \{E, \rho E\}$ for every $E \in Z$.*

This holds in all 94 classes with $N \leq 30$; no mixed-type solution was found. Combined with Theorem VI.8, Conjecture VI.9 gives the following. If R2 fails on an orbit O of any size, then $\sigma(O)$ consists entirely of non-even real solutions with $E(0) < 0$.

Conjecture VI.9 together with the sign statement (no non-even real solution with $E(0) < 0$) is equivalent to unitarity of all non-even solutions in Z_-. That is the full dichotomy, and it is of Zauner strength. The point of separating the two statements is that each alone is strictly weaker.

a. A reduced system, and total reality. Put $x_u = s_u E(u)$, where $s_u^2 = \langle u \rangle$ and $s_{-u} = s_u$. In these coordinates:

- $\tilde{c}$ is ordinary complex conjugation, and ρ is $x_u \mapsto x_{-u}$;
- the pairing relation becomes $x_u x_{-u} = 1$ for $u \neq 0$;
- the eigenvalue condition $\hat{E} = \lambda \langle \cdot \rangle E$ says that x lies in a subspace X_λ, the image of the eigenspace of the order-three unitary $W = \langle \cdot \rangle^{-1} \mathcal{F}$ (essentially Izumi's $R_{c'}$ (Ref. 19, Rem. 9.2)).

Both $\tilde{c}$ and ρ preserve X_λ. Real type means $x \in \mathbb{R}^G$, and unitary type means $|x_u| = 1$ for all $u \neq 0$. Either way, every $t_u = x_u + x_u^{-1}$ is real.

Conjecture VI.10 (total reality)**.** *Let $(G, \langle \cdot \rangle)$ be a metric group and $\lambda^3 = \int \langle x \rangle^{-1} dx$. Every $x \in X_\lambda$ with $x_0^2 = \sqrt{N} x_0 + 1$ and $x_u x_{-u} = 1$ for all $u \neq 0$ is real or unimodular. Equivalently, the even part t of x is real.*

Caution on interpretation. Despite its linear-algebraic appearance, Conjecture VI.10 is *not* a property of the Weil eigenspace X_λ. The control experiments below show that total reality fails for random subspaces with the same symmetries, and also fails for the true eigenspace once any of the pairing equations is dropped. TR is a property of the full, overdetermined solution variety, which is nonempty only because of the RW identities. Proposition VI.11 below shows that it is in fact an arithmetic statement.

The reduced system omits the pentagon equation. In every computed case it has the

same solutions as the full RW system, apart from two classes, $N = 9$ cycle $(2, 2, 5)$ and $N = 16$ cycle $(3, 3, 3)$, where extra solutions appear; these are also totally real.

We tested Conjecture VI.10 on all 268 nondegenerate cyclic metrics $\langle x\rangle = \zeta_{2N}^{ax(x+N)}$ with $5 \le N \le 21$, and on the 54 $SL_2(\mathbb{Z})$ classes with $N \le 20$. Every solution is real or unimodular, and none is of mixed type. Conjecture VI.10 implies Conjecture VI.9 coordinatewise.

Proposition VI.11 (arithmetic form of TR)**.** *Let $\mathcal{S}$ be the union of the solution sets of Conjecture VI.10 over all Galois conjugates of the metric datum $(\langle\cdot\rangle, \sqrt{N}, \lambda)$. Then $\mathcal{S}$ is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable. Conjecture VI.10 holds on all of $\mathcal{S}$ if and only if, for every solution, the field $\mathbb{Q}(t_u : u \in G)$ is totally real. Equivalently, every Galois conjugate of every x_u is real or of absolute value* 1*.*

Proof. Galois automorphisms act coordinatewise and map the data s_u, $\langle\cdot\rangle$ and λ to conjugate data, and hence $\mathcal{S}$ is stable and $\sigma(t_u(E)) = t_u(\sigma E)$. Hence every conjugate of t_u is again a t-coordinate of a solution, and it is real exactly when TR holds there. Finally, $x + x^{-1} \in \mathbb{R}$ if and only if $x \in \mathbb{R}$ or $|x| = 1$. □

For the Stark solution E_*, the rank-one Stark conjecture predicts exactly such a pattern: each conjugate is real at the distinguished real place and of absolute value 1 elsewhere. Thus, Conjecture VI.10, across all conjugate metric data, is a *weakened archimedean Stark statement*. It asserts that every conjugate is real or unimodular, without saying which. The sign–type correlation is precisely the missing information about which conjugates are which. TR is therefore not independent of Stark arithmetic, but it is strictly weaker than the full archimedean Stark statement.

Salem-type behaviour. The condition in Conjecture VI.10 has the shape of the defining property of Salem numbers: all conjugates real or unimodular, with totally real trace $\tau+\tau^{-1}$. We computed the minimal polynomials of t_u for the Stark solutions with $N = 6, 7, 9$. In every case they are totally real, which confirms TR at the level of minimal polynomials, independently of solver completeness. Exactly half of the roots lie outside $[-2, 2]$. Thus, the units x_u are Salem-type but not Salem: Salem numbers would have exactly one root outside. The half–half split is what the rank-one Stark conjecture predicts, since the places of the ray class field lie equally over the two infinite places of the real quadratic base field. Section VI D makes this precise: the base field is $F = \mathbb{Q}(\sqrt{N(N+4)})$, and the split is by the place of F (Definition VI.19).

Conjecture VI.12 (half split, HS)**.** *For the Stark solution E_* and every $u \neq 0$, all Galois conjugates of $t_u = x_u + x_u^{-1}$ are real. Moreover, exactly half of them satisfy $|t| > 2$ and the other half satisfy $|t| < 2$.*

Conjecture VI.12 strengthens Conjecture VI.10 for E_*. Up to the boundary case $|t| = 2$, the two halves are the conjugates at which x_u is real and those at which it is unimodular. HS records the archimedean *signature* predicted by Stark, but not which embedding is the distinguished one. Section VI D identifies the two halves: they are the embeddings above the two infinite places of the real quadratic field of γ. Its relation to the other statements is as follows.

Proposition VI.13. *Suppose $\sqrt{N+4} \notin K_0$. If Conjecture VI.12 holds for E_* and Conjecture VI.9 holds for all Galois-conjugate metric data, then some Galois conjugate of E_* is of unitary type.*

Proof. By HS, some $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ gives $|\sigma(t_u)| < 2$ for some u. Then $\sigma(x_u)$ is unimodular and not real. The conjugate σE_* solves the system for the conjugate metric datum, and it is not of real type because $\sigma(x_u) \notin \mathbb{R}$. By Conjecture VI.9 it is therefore of unitary type. □

Theorem VI.14 (HS and RU give SICs)**.** *Let $d \geq 4$ with $d+1$ neither a square nor twice a square, that is, with $\sqrt{N+4} \notin K_0$ (Proposition VI.57). If Conjecture VI.12 holds for E_* and Conjecture VI.9 holds for all Galois-conjugate metric data of G_d, then a WH-covariant SIC exists in dimension d.*

Proof. By Proposition VI.13 there is σ with σE_* of unitary type. For $u \neq 0$, unitary type gives

$$|\sigma E_*(u)|^2 = \langle u \rangle' \, \sigma E_*(u) \, \sigma E_*(-u) = 1,$$

where $\langle \cdot \rangle'$ is the conjugate metric. By Lemma VI.3, $\sigma(\varepsilon_p) = \sigma(\zeta_p)\, \sigma(E_*(u_p))^{-1}$ with $u_p \neq 0$. Hence $|\sigma(\varepsilon_p)| = 1$ for every $p \not\equiv 0$, and Corollary VI.2 gives the SIC. □

Conversely, Stark's archimedean conjecture implies HS. HS is therefore not "easier than Zauner" in any absolute sense: it is Zauner-strength modulo the dichotomy RU. What HS isolates is the counting part of the archimedean information, and RU isolates the coordinatewise purity part.

Control experiments. Total reality is not a generic feature of such systems. We ran two controls.

1. Replace X_λ by a random ρ- and conjugation-invariant subspace with dimensions chosen such that the pairing system is square. Then non-real solutions are common, for example only 16 of 64 solutions are real or unimodular for $N = 13$.
2. Keep the true Weil eigenspace but impose only a random square subset of the pairing equations. Again non-real solutions appear in most cases, along with mixed solutions in which some coordinates are real and others unimodular.

Thus, the phenomenon requires both the Weil eigenspace and the *full* overdetermined set of pairing equations. Any proof must use the whole system, and no square subsystem can be totally real in general. In particular, a representation-theoretic proof based on the order-three Weil operator alone cannot work. A total-reality mechanism of Shapiro type, such as a positive-definite trace form on the coordinate ring, would have to be built from the complete overdetermined system.

Izumi already observed this phenomenon in small cases (Ref. 19, Rem. 9.2). He noted that his C^* equations split into a Galois-stable part and two extra equations, $Jb = b$ and $d > 0$, which "come from the unitarity". He also noted that the Galois conjugates with $d < 0$ satisfy all equations except those two.

Proposition VI.15 (categorical form of R2)**.** *Let $\mathcal{C}_E$ be the Izumi near-group category attached to E (Ref. 12, Thm. 12), and let $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$. Then $h(E) = \rho E$ if and only if the Galois-conjugate category $h(\mathcal{C}_E)$ is equivalent to $\mathcal{C}_E$ by an equivalence that acts on the invertible objects by $g \mapsto -g$.*

Proof. By Ref. 12 (§5.5), E is recovered from the category up to relabelling G by an automorphism, and the relabelling is read off from the action on the invertible objects. The category $h(\mathcal{C}_E)$ is built from the data $(\langle\cdot\rangle, h(E))$, since h fixes K_0. Relabelling by -1 preserves $\langle\cdot\rangle$, and hence $\mathcal{C}_{\rho E} \simeq \mathcal{C}_E$ via $g \mapsto -g$. □

Thus, R2 says that $\mathcal{C}_E$ is Galois-stable with a prescribed action on its invertibles. By Ref. 19 (Cor. 13.5), $\mathrm{Aut}(\mathcal{C}_E)$ consists of the automorphisms of G that preserve $(\langle\cdot\rangle, E)$. Hence for non-even E the relabelling -1 is not an autoequivalence of $\mathcal{C}_E$, and the equivalence in Proposition VI.15 is necessarily Galois-twisted. Note that reflection does not produce a new category: $\mathcal{C}_{\rho E} \simeq \mathcal{C}_E$. The arithmetic content lies entirely in whether a Galois automorphism realizes this autoequivalence. In the same way, real type corresponds to $\tilde{c}$ fixing E, which is

the complex conjugate datum, and unitary type to $\tilde{c}$ acting as $g \mapsto -g$. This is the form in which Izumi's C^*-condition appears.

Remark VI.16 (relation to the Haagerup–Izumi programme)**.** Gannon, Schopieray and Yadav[20] construct cyclic Haagerup–Izumi (HI) categories from Barnes double-sine values. They work on the branch $\dim \rho = d_-$, and their pseudo-unitary d_+ categories are Galois conjugates. Such a conjugate is unitary exactly when its HI matrix is Hermitian (Ref. 21, Thm. 2(c)). Exact Hermiticity checks verify this for every odd $3 \le N \le 31$ (Ref. 20, Rem. 3.8), and the introduction of Ref. 20 reports verification for all odd $N \le 43$. The uniform statement remains open (Ref. 20, Rem. 3.8).

The near-group situation here is closely parallel. Unitary-type solutions with $E(0) < 0$ are Izumi's data for unitary near-group categories. The dichotomy "Z_- is unitary" is the near-group analogue of the open HI unitarity question. Moreover, for every odd $N \ge 3$, the equivariantization of an untwisted cyclic HI category for $\mathbb{Z}/N$ is a near-group category of type $(G \times \widehat{G},\, N^2)$ with $G = \mathbb{Z}/N$ and quadratic form $q(x, \chi) = \chi(x)$ (Ref. 20, Cor. 6.5). This is the Izumi type $(G, |G|)$ considered here. It does not, however, reach the SIC groups directly. The SIC metric group G_d has order $d^2(d-3)$, which is a square n^2 only when $d - 3 = k^2$, and then $n = dk$ is even, since $d = k^2 + 3$ is even whenever k is odd. Thus, the odd-N family of Ref. 20 never contains a SIC group. Transferring HI unitarity results to Z_- for G_d would require extending those methods to other metric groups, including groups of even order. Section VI I makes the near-group side of this analogy precise: for a pseudo-unitary conjugate, unitarity reduces to Izumi's Hermiticity condition, which is exactly unitary type (Proposition VI.52).

Remark VI.17 (on the strength of this evidence)**.** In every computed case, Z_- consists of unitary solutions. By Theorem VI.7 this already forces the conclusion. Thus, the computations are consistent with Conjecture V.2, but they do not show that GR is easier than Zauner's conjecture. What does separate the two is the weaker hypothesis of Theorem VI.6: when $|Z_+| = 2$, R2 follows as soon as Z_- contains no non-even real solution. An earlier hope that $|Z_+| = 2$ always holds is false: $|Z_+| \in \{2, 4, 6, 8, 10, 12\}$ all occur (Table IX).

b. The sign statement and the sign–type correlation. Two further statements recur below. Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ with $\sigma(\sqrt{N+4}) = -\sqrt{N+4}$, as in Lemma VI.5.

- The *sign statement*: no non-even real solution has $E(0) < 0$. This is the hypothesis under which Theorem VI.6 yields R2.
- The *sign–type correlation*: every solution with $E(0) < 0$ is of unitary type, or at least the specific conjugate σE_* is. This implies Z(d) directly, through Lemma VI.3 and Corollary VI.2, since σE_* always exists. Given TR, it also implies HS.

c. How these conditions are related. Section VI D introduces the place law PL and shows that it implies each of the conditions of this subsection. The implications are:

PL $\Longrightarrow$ HS $\Longrightarrow$ TR	(Prop. VI.21)
PL $\Longrightarrow$ sign–type correlation	(Thm. VI.22, Lemma VI.18)
PL $\Longrightarrow$ RU on the Galois orbit of E_*	(Thm. VI.22)
PL $\Longrightarrow$ GR	(Thm. VI.7)

In addition, the parity of PL is detected 2-adically (Corollary VI.27). HS, TR, RU and the sign–type correlation are therefore best read as shadows of PL: HS records the count, TR the reality, RU the purity, and the sign–type correlation the restriction to automorphisms fixing K_0.

d. A counting form of HS. Put $B_u = \{\sigma : |\sigma(t_u)| > 2\}$, and let A be the set of σ at which σE_* has negative categorical dimension. By Lemma VI.18, $A = \{\sigma : \sigma(\sqrt{D}) > 0\}$. On $\mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ this is the set where $\sigma E_*(0) > 0$, but over the full Galois group the sign of $E(0)$ is *not* the right invariant (Sec. VI D).

- $|A|$ is exactly half of the Galois group, for trivial reasons. Thus, given TR, the whole content of HS is that $|B_u|$ is also half: $|A| = |B_u|$.
- The place law is the equality $A = B_u$ for every u.

HS is thus the counting shadow of PL, and the converse fails in general. Showing that two counts agree is the kind of statement global Galois or regulator arguments can often reach. However, HS alone does not give SICs without RU, whereas PL does (Theorem VI.23). In the data, the equality $A = B_u$ holds in every case (Appendix A 2).

D. The place law and its single-unit and 2-adic forms

This subsection states the place law and proves that it gives SICs. It then reduces the archimedean input to a single unit, and finally shows which part of that input is visible 2-adically.

a. Which conjugates are unitary. The statements HS, RU and the sign–type correlation all concern the same question: at which embeddings is a conjugate of E_* real, and at which is it unitary? This subsection answers it in a Galois-invariant form. The answer is governed by the real quadratic field of γ, not by the sign of $E(0)$ alone. It yields a single statement that implies Zauner's conjecture without RU (Theorem VI.23), and a reduction of the archimedean input to one unit (Theorem VI.24).

Throughout, $E_* \in Z_+$ is the real-type Stark solution of Sec. VI. We fix $\overline{\mathbb{Q}} \subset \mathbb{C}$ with $\sqrt{N} > 0$ and $\sqrt{N+4} > 0$, and we regard every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ as an embedding $\overline{\mathbb{Q}} \to \mathbb{C}$. Put

$$D = N(N+4) = (\operatorname{tr}\gamma)^2 - 4, \qquad F = \mathbb{Q}(\sqrt{D}), \qquad \sqrt{D} = \sqrt{N}\sqrt{N+4} > 0.$$

Thus, F is the real quadratic field generated by the fixed point of γ. Since $(N+2)^2 - 4$ is never a square for $N \geq 1$, F is a genuine quadratic field. Its two infinite places $\infty_\pm$ are $\sqrt{D} \mapsto \pm\sqrt{D}$. As before, $x_u = s_u E_*(u)$ and $t_u = x_u + x_u^{-1}$. Recall $x_{-u} = x_u^{-1}$, and $x_u = \pm 1$ when $2u = 0$.

b. Types of conjugates. A conjugate σE_* solves the RW system for the conjugate datum, in which $\langle\cdot\rangle$, $\sqrt{N}$ and $E(0)$ are replaced by $\sigma\langle\cdot\rangle$, $\sigma(\sqrt{N}) = \pm\sqrt{N}$ and $\sigma(E_*(0))$. Since $|\sigma(s_u)| = 1$ and $\sigma(s_u)^2 = \sigma\langle u\rangle$, a direct computation from the definitions gives:

- σE_* is of real type if and only if $\sigma(x_u) \in \mathbb{R}$ for all u;
- σE_* is of unitary type if and only if $|\sigma(x_u)| = 1$ for all $u \neq 0$.

For example, if $|\sigma(x_u)| = 1$, then

$$\overline{\sigma E_*(u)} = \frac{\sigma(s_u)}{\sigma(x_u)} = \sigma\langle u\rangle \frac{\sigma(x_{-u})}{\sigma(s_{-u})} = \sigma\langle u\rangle\, \sigma E_*(-u).$$

Lemma VI.18 (sign of the categorical dimension)**.** *For $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the categorical dimension $d_\rho(\sigma E_*) = -\sigma(\sqrt{N})/\sigma(E_*(0))$ of the conjugate datum satisfies*

$$\operatorname{sign} d_\rho(\sigma E_*) = -\operatorname{sign}\sigma(\sqrt{D}).$$

In particular, for $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ one has $d_\rho(\sigma E_*) > 0$ if and only if $\sigma(E_*(0)) < 0$.

Proof. $\sigma(E_*(0)) = (\sigma(\sqrt{N}) + \sigma(\sqrt{N+4}))/2$ has the sign of $\sigma(\sqrt{N+4})$, because $\sqrt{N+4} > \sqrt{N}$. Hence $\operatorname{sign} d_\rho = -\operatorname{sign}\sigma(\sqrt{N})\operatorname{sign}\sigma(\sqrt{N+4}) = -\operatorname{sign}\sigma(\sqrt{D})$. If σ fixes $K_0 \ni \sqrt{N}$, the first factor is $+1$. □

Lemma VI.18 shows that the condition "$E(0) < 0$" of the earlier sections is the correct one only for automorphisms fixing K_0. Lemma VI.50 below gives the exact value $d_\rho(\sigma E_*) = (N - \sigma(\sqrt{D}))/2$, and shows that positive categorical dimension is the same as pseudo-unitarity of the conjugate near-group category. Over the full Galois group, the invariant quantity is the sign of the categorical dimension, or equivalently the place of F below σ. This matters as soon as σ moves $\sqrt{N}$, and the data confirm it. For $N = 6$ and $u = 2$, $\sqrt{N+4}$ and $\sqrt{N}$ both lie in $\mathbb{Q}(t_u)$, but each splits the conjugates of t_u 4/4 against the pattern $|t| \gtrless 2$, whereas $\sqrt{D}$ splits them 8/0 (Appendix A 2).

Definition VI.19 (place law)**.** Let $u \in G_d$ with $2u \neq 0$ and $t_u \neq \pm 2$. We say that E_* *satisfies the place law at* u, written PL(u), if for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ the number $\sigma(t_u)$ is real and

$$\big(\sigma(t_u)^2 - 4\big)\,\sigma(\sqrt{D}) > 0.$$

That is, $|\sigma(t_u)| > 2$ above ∞_+ and $|\sigma(t_u)| < 2$ above ∞_-.

Conjecture VI.20 (place law)**.** *For every $d \geq 4$, the Stark solution E_* of the SIC group G_d satisfies* PL(u) *for every $u \in G_d$ with $2u \neq 0$.*

By Theorem VI.23 below, Conjecture VI.20 implies Zauner's existence conjecture for WH-covariant SICs in every dimension $d \geq 4$. It is implied by the archimedean part of the rank-one Stark conjecture.

Proposition VI.21 (equivalent forms of PL)**.** *Let $2u \neq 0$ and $t_u \neq \pm 2$, and put $y_u = D^{1/4}(x_u - x_u^{-1})$ for a fixed real fourth root $D^{1/4}$. The following are equivalent.*

(i) PL(u).

(ii) $\mathbb{Q}(t_u)$ *is totally real,* $\sqrt{D} \in \mathbb{Q}(t_u)$*, and* $(t_u^2 - 4)\sqrt{D}$ *is totally positive in* $\mathbb{Q}(t_u)$*.*

(iii) The field $\mathbb{Q}(t_u, y_u)$ *is totally real.*

If they hold, then exactly $m_u = [\mathbb{Q}(t_u) : F]$ conjugates of t_u satisfy $|t| > 2$, and exactly m_u satisfy $|t| < 2$. In particular PL(u) *implies HS at u.*

Proof. (i)⇒(ii). If $\sqrt{D} \notin \mathbb{Q}(t_u)$, some σ, σ' agree on t_u but $\sigma'(\sqrt{D}) = -\sigma(\sqrt{D})$, which contradicts (i). The rest of (ii) is a restatement of (i). (ii)⇒(i) is immediate.

(i)⇔(iii). We have $y_u^2 = (t_u^2 - 4)\sqrt{D}$. If (i) holds, every $\sigma(y_u)^2$ is positive and every $\sigma(t_u)$ is real, and hence $\mathbb{Q}(t_u, y_u)$ is totally real. Conversely, if $\mathbb{Q}(t_u, y_u)$ is totally real, then

$\sigma(y_u)^2 > 0$, since $y_u \neq 0$. Also $\sigma(\sqrt{D}) = \sigma(y_u)^2/\sigma(t_u^2 - 4)$ is real and nonzero, and hence $(\sigma(t_u)^2 - 4)\sigma(\sqrt{D}) = \sigma(y_u)^2 > 0$.

For the count, $F \subset \mathbb{Q}(t_u)$, and exactly half of the $[\mathbb{Q}(t_u) : \mathbb{Q}] = 2m_u$ embeddings of $\mathbb{Q}(t_u)$ lie above each of $\infty_\pm$. □

Form (iii) shows that the place law is a total-reality statement of the same kind as TR (Proposition VI.11). TR asks that the even part $x_u + x_u^{-1}$ generate a totally real field. PL asks the same, in addition, for the odd part $x_u - x_u^{-1}$ twisted by $D^{1/4}$.

Theorem VI.22 (the place law determines every type)**.** *Suppose $t_u \neq \pm 2$ whenever $2u \neq 0$. Then* PL(u) *holds for all u with $2u \neq 0$ if and only if, for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$:*

- *σE_* is of real type when $\sigma(\sqrt{D}) > 0$;*
- *σE_* is of unitary type when $\sigma(\sqrt{D}) < 0$.*

Equivalently, by Lemma VI.18, a conjugate of E_* is unitary if and only if its categorical dimension is positive.

Proof. Suppose PL holds for all u, and fix σ.

- If $\sigma(\sqrt{D}) > 0$, then each $\sigma(t_u)$ is real with $|\sigma(t_u)| > 2$. Thus, $\sigma(x_u)$, a root of $X^2 - \sigma(t_u)X + 1$, is real. Hence σE_* is of real type.
- If $\sigma(\sqrt{D}) < 0$, then $\sigma(t_u) \in (-2, 2)$, and hence $|\sigma(x_u)| = 1$ for every $u \neq 0$. Hence σE_* is of unitary type.

The two indices u with $2u = 0$ and $u = 0$ impose no condition. The type is decided by $\sigma|_F$ alone, and hence it is the same for all u simultaneously.

Conversely, real type gives $\sigma(x_u) \in \mathbb{R}$ and hence $|\sigma(t_u)| \geq 2$. Unitary type gives $|\sigma(x_u)| = 1$ and hence $\sigma(t_u) \in [-2, 2]$. The value ± 2 is excluded by hypothesis, and hence the inequalities are strict and match the sign of $\sigma(\sqrt{D})$. □

Theorem VI.23 (the place law gives SICs)**.** *Let $d \geq 4$ and let E_* be the Stark solution for G_d. If* PL(u) *holds for every $u \in G_d$ with $2u \neq 0$, then a WH-covariant SIC exists in dimension d. No assumption of RU, and no assumption on $d+1$, is needed.*

Proof. $\sqrt{D}$ is irrational, and hence some σ satisfies $\sigma(\sqrt{D}) < 0$. By Theorem VI.22, σE_* is of unitary type, and hence $|\sigma(E_*(u))| = 1$ for $u \neq 0$. The proof of Theorem VI.14 (Lemma VI.3 and Corollary VI.2) then gives the SIC. □

c. Relation to the earlier statements. The place law is the Galois-invariant completion of the conjectures of this section.

- *TR.* PL implies TR for every conjugate of E_*, by form (ii).
- *HS.* PL implies HS, and it identifies the two halves: they are the embeddings above ∞_+ and above ∞_- (Proposition VI.21).
- *RU.* By Theorem VI.22, PL for all u implies RU on the Galois orbit of E_*, which is all that Theorem VI.14 uses. The danger that different u choose different types, which RU was introduced to exclude, cannot occur, because under PL the type depends only on $\sigma|_F$.
- *Sign–type correlation.* Restricted to $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$, Theorem VI.22 is exactly the sign–type correlation of Sec. VII, by Lemma VI.18. PL is that correlation extended to automorphisms moving $\sqrt{N}$.
- *R2 and GR.* Suppose $\sqrt{N+4} \notin K_0$. Take $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ with $\sigma(\sqrt{N+4}) = -\sqrt{N+4}$. Then $\sigma(\sqrt{D}) < 0$, and hence PL makes σE_* unitary, and Theorem VI.7 gives R2 with the explicit element $h = (\sigma^{-1}c\sigma)c$. Thus PL implies Conjecture V.2 as well.
- *Stark.* PL says that each x_u is real at every place above ∞_+ and has absolute value 1 at every place above ∞_-. This is exactly the shape of the archimedean part of the rank-one Stark conjecture over F. Thus, PL is not weaker than Stark's archimedean statement; it is that statement, with the distinguished place made explicit.

Accordingly, Theorem VI.23 replaces the pair HS + RU of Theorem VI.14 by a single statement. Moreover, each PL(u) concerns one algebraic number at a time.

d. Evidence. For 39 of the 94 classes of Table XVIII ($5 \le N \le 18$), including the $d = 4$ SIC group ($N = 16$, cycle $(3, 3, 3)$), we identified the minimal polynomials of the traces t_u of the real solutions in Z_+. These are 100 distinct polynomials, of degree up to 32 (Table XVI). Each was tested exactly in PARI/GP for three properties:

- total reality;
- whether $\sqrt{D}$ lies in $\mathbb{Q}(t_u)$;
- the sign of $(\sigma(t_u)^2 - 4)\sigma(\sqrt{D})$ at every embedding σ.

In all 100 cases the place law holds, with no exception. It also holds in the SIC dimension $d = 5$ ($N = 50$), at all 16 embeddings (Sec. VI D, end). In contrast, $\mathbb{Q}(\sqrt{N+4})$ and $\mathbb{Q}(\sqrt{N})$ are often not contained in $\mathbb{Q}(t_u)$, and when they are, they do not predict the split.

e. Reduction of the archimedean input to a single unit. Corollary VI.2 needs *all* overlap units to be unimodular under one embedding. Theorem VI.23 obtains this from PL at every u. The next result shows that algebraic information transports unimodularity from one unit to all of them.

Theorem VI.24 (single-unit reduction)**.** *Let $u_0 \in G_d$, and suppose:*

(G1) (generation) $x_v \in \mathbb{Q}(x_{u_0})$ *for every* $v \in G_d$*;*

(G2) (inversion) *every field embedding* $\kappa : \mathbb{Q}(x_{u_0}) \to \overline{\mathbb{Q}}$ *with* $\kappa(x_{u_0}) = x_{u_0}^{-1}$ *satisfies* $\kappa(x_v) = x_v^{-1}$ *for all* v*.*

If some Galois conjugate of x_{u_0} lies on the unit circle, then the corresponding conjugate σE_* is of unitary type, and a WH-covariant SIC exists in dimension d.

Proof. Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with $|\sigma(x_{u_0})| = 1$, and put $c_\sigma = \sigma^{-1} c \sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Then

$$c_\sigma(x_{u_0}) = \sigma^{-1}\big(\overline{\sigma(x_{u_0})}\big) = \sigma^{-1}\big(\sigma(x_{u_0})^{-1}\big) = x_{u_0}^{-1}.$$

Thus, c_σ restricts to an embedding κ of $\mathbb{Q}(x_{u_0})$ as in (G2), and $c_\sigma(x_v) = x_v^{-1}$ for every v, by (G1) and (G2). Applying σ gives $\overline{\sigma(x_v)} = \sigma(c_\sigma x_v) = \sigma(x_v)^{-1}$, that is, $|\sigma(x_v)| = 1$ for all $v \neq 0$. Hence σE_* is of unitary type, and the SIC follows as in Theorem VI.23. □

Corollary VI.25 (R2 version)**.** *Assume R2, that is, $h(E_*) = \rho E_*$ for some $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$, and assume (G1) for some u_0. If some conjugate of x_{u_0} is unimodular, then a SIC exists in dimension d.*

Proof. Since h fixes $s_v \in K_0$, we have $h(x_v) = s_v E_*(-v) = x_{-v} = x_v^{-1}$ for all v. An embedding of $\mathbb{Q}(x_{u_0})$ is determined by the image of x_{u_0}, and hence every κ as in (G2) equals $h|_{\mathbb{Q}(x_{u_0})}$, and (G2) holds. □

Theorem VI.7 gives the converse direction, Z ⇒ R2. Thus, modulo the algebraic conditions (G1)–(G2), the archimedean content of Zauner's conjecture is concentrated in one algebraic number: *does x_{u_0} have a conjugate on the unit circle?* The next corollary turns this into the sign of a rational integer.

Corollary VI.26 (integer criterion)**.** *Let P be the minimal polynomial of t_{u_0}, of degree n, and suppose $[\mathbb{Q}(x_{u_0}) : \mathbb{Q}(t_{u_0})] = 2$. Then*

$$P(2)P(-2) = \prod_i (t_i^2 - 4) = (-1)^n \, N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}\big(x_{u_0}^2 - 1\big),$$

where t_i runs over the roots of P. If $P(2)P(-2) < 0$, then P has a real root in $(-2,2)$, and hence x_{u_0} has a non-real unimodular conjugate. Under (G1)–(G2), a SIC then exists in dimension d. Under $\mathrm{PL}(u_0)$*, the number of roots of P in $(-2,2)$ is $m_{u_0} = n/2$, and hence $P(2)P(-2) < 0$ holds exactly when m_{u_0} is odd.*

Proof. $P(2)P(-2) = \prod(2 - t_i)(-2 - t_i) = \prod(t_i^2 - 4)$. Non-real roots occur in conjugate pairs and contribute positive factors, real roots outside $[-2,2]$ contribute positive factors, and real roots inside contribute negative ones. Thus, the sign is $(-1)^{\#\{i:\, t_i \in (-2,2)\}}$. For the norm identity, note that $N_{\mathbb{Q}(x)/\mathbb{Q}(t)}(x-1) = (x-1)(x^{-1}-1) = 2-t$ and $N_{\mathbb{Q}(x)/\mathbb{Q}(t)}(x+1) = 2+t$. Hence $P(2) = N_{\mathbb{Q}(x)/\mathbb{Q}}(x-1)$ and $P(-2) = (-1)^n N_{\mathbb{Q}(x)/\mathbb{Q}}(x+1)$. □

f. Exact checks. The conditions (G1) and (G2) are exact statements about the minimal polynomial of x_{u_0}. We verified them in PARI/GP for 18 classes, including the $d = 4$ SIC group (Table XVII). In every class some u_0 satisfies (G1), and (G2) holds for every such u_0.

- For $d = 4$, all six admissible u_0 generate, with $[\mathbb{Q}(x_{u_0}) : \mathbb{Q}] = 8$ and $m_{u_0} = 2$. Thus, the integer criterion does not apply there, but a Sturm count of the roots of P in $(-2,2)$ does.
- In four classes ($N = 9$ with cycles $(2,2,5)$ and (11), and $N = 12$ with cycles $(2,3,2,3)$ and $(4,4)$), a generating u_0 has m_{u_0} odd and $P(2)P(-2) < 0$. There, Theorem VI.24 certifies the existence of a unitary-type conjugate from the sign of a single integer.

g. A 2-adic shadow of the place law. The single-unit criterion asks whether one totally real algebraic integer has a conjugate in $(-2,2)$. The parity of the number of such conjugates can be read off from two integers.

Corollary VI.27 (2-adic signature criterion)**.** *Let $P \in \mathbb{Z}[X]$ be monic with $P(0)$ odd, and let $t_1, \ldots, t_n$ be its roots. Then $P(2)P(-2) \equiv 1 \pmod 4$, and consequently*

$$\#\{i :\ t_i \in (-2,2)\} \ \equiv\ \frac{|P(2)|\,|P(-2)| - 1}{2} \pmod 2.$$

In particular, let P be the minimal polynomial of t_{u_0}, with $[\mathbb{Q}(x_{u_0}) : \mathbb{Q}(t_{u_0})] = 2$ as in Corollary VI.26, and hence $|P(2)| = |N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}(x_{u_0}-1)|$ and $|P(-2)| = |N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}(x_{u_0}+1)|$. Suppose

(i) (G1) and (G2) hold for u_0 (for instance, R2 and (G1));

(ii) $N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}(x_{u_0} - 1)$ is odd;

(iii) $|N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}(x_{u_0} - 1)| \cdot |N_{\mathbb{Q}(x_{u_0})/\mathbb{Q}}(x_{u_0} + 1)| \equiv 3 \pmod 4$.

Then a WH-covariant SIC exists in dimension d.

Proof. Write $P(X) = Q(X^2) + X\,R(X^2)$ with $Q, R \in \mathbb{Z}[X]$. Then

$$P(2)P(-2) = Q(4)^2 - 4R(4)^2 \equiv Q(4)^2 \pmod 4.$$

Also $Q(4) \equiv Q(0) = P(0) \pmod 4$ is odd, and hence $Q(4)^2 \equiv 1 \pmod 8$, and $P(2)P(-2) \equiv 1 \pmod 4$. An odd integer $M \equiv 1 \pmod 4$ is negative exactly when $|M| \equiv 3 \pmod 4$. The parity formula then follows from the sign rule in the proof of Corollary VI.26.

For the second part, $P(0) = \pm N_{\mathbb{Q}(t)/\mathbb{Q}}(t)$ and $t = (x^2+1)/x$ with x a unit, and hence $P(0)$ is odd if and only if $x^2 + 1 \equiv (x-1)^2$ is prime to 2, that is, if and only if $N(x-1)$ is odd. By (iii), $P(2)P(-2) < 0$, and Corollary VI.26 gives the SIC. □

h. What the criterion detects. Corollary VI.27 is the first statement in this paper in which the surviving archimedean obstruction leaves a non-archimedean footprint. Hypotheses (ii) and (iii) involve only the prime-ideal factorizations of $(x_{u_0} - 1)$ and $(x_{u_0} + 1)$ and a condition at 2. No embedding into $\mathbb{R}$ or $\mathbb{C}$ is used. Yet they determine one bit of the archimedean signature, namely the parity of the number of unimodular conjugates of x_{u_0}. This is an elementary instance of Hilbert reciprocity: the sign of a rational number is determined by its local data at the finite primes.

The criterion detects only this parity. Under PL the number of unimodular conjugates is $m_{u_0} = [\mathbb{Q}(t_{u_0}) : F]$, and hence the criterion can succeed only when m_{u_0} is odd. It says nothing about how the conjugates are distributed among the places of F, which is the content of PL. The two statements live at different levels:

$$\text{finite primes} \longrightarrow |N(x_u \pm 1)| \longrightarrow \text{parity of the unitary count},$$
$$\text{PL} \longrightarrow \text{full distribution among the infinite places.}$$

When $P(0)$ is even, the sign of $P(2)P(-2)$ is still determined by 2-adic data: by $|P(2)P(-2)|$ together with $P(2)P(-2)$ modulo $2^{v_2(P(2)P(-2))+2}$, that is, by congruences of t_{u_0} modulo a power of 2. In that case, however, it is no longer a function of the two norms alone.

i. Data. Across all 100 polynomials, $P(2)P(-2) < 0$ holds exactly for the 13 with m_u odd, as Corollary VI.26 predicts under PL.

- $P(0)$ is odd for every polynomial with N odd, for 35 of the 38 with $N \equiv 2 \pmod 4$, and for none with $N \equiv 0 \pmod 4$.

- The congruence $P(2)P(-2) \equiv 1 \pmod 4$ holds for all 57 polynomials with $P(0)$ odd, as it must.
- For the SIC groups, $N = d^2(d-3) \equiv 2 \pmod 4$ exactly when $d \equiv 1 \pmod 4$ ($d = 5, 9, 13, \dots$). This is where hypothesis (ii) is expected to hold, and it does for $d = 5$ (below). For $d = 4$, $N = 16$ and $P(0)$ is even.

The parity that the criterion reads off is in fact an invariant of a field, independent of the chosen unit.

Proposition VI.28 (the parity is a field invariant)**.** *Let $M = \mathbb{Q}(x_v : v \in G_d)$, and suppose (G1)–(G2) hold for u_0, with $x_{u_0} \neq x_{u_0}^{-1}$. Then κ is an automorphism of M of order 2, and $\mathbb{Q}(t_{u_0}) = M^\kappa$ has degree $[M : \mathbb{Q}]/2$. If moreover* PL(u_0) *holds, the number of conjugates of t_{u_0} in $(-2, 2)$ is $[M : \mathbb{Q}]/4$. Consequently, under PL, Corollaries VI.26 and VI.27 can succeed for a generating unit only if $[M : \mathbb{Q}] \equiv 4 \pmod 8$. Whether they succeed does not depend on which generating unit is used.*

Proof. By (G1) we have $M = \mathbb{Q}(x_{u_0})$, and by (G2) $\kappa(x_v) = x_v^{-1} \in M$ for all v. Thus, $\kappa(M) = M$ and $\kappa^2 = \mathrm{id}$. The element t_{u_0} is κ-invariant, and x_{u_0} is a root of $X^2 - t_{u_0}X + 1$, and hence $[M : \mathbb{Q}(t_{u_0})] \leq 2$. Since $\kappa \neq \mathrm{id}$, this forces $\mathbb{Q}(t_{u_0}) = M^\kappa$. Under PL, exactly half of the $[M : \mathbb{Q}]/2$ embeddings of $\mathbb{Q}(t_{u_0})$ lie above ∞_-, and these are the conjugates in $(-2, 2)$. □

j. The case $d = 5$. The RW system for $N = 50$ is too large for our complete solver, and hence we used the ghost side instead. We took one of the 80 U_P-Hermitian ghosts in the complete solution set of V_5 (Table II) and refined it to 400 digits. Its normalized overlaps ε_p are real by Lemma V.1. Since $x_{u_p} = s_{u_p}\zeta_p\varepsilon_p^{-1}$ (Lemma VI.3) and both x_{u_p} and ε_p are real, $s_{u_p}\zeta_p = \pm 1$. Hence $t_{u_p} = \pm(\varepsilon_p + \varepsilon_p^{-1})$, and the minimal polynomial is determined up to $X \mapsto -X$.

We validated this method on $d = 4$: the ghost reproduces exactly the polynomial $X^4 - 6X^2 + 4$ obtained from the RW solution for cycle $(3, 3, 3)$, together with (G1) and (G2) for every generating u_0.

For $d = 5$ we have $D = 2700$ and $F = \mathbb{Q}(\sqrt{3})$. The results are as follows.

- All 24 overlaps give one minimal polynomial,

$$P(X) = X^{16} - 32X^{14} + 423X^{12} - 2984X^{10} + 12105X^8 \\ - 28304X^6 + 36023X^4 - 21432X^2 + 3721.$$

- P is totally real, with 8 roots outside and 8 inside $[-2, 2]$.
- $\sqrt{3} \in \mathbb{Q}(t)$, and the place law holds at all 16 embeddings.
- $P(0) = 61^2$ is odd, as predicted for $d \equiv 1 \pmod 4$.
- $P(2) = P(-2) = 5^2$.
- (G1) and (G2) hold exactly for $u_0 = (0, 1)$, with $[\mathbb{Q}(x_{u_0}) : \mathbb{Q}] = 32$ (Table XVII).

Thus $m = 8$ and $[M : \mathbb{Q}] = 32$, and by Proposition VI.28 the parity criteria cannot apply in dimension 5, whichever unit is used.

The $d = 5$ computation therefore confirms PL in the first SIC dimension beyond $d = 4$. It also shows a limitation: the 2-adic criterion is controlled by the 2-part of $[M : F]$. The method succeeds only when $[M : F] \equiv 2 \pmod 4$, and this fails for $d = 4$ ($[M : F] = 4$) and $d = 5$ ($[M : F] = 16$).

The norms $|N(x_u \pm 1)|$ are not prime powers. Examples are $5^4 \cdot 13$ ($N = 13$), $17 \cdot 101^4$ ($N = 17$) and $7^2 \cdot 839^2$ ($N = 14$). They are nevertheless highly structured. Apart from primes dividing N, almost every prime factor satisfies $p \equiv \pm 1 \pmod N$: the primes 53, 19, 83, 839, 61, 31, 1471, 101, 8059, 107 and 17 occur for $N = 9, 10, 14, 14, 15, 16, 16, 17, 17, 18, 18$. Such primes split completely in $\mathbb{Q}(\zeta_N)^+$. The exceptions are 7 for $N = 16$ (with $7 \equiv -1 \pmod 8$, occurring as 7^4) and 3 for $N = 5$. This is the pattern expected if the norms are governed by splitting in the relevant ray class field. A formula for $|N(x_u \pm 1)|$, if one exists, is therefore more likely an Euler-product or Gross–Stark-type description over split primes than a closed expression. Together with (G1)–(G2), such a description would prove Z(d) for every d satisfying hypotheses (ii)–(iii) of Corollary VI.27 for some generating u_0. The full table of $P(\pm 2)$ is included in the supplementary material.

E. A class-field form of the place law

The place law is an archimedean statement, but Proposition VI.28 shows that norms of $x_u \pm 1$ see only one bit of it. We now show that, under the algebraic part of the Stark picture, the *whole* place law is equivalent to a congruence at a single prime. We then verify

this congruence for the SIC groups $d = 4$ and $d = 5$.

Let $M = \mathbb{Q}(x_v : v \in G_d)$ and assume $t_u \neq \pm 2$ whenever $2u \neq 0$. By Proposition VI.21, PL forces $F \subset M$; we assume this throughout.

Proposition VI.29 (class-field form of PL)**.** *Suppose that M/F is abelian. Then the following hold.*

(a) Every place of M above ∞_+ is real. All complex conjugations at places of M above ∞_- restrict to a single element $c_- \in \mathrm{Gal}(M/F)$ with $c_-^2 = 1$.

(b) PL(u) *holds for every u with $2u \neq 0$ if and only if $c_-(x_v) = x_v^{-1}$ for every v, that is, if and only if $c_- = \kappa$ on M.*

(c) Let $\mathfrak{f}$ be the finite part of the conductor of M/F, and let $\beta \in \mathcal{O}_F$ satisfy $\beta \equiv 1 \pmod{}^{\times} \mathfrak{f}$, $\beta > 0$ at ∞_+ and $\beta < 0$ at ∞_-. Then c_- is the Artin symbol of the ideal (β). Consequently PL *holds if and only if, for one (equivalently every) prime $\mathfrak{p} \nmid \mathfrak{f}$ of F in the ray class of (β) at which the reduction of the minimal polynomial of x_{u_0} is separable, and every prime $\mathfrak{P} \mid \mathfrak{p}$ of M,*

$$x_v^{N\mathfrak{p}+1} \equiv 1 \pmod{\mathfrak{P}} \qquad \text{for all } v.$$

Under (G1), it suffices to check $v = u_0$.

Proof. (a) The identity embedding of M is real, because E_* is of real type, and hence every x_v is real. It lies above ∞_+, since $\sqrt{D} > 0$ there. Because M/F is Galois, every place above ∞_+ is a conjugate of the identity place, and its decomposition group is conjugate to the trivial one. Hence every such place is real. The complex conjugation at a place above ∞_- fixes F, and hence it lies in $\mathrm{Gal}(M/F)$. These elements are conjugate in $\mathrm{Gal}(M/F)$, which is abelian, and hence they coincide.

(b) Let $\sigma : M \to \mathbb{C}$ lie above ∞_-. Then $\overline{\sigma(y)} = \sigma(c_- y)$ for $y \in M$. If $c_-(x_v) = x_v^{-1}$, then $|\sigma(x_v)| = 1$, and hence $\sigma(t_v) \in [-2, 2]$, and in fact $\sigma(t_v) \in (-2, 2)$ because $t_v \neq \pm 2$. Above ∞_+, $\sigma(x_v)$ is real by (a), and hence $|\sigma(t_v)| > 2$. This is PL. Conversely, PL gives $|\sigma(x_v)| = 1$ above ∞_-. Therefore $\sigma(c_- x_v) = \overline{\sigma(x_v)} = \sigma(x_v)^{-1}$, and hence $c_-(x_v) = x_v^{-1}$.

(c) The idele whose only nontrivial component is -1 at ∞_- maps to c_- under the Artin map. The principal idele β has trivial Artin image. Its components at the primes dividing $\mathfrak{f}$ are $\equiv 1$, and its component at ∞_+ is positive. Hence the Artin symbol of the ideal (β) equals $c_-^{-1} = c_-$, and every prime $\mathfrak{p}$ in the ray class of (β) has $\mathrm{Frob}_{\mathfrak{p}} = c_-$. Since $\mathrm{Frob}_{\mathfrak{P}}(y) \equiv y^{N\mathfrak{p}}$

(mod $\mathfrak{P}$), the identity $c_- = \kappa$ is equivalent to $x_v^{N\mathfrak{p}} \equiv x_v^{-1} \pmod{\mathfrak{P}}$, given separability. Combine this with (b). □

Proposition VI.29 moves the place law from the infinite places to the finite ones. Under the algebraic hypothesis that M/F is abelian, the full archimedean distribution asserted by PL, and not merely its parity, is equivalent to a single congruence: *at primes of F in the ray class of (β), the overlap units reduce into the norm-one torus.* This is the finite-field analogue of the unit circle. It refines the finite-field criterion of Theorem IV.6 by naming the relevant primes through the sign of a generator at the two infinite places of F.

Proposition VI.29 is a translation, not a new conjecture. It restates, for the overlap units, the classical archimedean Stark relation $\varepsilon^{1+c_-} = 1$, where c_- is complex conjugation at the ramified infinite place. The involution $\kappa : x_v \mapsto x_v^{-1}$ plays the role of $\varepsilon \mapsto \varepsilon^{-1}$, and hence the place law becomes the identity $c_- = \kappa$. The hypothesis that M/F is abelian is the algebraic (non-archimedean) part of Stark's conjecture in this setting. It is not proved here.

a. Test for $d = 4$ and $d = 5$. We used the generating unit x_{u_0} of Sec. VI D (from the U_P-ghost, Table XVII), computed its relative minimal polynomial over F, and applied PARI/GP (`nfgaloisconj`, `rnfconductor`, `bnrisprincipal`). For every degree-one prime $\mathfrak{p}$ of F with $N\mathfrak{p} \le 5000$ not dividing the conductor, we compared two independent quantities:

- the action of $\mathrm{Frob}_\mathfrak{p}$ on x_{u_0}, read off from the factorization of the relative minimal polynomial modulo $\mathfrak{p}$;
- the ray class of $\mathfrak{p}$.

The results are in Table V.

- In both dimensions, M/F is abelian, M is not Galois over $\mathbb{Q}$, and the conductor is ramified at ∞_- only.
- The subgroup cut out by M is trivial, and hence M is the *full* ray class field of F modulo $\mathfrak{f}\infty_-$, with $\mathfrak{f} = (8) = (2)^3$ for $d = 4$ and $\mathfrak{f} = \mathfrak{p}_2^4(5)$ for $d = 5$ (Table V).
- At every one of the 1305 primes tested, $\mathrm{Frob}_\mathfrak{p}$ acts as κ exactly when $\mathfrak{p}$ lies in the class of (β), acts trivially exactly in the trivial class, and acts in neither way otherwise.
- The opposite sign pattern ($\beta' \equiv 1$, $\beta' < 0$ at ∞_+, $\beta' > 0$ at ∞_-) gives the trivial class, as it must, since ∞_+ is unramified.

Thus, the congruence distinguishes the two infinite places of F. Together with Proposition VI.29, this confirms that $c_- = \kappa$, which is the place law, in both SIC dimensions. It

confirms it through finite primes alone.

TABLE V. Class-field test of the place law in the SIC dimensions $d = 4, 5, 9, 13$. In every case the unit field is abelian over F and ramified at ∞_- only. Here M is the field of the overlap units and M_2 the field of their squares (the sign-free unit field). Panel (b) tests $c_- = \kappa$ (Proposition VI.29) at the degree-one primes $\mathfrak{p} \nmid \mathfrak{f}$ of F; entries are matches/primes tested, and there are no mismatches. The class of c_- is the ray class of (β).

(a) Structure of the unit field

d	F	field	Galois group over F	conductor	identification
4	$\mathbb{Q}(\sqrt{5})$	M	$(\mathbb{Z}/2)^2$	$(8)\,\infty_-$	exact
5	$\mathbb{Q}(\sqrt{3})$	M	$\mathbb{Z}/8 \times \mathbb{Z}/2$	$\mathfrak{p}_2^4(5)\,\infty_-$	exact
9	$\mathbb{Q}(\sqrt{15})$	M_2	$\mathbb{Z}/12 \times \mathbb{Z}/3$	$(9)\,\infty_-$	exact
13	$\mathbb{Q}(\sqrt{35})$	M_2	$\mathbb{Z}/24 \times \mathbb{Z}/4$	$(13)\,\infty_-$	Frobenius orders[†]

(b) Frobenius test of $c_- = \kappa$

d	primes tested	class of c_-: $\mathrm{Frob}_\mathfrak{p} = \kappa$	trivial class: $\mathrm{Frob}_\mathfrak{p} = 1$	other classes: $\mathrm{Frob}_\mathfrak{p} \notin \{1, \kappa\}$
4	653 ($N\mathfrak{p} \le 5000$)	164/164	156/156	333/333
5	652 ($N\mathfrak{p} \le 5000$)	36/36	34/34	582/582
9	—	not needed: $c_- = \kappa$ follows from Proposition VI.30		
13	2230 ($N\mathfrak{p} \le 20000$)	15/15	order of $\mathrm{Frob}_\mathfrak{p}$ = order of class: 2230/2230	

[†]For $d = 13$ the trace field M_2^κ (conductor (13), group $\mathbb{Z}/12 \times \mathbb{Z}/4$) was computed exactly; M_2 was identified by comparing the order of $\mathrm{Frob}_\mathfrak{p}$ with the order of the class of $\mathfrak{p}$ in $\mathrm{Cl}((13)\infty_-)$. Ghost overlaps were refined to 400 ($d = 5$), 2000 ($d = 9$) and 20500 ($d = 13$) digits.

b. Conductor. For $d = 5$, the finite conductor $\mathfrak{p}_2^4(5)$ of $\mathbb{Q}(\varepsilon_p)$ has a 2-part that comes only from the sign ambiguity of the units, which reflects the phase convention for the displacement operators. The sign-free combinations ε_p^2, $\varepsilon_p\varepsilon_q$ and $\varepsilon_p/\varepsilon_q$ generate the full ray class field of $\mathbb{Q}(\sqrt{3})$ modulo $(5)\infty_-$, with Galois group $\mathbb{Z}/8$. For $d = 4$, the units themselves give $(8)\infty_-$, and their squares give $(4)\infty_-$. In both cases the conductor is $d'\infty_-$ for the appropriate version of the unit field, where $d' = d$ for d odd and $d' = 2d$ for d even is the integer that appears in the AFK construction[11]. Thus, for $d = 4, 5$, the overlap-unit field of the RW/Izumi construction and the conductor of the AFK construction describe the same ray class field, ramified at ∞_- alone.

c. The next SIC dimensions, $d = 9$ and $d = 13$. The same pattern persists (Table V). For $d = 9$, the sign-free unit field M_2 is exactly the ray class field of $\mathbb{Q}(\sqrt{15})$ modulo $(9)\infty_-$, with $\mathrm{Gal}(M_2/F) \cong \mathbb{Z}/12 \times \mathbb{Z}/3 = \mathrm{Cl}((9)\infty_-)$. For $d = 13$, the trace field M_2^κ is exactly

the ray class field of $\mathbb{Q}(\sqrt{35})$ modulo (13), with Galois group $\mathbb{Z}/12 \times \mathbb{Z}/4$ and no infinite ramification. The Frobenius test identifies M_2 itself with the ray class field modulo $(13)\infty_-$, of degree 96 over F. At all 15 tested primes in the class of c_-, the Frobenius acts as κ. In both dimensions PL was also checked directly: for every conjugate of $t = x_{u_0}^2 + x_{u_0}^{-2}$ (36 for $d = 9$, 96 for $d = 13$), the sign of $t^2 - 4$ agrees with the sign of the corresponding conjugate of $\sqrt{D}$, and exactly half of the conjugates lie in $(-2, 2)$. Refining the norm pattern observed above, the prime factors of $P_t(\pm 2)$ outside the conductor all satisfy $p \equiv \pm 1 \pmod{4d}$. For example, they are $37, 71, 181, 431, 1151, 295129$ for $d = 9$. Thus, in every SIC dimension computed, the sign-free unit field is the full ray class field of F of conductor $d'\infty_-$, and its κ-fixed subfield is the ray class field of conductor d'. For $d = 17$ the units were not identified at 20500 digits, and we estimate that about 45000 digits are needed.

The place law depends only on $|\sigma(x_v)|$, and hence Proposition VI.29 and its proof apply verbatim to M_2 in place of M. In particular, PL is equivalent to $c_- = \kappa$ on M_2 whenever M_2/F is abelian.

Proposition VI.30 (unique involution)**.** *Let M' be M or M_2, and suppose the following hold:*

(i) M'/F is Galois;

(ii) inversion $\kappa : x_v \mapsto x_v^{-1}$ is an automorphism of M' fixing F, and $\kappa \neq \mathrm{id}$;

(iii) ∞_- ramifies in M';

(iv) $\mathrm{Gal}(M'/F)$ has exactly one element of order 2.

Then every complex conjugation of M' above ∞_- equals κ. Hence PL *holds, and therefore* Z(d) *holds.*

Proof. A complex conjugation at a place of M' above ∞_- fixes F. By (iii) it is nontrivial, and hence it is an element of order 2 in $\mathrm{Gal}(M'/F)$. By (ii), κ is also an element of order 2 in $\mathrm{Gal}(M'/F)$. By (iv) they coincide. The proof of Proposition VI.29(a) uses only that M'/F is Galois, and hence every place above ∞_+ is real. The argument of Proposition VI.29(b) then gives PL, and Theorem VI.23 gives Z(d). □

Proposition VI.30 removes the archimedean input entirely in such dimensions. What remains, (i) and (ii), is purely algebraic: it is the algebraic part of the Stark picture together with R2. Hypothesis (ii) also shows that $\mathrm{Gal}(M'/F)$ always has even order whenever

inversion is a field automorphism, and hence families with $\mathrm{Gal}(M'/F)$ of odd order cannot occur.

When does (iv) hold? Suppose, as in all computed cases, that M_2 is the full ray class field of F modulo $d'\infty_-$. Then (iii) holds exactly when $|\mathrm{Cl}(d'\infty_-)| = 2\,|\mathrm{Cl}(d')|$. Condition (iv) then says that the 2-Sylow subgroup of $\mathrm{Cl}(d'\infty_-)$ is cyclic. Among $4 \le d \le 2000$ with ∞_- ramified, this happens for exactly 24 dimensions:

$$5,\ 7,\ 9,\ 11,\ 15,\ 17,\ 19,\ 27,\ 47,\ 53,\ 67,\ 71,\ 99,$$
$$191,\ 199,\ 431,\ 449,\ 467,\ 787,\ 881,\ 971,\ 1061,\ 1619,\ 1681.$$

The counts by range are 15/247 for $4 \le d \le 250$, then 3, 0, 3, 1, 0, 2, 0 in successive blocks of 250 up to 2000. For $d = 4$ and $d = 13$ the 2-Sylow is not cyclic, $(\mathbb{Z}/2)^2$ and $\mathbb{Z}/8 \times \mathbb{Z}/4$ respectively, but $c_- = \kappa$ holds there numerically (Table V). Genus theory, the contribution of $(\mathcal{O}_F/d')^\times$, and the sign at ∞_- force 2-torsion whenever $\mathrm{disc}\, F$ or d' has many prime factors. By the Erdős–Kac theorem the number of prime factors of the square-free part of $(d-3)(d+1)$ grows like $\log\log d$, so the set of d with cyclic 2-Sylow is expected to have density zero (unlike the odd part of class groups, which Cohen–Lenstra heuristics model). Infinitude is still plausible under Bateman–Horn or Schinzel-type conjectures on prime-pattern families (for instance prime d with $d+1 = 2m^2$ and m^2-2 prime), but it is unproved, and even an infinite such set would still require the algebraic hypotheses (i)–(ii) of Proposition VI.30 in those dimensions.

Proposition VI.31 (the unit field is not cyclotomic over F)**.** *Let K be a finite extension of F contained in $F \cdot \mathbb{Q}^{\mathrm{ab}}$, the compositum of F with all cyclotomic fields. Then the complex conjugations of K above ∞_+ and above ∞_- coincide. In particular, K is ramified at ∞_+ if and only if it is ramified at ∞_-. Consequently, if M is ramified at ∞_- but not at ∞_+, as for $d = 4, 5$, then $M \not\subset F \cdot \mathbb{Q}^{\mathrm{ab}}$.*

Proof. Let c be complex conjugation for the identity embedding, which lies above ∞_+, and let $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with $g(\sqrt{D}) = -\sqrt{D}$. A complex conjugation above ∞_- is gcg^{-1}. On $F \cdot \mathbb{Q}^{\mathrm{ab}}$, both c and gcg^{-1} fix F, because F is real at both places. Restricting further to $\mathbb{Q}^{\mathrm{ab}}$, where the Galois group is abelian, gives $gcg^{-1}|_{\mathbb{Q}^{\mathrm{ab}}} = c|_{\mathbb{Q}^{\mathrm{ab}}}$. An element of $\mathrm{Gal}(F\mathbb{Q}^{\mathrm{ab}}/F)$ is determined by its restriction to $\mathbb{Q}^{\mathrm{ab}}$. Therefore c and gcg^{-1} agree on $F\mathbb{Q}^{\mathrm{ab}}$, and hence on K. □

Proposition VI.31 explains why the remaining step cannot be elementary. The field M, and with it the identity $c_- = \kappa$, cannot be reached through roots of unity, Gauss sums, or Kronecker–Weber. Any proof must construct abelian extensions of the real quadratic field F that are ramified at exactly one real place, which is the content of Hilbert's twelfth problem for F.

d. What this changes. For a given d, the place law, and hence $\mathrm{Z}(d)$ by Theorem VI.23, follows from two inputs:

(i) the *algebraic* statement that the overlap-unit field M is abelian over F, which is expected to be a ray class field of conductor $\mathfrak{f}\infty_-$;

(ii) the *congruence* $x_{u_0}^{N\mathfrak{p}+1} \equiv 1 \pmod{\mathfrak{P}}$ at a single prime $\mathfrak{p}$ in the class of c_-.

When $\mathrm{Gal}(M/F)$ has a unique element of order 2, input (ii) is automatic once κ is an automorphism of M fixing F and ∞_- ramifies (Proposition VI.30). Neither input refers to an embedding into $\mathbb{C}$. The remaining difficulty is not removed: proving (ii) uniformly in d requires knowing the reductions of the RW special values modulo such primes. It is, however, moved to the setting of Gross–Stark-type and p-adic methods, where comparable statements have been proved.

F. Total reality in place of the unique involution

Proposition VI.30 reaches $c_- = \kappa$ without any archimedean input, but it pays for this with hypothesis (iv), a condition on $\mathrm{Cl}(d'\infty_-)$ that holds for only 24 dimensions below 2000 and is expected to have density zero. We now show that (iv) can be traded for total reality (TR), the weakest-looking archimedean statement of Sec. VI C. Under the algebraic hypotheses this does not weaken the archimedean input, because TR then turns out to be equivalent to PL (Corollary VI.33); what changes is that no condition on the class group remains, and that the ramification hypothesis becomes unconditional on a set of dimensions of positive density.

Proposition VI.32 (total reality replaces the unique involution)**.** *Let M' be M or M_2, and suppose:*

(i) M'/F is Galois;

(ii) inversion κ is an automorphism of M' fixing F, and $\kappa \neq \mathrm{id}$;

(iii) ∞_- ramifies in M';

(iv) the fixed field M'^{κ} is totally real.

Then every complex conjugation of M' at a place above ∞_- equals κ. Hence PL *holds, and therefore* Z(d) *holds.*

Proof. Let w be a place of M' above ∞_-, let $\sigma : M' \to \mathbb{C}$ induce it, and let c_w be the associated complex conjugation. Since F is real at ∞_-, c_w fixes F, and hence $c_w \in \mathrm{Gal}(M'/F)$. Because M'/F is Galois, all places above ∞_- are conjugate, and hence by (iii) all of them are complex; in particular $c_w \neq \mathrm{id}$.

By (iv), $\sigma(M'^{\kappa}) \subset \mathbb{R}$, and hence c_w acts trivially on M'^{κ}. Therefore $c_w \in \mathrm{Gal}(M'/M'^{\kappa})$, and $\mathrm{Gal}(M'/M'^{\kappa}) = \langle\kappa\rangle$ by Artin's lemma, a group of order 2 by (ii). Since $c_w \neq \mathrm{id}$, we get $c_w = \kappa$. Consequently $\overline{\sigma(x_v)} = \sigma(\kappa x_v) = \sigma(x_v)^{-1}$, that is, $|\sigma(x_v)| = 1$ for every v. The proof of Proposition VI.29(a) uses only that M'/F is Galois, and hence every place above ∞_+ is real; the argument of Proposition VI.29(b) then gives PL, and Theorem VI.23 gives Z(d). □

Hypothesis (iv) is exactly Conjecture VI.10. Indeed, under (G1) one has $[M : \mathbb{Q}(t_{u_0})] = 2$ and $t_v \in M^{\kappa}$ for every v, so that

$$M^{\kappa} = \mathbb{Q}(t_v : v \in G_d) = \mathbb{Q}(t_{u_0}),$$

and total reality of that field is TR in the form of Proposition VI.11. For M_2 the same applies with $t_v^2 - 2$ in place of t_v.

Corollary VI.33 (the archimedean hierarchy collapses)**.** *Assume (i)–(iii) of Proposition VI.32 together with (G1). Then*

$$\mathrm{TR} \iff \mathrm{HS} \iff \mathrm{PL}.$$

Proof. PL ⇒ HS ⇒ TR is Proposition VI.21. The converse is Proposition VI.32. □

Remark VI.34. Propositions VI.30 and VI.32 are not comparable hypothesis by hypothesis: the former removes the archimedean input altogether, whereas the latter retains TR. What is gained is reach, not a weaker archimedean hypothesis: under (i)–(iii), TR is equivalent to PL by Corollary VI.33. TR was verified for all 268 nondegenerate cyclic metrics with $5 \leq N \leq 21$ and all 54 $SL_2(\mathbb{Z})$ classes with $N \leq 20$, whereas the cyclic 2-Sylow condition of Proposition VI.30 holds for only 24 dimensions $d \leq 2000$. Under Corollary VI.33 the separation between TR, HS and PL recorded in Sec. VI C disappears: the three differ only

in the absence of the algebraic hypotheses. In particular, proving TR in these dimensions is exactly as hard as proving PL.

a. The remaining hypothesis is about units, not class groups. Assume, as in all computed cases, that M_2 is the ray class field of F modulo $d'\infty_-$. Hypothesis (iii) is then the statement that this conductor really involves ∞_-, that is, $|\mathrm{Cl}(d'\infty_-)| = 2\,|\mathrm{Cl}(d')|$. It admits a criterion free of class numbers.

Proposition VI.35 (unit criterion for ramification at ∞_-)**.** *Put* $\mathfrak{m}_0 = d'\mathcal{O}_F$ *and* $\mathfrak{m} = \mathfrak{m}_0\infty_-$*, and let*

$$U_1 = \{\eta \in \mathcal{O}_F^\times : \eta \equiv 1 \pmod{d'}\}.$$

The natural surjection $\mathrm{Cl}(\mathfrak{m}) \to \mathrm{Cl}(\mathfrak{m}_0)$ *has kernel of order* 2 *if every* $\eta \in U_1$ *is positive at* ∞_-*, and is an isomorphism otherwise. Hence* ∞_- *ramifies in the ray class field of conductor* $\mathfrak{m}$ *if and only if*

$$(\mathrm{C}_d) \qquad \textit{no unit of } F \textit{ congruent to } 1 \textit{ modulo } d' \textit{ is negative at } \infty_-.$$

Proof. The kernel is $P_{\mathfrak{m}_0,1}/P_{\mathfrak{m},1}$, where $P_{\mathfrak{m}_0,1}$ consists of the principal ideals with a generator $\equiv 1 \pmod{d'}$, and $P_{\mathfrak{m},1}$ requires in addition a generator positive at ∞_-. Two generators of one ideal that are both $\equiv 1 \pmod{d'}$ differ by an element of U_1, and hence $(\alpha) \mapsto \mathrm{sgn}_{\infty_-}(\alpha)$ induces an isomorphism $P_{\mathfrak{m}_0,1}/P_{\mathfrak{m},1} \cong \{\pm 1\}/\mathrm{sgn}_{\infty_-}(U_1)$. The sign map is onto $\{\pm 1\}$, because $\alpha = 1 - nd'$ is $\equiv 1 \pmod{d'}$ and negative at ∞_- for n large. By class field theory the ray class field of conductor $\mathfrak{m}$ strictly contains that of conductor $\mathfrak{m}_0$ exactly when the kernel is nontrivial, and the additional ramification is then at ∞_-. □

Throughout this part put $\Delta = (d-3)(d+1) = (d-1)^2 - 4$; then $F = \mathbb{Q}(\sqrt{\Delta})$, since $D = d^2(d-2)^2\Delta$. The point is that the unit $\varepsilon_\alpha = \frac{1}{2}(d - 1 + \sqrt{\Delta})$, the eigenvalue of α that exceeds 1 at ∞_+, reduces modulo d to a root of unity of *odd* order.

Lemma VI.36 (ε_α is a cube root of unity modulo d)**.** $\varepsilon_\alpha \in \mathcal{O}_F$*,* $N_{F/\mathbb{Q}}(\varepsilon_\alpha) = 1$*, and* $\varepsilon_\alpha > 0$ *at both infinite places. Moreover* $\varepsilon_\alpha^2 = (d-1)\varepsilon_\alpha - 1$*, and hence*

$$\varepsilon_\alpha^3 = d(d-2)\,\varepsilon_\alpha + (1-d).$$

Consequently:

(a) *for d odd, $\varepsilon_\alpha^3 \equiv 1 \pmod d$; in fact $\Delta \equiv -3 \pmod d$, so the image of ε_α in $\mathcal{O}_F/d$ is $\frac{1}{2}(-1+\sqrt{-3})$, a primitive cube root of unity;*

(b) *for d even, $\varepsilon_\alpha^3 \equiv 1-d \pmod{2d}$ and $(1-d)^2 \equiv 1 \pmod{2d}$, so ε_α has order dividing 6 in $(\mathcal{O}_F/2d)^\times$.*

Proof. Since $2\varepsilon_\alpha-(d-1)=\sqrt{\Delta}$ and $\Delta=(d-1)^2-4\equiv(d-1)^2 \pmod 4$, ε_α is an algebraic integer, with $N(\varepsilon_\alpha)=\frac{(d-1)^2-\Delta}{4}=1$; and $\Delta<(d-1)^2$ gives $\frac{1}{2}(d-1-\sqrt{\Delta})>0$. The quadratic relation is the characteristic equation of α, and cubing it gives $\varepsilon_\alpha^3=((d-1)^2-1)\varepsilon_\alpha-(d-1)=d(d-2)\varepsilon_\alpha+(1-d)$. For d odd, $d \mid d(d-2)$ and $1-d\equiv 1 \pmod d$, which is (a); and $\Delta=(d-3)(d+1)\equiv(-3)(1)=-3 \pmod d$, so modulo d the element $\sqrt{\Delta}$ is a square root of -3 and $\varepsilon_\alpha=\frac{1}{2}(-1+\sqrt{-3})$, whose cube is 1. For d even, $d-2$ is even, so $2d \mid d(d-2)$, and $2d \mid d^2$, so $(1-d)^2=1-2d+d^2\equiv 1 \pmod{2d}$, which is (b). □

Theorem VI.37 (ramification from the unit group)**.** *If $\mathcal{O}_F^\times=\langle -1,\varepsilon_\alpha\rangle$, then (C_d) holds for every $d\geq 4$; equivalently, ∞_- ramifies in the ray class field of F of conductor $d'\infty_-$.*

Proof. Every unit is $\pm\varepsilon_\alpha^n$. By Lemma VI.36, $\varepsilon_\alpha>0$ at ∞_-, and hence so is ε_α^n; such units never violate (C_d). A unit negative at ∞_- is therefore of the form $-\varepsilon_\alpha^n$, and it is $\equiv 1 \pmod{d'}$ precisely when $\varepsilon_\alpha^n\equiv -1 \pmod{d'}$. Cubing that congruence gives $(\varepsilon_\alpha^3)^n\equiv -1 \pmod{d'}$.

If d is odd, then $\varepsilon_\alpha^3\equiv 1 \pmod d$, so $1\equiv -1 \pmod d$, which is impossible for $d\geq 4$. If d is even, then $(\varepsilon_\alpha^3)^n\in\{1,1-d\}$ modulo $2d$ by Lemma VI.36(b), so either $-1\equiv 1 \pmod{2d}$, or $-1\equiv 1-d \pmod{2d}$, that is, $d\equiv 2 \pmod{2d}$; both are impossible for $d\geq 4$. □

Theorem VI.38 (an unconditional family of positive density)**.** *Let $d=2m+1$ with $m\geq 2$ even and m^2-1 squarefree. Then $F=\mathbb{Q}(\sqrt{m^2-1})$, $\mathcal{O}_F=\mathbb{Z}[\sqrt{m^2-1}]$, and*

$$\mathcal{O}_F^\times=\langle -1,\ m+\sqrt{m^2-1}\rangle=\langle -1,\varepsilon_\alpha\rangle.$$

Hence (C_d) holds and ∞_- ramifies in the ray class field of conductor $d\infty_-$. The set $\mathcal{D}$ of such d has natural density

$$\tfrac{1}{4}\prod_{p>2}\left(1-2p^{-2}\right)\approx 0.16132,$$

and in particular it is infinite.

Proof. Since d is odd, $\Delta=(d-3)(d+1)=4(m^2-1)$, so the squarefree part of Δ is m^2-1 and $\varepsilon_\alpha=\frac{1}{2}(2m+2\sqrt{m^2-1})=m+\sqrt{m^2-1}$. As m is even, $m^2-1\equiv 3 \pmod 4$, and hence

$\mathcal{O}_F = \mathbb{Z}[\sqrt{m^2-1}]$ and $x^2-(m^2-1)y^2 = -1$ is insoluble, since modulo 4 it reads $x^2+y^2 \equiv 3$. Therefore $\mathcal{O}_F^\times = \langle -1, \eta \rangle$ with η the fundamental solution of $x^2 - (m^2-1)y^2 = 1$. The continued fraction $\sqrt{m^2-1} = [\, m-1; \overline{1,\ 2m-2}\,]$ has period 2, with first period convergent $m/1$, and hence $\eta = m+\sqrt{m^2-1} = \varepsilon_\alpha$. Theorem VI.37 applies. This is the Richaud–Degert case $D_0 = n^2-1$.

For the density, $d \mapsto m = (d-1)/2$ is a bijection from the odd integers onto $\mathbb{Z}$, and the conditions are that m be even, of density $\frac{1}{4}$ in d, and that $(m-1)(m+1)$ be squarefree. For m even the factors $m \pm 1$ are odd and coprime, so $p^2 \mid m^2-1$ for an odd prime p if and only if $m \equiv \pm 1 \pmod{p^2}$, which excludes two residues modulo p^2; and $p^2 \le m^2-1$ forces $p \le m$, so the sieve is finite. Inclusion–exclusion gives the stated density, by Mirsky's theorem on squarefree values of products of distinct linear forms. □

Corollary VI.39 (a reformulation on the family $\mathcal{D}$)**.** *Let $d \in \mathcal{D}$ and suppose:*

(S1) M_2 *is the ray class field of* F *of conductor* $d\infty_-$*, the algebraic part of the Stark picture;*

(S2) R2 *holds, and (G1) holds for some* u_0*;*

(S3) TR *holds for* E_**, that is,* $\mathbb{Q}(t_u : u \in G_d)$ *is totally real.*

Then Z(d) *holds. Since* $\mathcal{D}$ *is infinite, (S1)–(S3) for all* $d \in \mathcal{D}$ *would imply Problem 1.*

Proof. (S1) makes M_2/F abelian, hypothesis (i), and by Theorem VI.38 its conductor involves ∞_-, hypothesis (iii). (S2) gives $\kappa \in \mathrm{Gal}(M_2/F)$, hypothesis (ii), and identifies M_2^κ with the trace field; (S3) is hypothesis (iv). Apply Proposition VI.32. □

Corollary VI.39 is conditional and does not reduce the archimedean difficulty. By Corollary VI.33, under (S1) and (S2) hypothesis (S3) is equivalent to the place law itself, so the corollary is a reformulation of Theorem VI.23 on the family $\mathcal{D}$, not a proof of Problem 1 from weaker input. What it does achieve is the following. No condition on $\mathrm{Cl}(d'\infty_-)$ survives, no congruence has to be checked, and the one hypothesis of Proposition VI.30 that is not algebraic in nature, ramification at ∞_-, is proved unconditionally on a set of positive density rather than assumed. Compared with Proposition VI.30, which needs no archimedean input but applies only to 24 dimensions below 2000, the present route applies on a set of positive density but needs the full archimedean input, in the form TR.

b. Verification. Proposition VI.35, Lemma VI.36 and Theorem VI.37 were checked by exact integer arithmetic in $\mathcal{O}_F = \mathbb{Z}[w]$, with $w = \frac{1}{2}(1+\sqrt{D_0})$ or $\sqrt{D_0}$ for D_0 the squarefree part of Δ, for every $4 \le d \le 20000$ (Table VI). The fundamental unit of F was

computed from the continued fraction of $\sqrt{D_0}$, corrected by the index-3 test $a(a^2 \mp 3) = 2x_1$ when $D_0 \equiv 1 \pmod 4$, and spot-checked against $\varepsilon(\sqrt{5}) = \frac{1}{2}(1+\sqrt{5})$, $\varepsilon(\sqrt{15}) = 4+\sqrt{15}$, $\varepsilon(\sqrt{35}) = 6+\sqrt{35}$ and $\varepsilon(\sqrt{21}) = \frac{1}{2}(5+\sqrt{21})$. Condition ($\mathrm{C}_d$) held in every one of the 19 997 dimensions, with no exception, which also extends to $d \le 20000$ the census of Proposition VI.30(iii) reported above for $d \le 2000$. Independently, for every $4 \le d \le 2000$ we confirmed $|\mathrm{Cl}(d'\infty_-)| = 2\,|\mathrm{Cl}(d')|$ directly with PARI/GP (`bnrclassno`), with no exception.

TABLE VI. Verification of the unit criterion (C_d) and of Lemma VI.36, by exact arithmetic in $\mathcal{O}_F$, for every $4 \le d \le 20000$. Here $\mathcal{D}$ is the Richaud–Degert family of Theorem VI.38, whose members need no verification.

Statement checked, $4 \le d \le 20000$	count	fraction
dimensions examined	19 997	
(C_d) holds, i.e. ∞_- ramifies	19 997	all
$\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle$ (Thm. VI.37 applies)	19 697	98.50%
$d \in \mathcal{D}$ (Thm. VI.38, unconditional)	3 229	16.15%
$\mathrm{ord}(\varepsilon_\alpha) = 3$ in $(\mathcal{O}_F/d)^\times$, d odd	9 998	all odd d
$\mathrm{ord}(\varepsilon_\alpha) = 6$ in $(\mathcal{O}_F/2d)^\times$, d even	9 999	all even d

The smallest members of $\mathcal{D}$ are 5, 9, 13, 25, 29, 33, 37, 41, 45, 61, 65, 69, 73, 77, 81, 85, 105, 109, 113, 117, 121, …, and they include $d = 5, 9, 13$, the odd dimensions in which the overlap-unit field has been identified exactly (Table V). The density of $\mathcal{D}$ observed up to 20000 is 0.16147, against the predicted 0.16132. Figure 1 compares the three sets of dimensions discussed in this subsection with the set of all d.

G. An explicit reciprocity law for the overlap units

Proposition VI.29 identifies the place law with the action of a single Galois element, c_-, on the overlap units. Class field theory describes every element of $\mathrm{Gal}(M_2/F)$ as an Artin symbol, and hence the place law would follow from a rule computing the Artin symbol of an ideal *directly* from a generator. For special values at $s = 0$ of partial zeta functions such a rule is known: Arakawa[22] (Thms. A and B), building on Coates–Sinnott and Deligne–Ribet, shows that the idele s acts on $\exp(2\pi i\,\zeta(0,\cdot))$ through the regular representation $F^\times \to GL_2(\mathbb{Q})$ of F, $z \mapsto s^{-1}z$, for ideles s that are *positive at both infinite places*. For the values at $s = 0$ of the derivatives, which are the Stark–Shintani units relevant here, the analytic half of the corresponding statement is known: the invariant of the class $\mathcal{RA}$

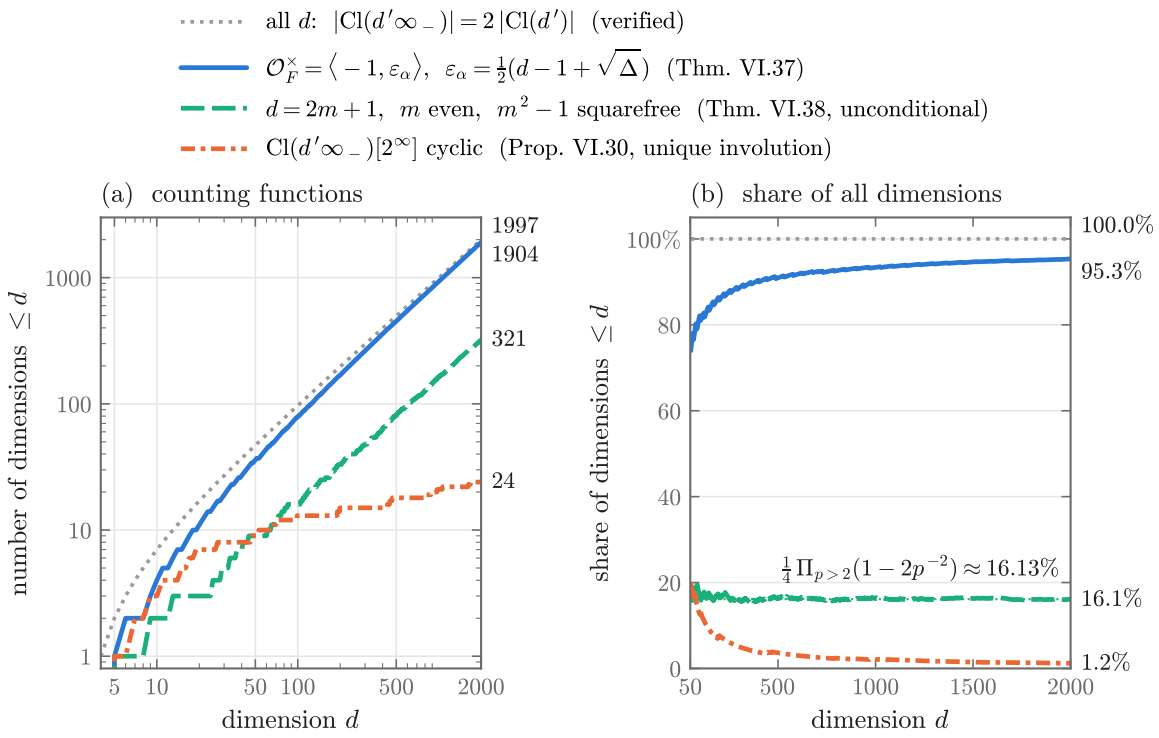


FIG. 1. Dimensions $4 \leq d \leq 2000$ covered by the hypotheses of this subsection. Dotted gray: all d; condition (C_d), and hence ramification at ∞_-, holds for each of them (Table VI). Solid blue: $\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle$, where Theorem VI.37 proves (C_d). Dashed green: the family $\mathcal{D}$ of Theorem VI.38, where (C_d) is proved unconditionally. Dash-dotted orange: the 24 dimensions with cyclic 2-Sylow subgroup of $\mathrm{Cl}(d'\infty_-)$, where Proposition VI.30 applies (census of Sec. VI E). (a) Counting functions on logarithmic axes: a set of positive density appears as a line of slope 1, parallel to the gray reference, and a set of density zero bends away from it. (b) The same sets as a share of all dimensions $\leq d$; the dotted green line is the predicted density $\frac{1}{4}\prod_{p>2}(1-2p^{-2}) \approx 0.1613$ of $\mathcal{D}$. The figure concerns hypothesis (iii) only; the archimedean input (TR, equivalently PL) is not affected.

is the inverse of the invariant of $\mathcal{A}$, where $\mathcal{R}$ is the class of c_- (Ref. 17, Cor. 8.3; Ref. 23, Thm. 6.1). The missing Galois half is a reciprocity law. In this subsection we state such a law in explicit form, show that it implies $c_- = \kappa$, and verify it exactly in the SIC dimensions $d = 5$ and $d = 9$.

a. The matrix of a generator. Let $d \geq 5$ be odd, and let $\tilde{\Pi}$ be a ghost in dimension d with normalized overlaps ε_p, $p \in \mathbb{Z}_d^2 \setminus \{0\}$, which are real with $\varepsilon_{-p} = \varepsilon_p^{-1}$. Put $e_p = \varepsilon_p^2$, and let $M_\varepsilon = \mathbb{Q}(e_p : p \neq 0)$ be the sign-free field of the ghost overlaps. The group $SL_2(\mathbb{Z}_d)$

acts on the labels by $p \mapsto gp$, with p regarded as a column vector. The ghosts constructed in Refs. 11 and 13 have Zauner symmetry: there is an element $\Lambda \in SL_2(\mathbb{Z}_d)$ of order 3 with $\operatorname{tr}\Lambda \equiv d-1 \pmod d$ and

$$e_{\Lambda p} = e_p \qquad \text{for all } p \neq 0.$$

Let

$$\epsilon_\alpha = \tfrac{1}{2}\big(d-1+\sqrt{(d-3)(d+1)}\big)$$

be the eigenvalue of α (trace $d-1$) that is larger than 1 at ∞_+, and let $\mathcal{O}_\alpha = \mathbb{Z}[\epsilon_\alpha]$. Since $\sqrt{D} = d(d-2)\sqrt{(d-3)(d+1)}$, $\mathcal{O}_\alpha$ is an order of F. The unit ϵ_α is totally positive with $\epsilon_\alpha + \epsilon_\alpha^{-1} = d-1$. The matrix Λ^{-1} satisfies the same relation, $\Lambda^{-2} - (d-1)\Lambda^{-1} + I = 0$, by the Cayley–Hamilton theorem. Hence there is a unique ring homomorphism

$$\phi : \mathcal{O}_\alpha \longrightarrow M_2(\mathbb{Z}_d), \qquad \phi(\epsilon_\alpha) = \Lambda^{-1}.$$

Concretely, $\phi(a + b\,\delta) = aI + b\,(2\Lambda^{-1} - (d-1)I)$ for $\delta = 2\epsilon_\alpha - (d-1) = \sqrt{(d-3)(d+1)}$. If $\varpi \in \mathcal{O}_\alpha$ is prime to d, then $\det\phi(\varpi) \equiv N_{F/\mathbb{Q}}(\varpi) \not\equiv 0$, and $\phi(\varpi) \in GL_2(\mathbb{Z}_d)$. In $d = 5$ and $d = 9$, $\mathcal{O}_\alpha = \mathcal{O}_F$, and $\phi(\sqrt{D_0}) = \Lambda^{-1} - \frac{d-1}{2}I$ with $D_0 = 3$, resp. 15.

Conjecture VI.40 (explicit reciprocity law)**.** *Let $d \geq 5$ be odd. Suppose that M_ε/F is abelian with conductor dividing $(d)\infty_-$. Then for every $\varpi \in \mathcal{O}_\alpha$ prime to d with $\varpi > 0$ at ∞_-, the Artin symbol of the ideal $\varpi\mathcal{O}_F$ acts on the squared overlaps by*

$$\left(\frac{M_\varepsilon/F}{\varpi\mathcal{O}_F}\right) e_p \;=\; e_{\phi(\varpi)p} \qquad \textit{for all } p \neq 0.$$

The normalization is at ∞_- only; the sign of ϖ at ∞_+ is free. This asymmetry is the archimedean content of the law. If ϖ is replaced by $-\varpi$, then $\phi(\varpi)$ changes to $-\phi(\varpi)$, and the right-hand side changes by $e_{-p} = e_p^{-1}$, that is, by κ. A normalization at ∞_+ would therefore assign the same ideal a different Galois element whenever ϖ has mixed signs.

Proposition VI.41 (consistency)**.** *The right-hand side of Conjecture VI.40 depends only on the ideal $\varpi\mathcal{O}_F$ whenever every unit of $\mathcal{O}_\alpha$ that is positive at ∞_- is a power of ϵ_α; this holds for $d = 5$ and $d = 9$. The map $\varpi \mapsto (e_p \mapsto e_{\phi(\varpi)p})$ is multiplicative, and for $\varpi = m \in \mathbb{Z}$, $m > 0$, it is $e_p \mapsto e_{mp}$.*

Proof. Two generators positive at ∞_- differ by a unit u positive at ∞_-, and by hypothesis $u = \epsilon_\alpha^k$. Then $\phi(u\varpi) = \Lambda^{-k}\phi(\varpi)$, and $e_{\Lambda^{-k}q} = e_q$ for all q. Multiplicativity holds because ϕ

is a ring homomorphism, and $\phi(m) = mI$. For $d = 5$ and $d = 9$, $\epsilon_\alpha = 2 + \sqrt{3}$ and $4 + \sqrt{15}$ are the fundamental units of $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{15})$; both have norm $+1$, and -1 is negative at ∞_-. □

Proposition VI.42 (the reciprocity law implies the place law)**.** *Let $d \geq 5$ be odd, and suppose that Conjecture VI.40 holds for d. Then complex conjugation c_- at every place of M_ε above ∞_- acts on M_ε as $\kappa : e_p \mapsto e_p^{-1}$. Consequently $|\sigma(\varepsilon_p)| = 1$ for every $p \neq 0$ and every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ above ∞_-, and $\sigma(\tilde{\Pi})$ is a WH-covariant SIC fiducial. In particular* Z(d) *holds.*

Proof. Put $\varpi = -1 - d\,\delta \in \mathcal{O}_\alpha$, with $\delta = \sqrt{(d-3)(d+1)}$. At ∞_+, $\varpi = -1 - d\delta < 0$. At ∞_-, $\varpi = -1 + d\delta > 0$, because $d\delta \geq 5\sqrt{12} > 1$. Moreover $\varpi \equiv -1 \pmod{d\mathcal{O}_\alpha}$, and hence ϖ is prime to d and $\phi(\varpi) = -I$.

Let $\beta = -\varpi$. Then $\beta \equiv 1 \pmod{d\mathcal{O}_F}$, $\beta > 0$ at ∞_+ and $\beta < 0$ at ∞_-. Since the conductor $\mathfrak{f}\infty_-$ of M_ε/F divides $(d)\infty_-$, the argument of Proposition VI.29(c), applied to M_ε, shows that the Artin symbol of $(\beta) = \varpi\mathcal{O}_F$ is c_-, the common complex conjugation at the places above ∞_- (Proposition VI.29(a)). Conjecture VI.40 gives

$$c_-(e_p) = e_{-p} = \varepsilon_{-p}^2 = e_p^{-1} \qquad \text{for all } p \neq 0.$$

Let σ lie above ∞_-. Then $\overline{\sigma(y)} = \sigma(c_- y)$ for $y \in M_\varepsilon$, and hence $\overline{\sigma(e_p)} = \sigma(e_p)^{-1}$, that is, $|\sigma(\varepsilon_p)|^2 = |\sigma(e_p)| = 1$. Corollary VI.2 gives the SIC. □

Proposition VI.42 uses the law only at the single element $\varpi = -1 - d\delta$, whose matrix is $-I$. Conversely, $c_- = \kappa$ is exactly the instance of Conjecture VI.40 at this element. The point of the full law is that it is a uniform statement about all ideals, of the type that reciprocity theorems prove, rather than an isolated archimedean identity. It is the analogue for Stark–Shintani units ($\zeta'(0)$) of Arakawa's law for $\exp(2\pi i\zeta(0))$: restricted to ϖ that are totally positive it has the same shape, and the new feature is that the idele -1 at ∞_- must be included. It also makes explicit an empirical observation of AFK (Ref. 11, Obs. 8.2): if $\tilde{E}$ is the field of the ghost overlaps and H the ring class field, then $\mathrm{Gal}(\tilde{E}/H)$ is isomorphic to $\mathcal{M}/\mathcal{S}$, where $\mathcal{S} \subset GL_2(\mathbb{Z}_d)$ is the symmetry group of the overlaps and $\mathcal{M}$ a maximal abelian subgroup containing it, with $h(\tilde{\mu}_p) = \tilde{\mu}_{F_h p}$. Conjecture VI.40 names this isomorphism: on the Artin symbol of a principal ideal $\varpi\mathcal{O}_F$ it is $\varpi \mapsto \phi(\varpi)\mathcal{S}$, with ϖ normalized at ∞_-,

and it extends the description from $\mathrm{Gal}(\tilde{E}/H)$ to the element c_-, which is what the place law needs.

b. Exact test in $d = 5$ and $d = 9$. We tested Conjecture VI.40 on the ghosts used for Table V.

- *Field arithmetic.* We found the minimal polynomial P of one squared overlap e_{p_0}, of degree 16 for $d = 5$ ($p_0 = (0, 1)$) and 72 for $d = 9$ ($p_0 = (1, 0)$), and wrote every e_p as an exact polynomial in e_{p_0} modulo P. For $d = 5$ the expressions were found by integer relation detection at 400 digits. For $d = 9$ this is not feasible at the available 2000 digits, because the coefficients are too large. Instead, the overlaps of degree 72 were matched with the 36 automorphisms of $\mathbb{Q}(e_{p_0})$ computed by `nfgaloisconj`, and those of degrees 8 and 24 were obtained by `nfroots`. Every expression was checked exactly against its own minimal polynomial. For $d = 9$ the 36 automorphisms form $\mathrm{Gal}(M_\varepsilon/F)$, in agreement with Table V.
- *Zauner symmetry.* The stabilizer of (e_p) in $GL_2(\mathbb{Z}_d)$ is $\{I, \Lambda, \Lambda^2\}$, with $\Lambda = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ for $d = 5$ and $\Lambda = \left(\begin{smallmatrix} 7 & 3 \\ 2 & 1 \end{smallmatrix}\right)$ for $d = 9$.
- *Frobenius.* For a prime $\mathfrak{P}$ of M_ε above a prime $\mathfrak{p} \nmid d\,\mathrm{disc}(P)$ of F, the action of $\mathrm{Frob}_\mathfrak{p}$ on the labels was read off exactly from $y \mapsto y^{N\mathfrak{p}} \pmod{\mathfrak{P}}$ applied to the reductions of the exact expressions; for $d = 9$ it was also identified with one of the 36 automorphisms.
- *Generator.* For split $q = N\mathfrak{p}$, a generator $\varpi = a + b\sqrt{D_0}$ of $\mathfrak{p}$ was found with `bnfisprincipal` and multiplied by -1 if necessary to make it positive at ∞_-. For inert q, $\varpi = q$.

The results are in Table VII. The law holds at every prime tested. Every tested alternative convention fails: normalizing ϖ at ∞_+ instead of ∞_-, replacing Λ^{-1} by Λ, transposing or inverting $\phi(\varpi)$, and, for inert primes, replacing qI by $-qI$.

c. The ghost multiplet in $d = 9$. The class number of $\mathbb{Q}(\sqrt{15})$ is 2, and this is visible in the overlaps. Of the 36 automorphisms of M_ε/F, exactly 18 permute the squared overlaps $\{e_p\}$ of the ghost. The other 18 send *every* e_p to a number that is not a squared overlap of this ghost: they map it to a Galois-conjugate point of V_9 whose overlap moduli differ from those of $\tilde{\Pi}$, and which is therefore not equivalent to it under the extended Clifford group. The Frobenius at a split prime preserves the ghost exactly when the prime is principal (122 of 122 against 0 of 136), and the inert primes, which are principal, preserve it. Thus the ideal class group, that is $\mathrm{Gal}(H/F)$ for the ring class field H of $\mathcal{O}_\alpha = \mathcal{O}_F$, acts on a multiplet of

TABLE VII. Exact test of the reciprocity law, Conjecture VI.40. Entries are primes at which $\mathrm{Frob}_{\mathfrak{p}}(e_p) = e_{\phi(\varpi)p}$ for all p / primes tested. Split primes are counted with nontrivial Frobenius; for $d = 5$ each prime $\mathfrak{P}$ of M_ε is counted, for $d = 9$ each prime $\mathfrak{p}$ of F. The last two columns test the alternative normalization ($\varpi > 0$ at ∞_+) and the sign $-qI$ at inert primes.

d	F	range	split, $\varpi > 0$ at ∞_-	inert, qI	split, $\varpi > 0$ at ∞_+	inert, $-qI$
5	$\mathbb{Q}(\sqrt{3})$	$q \le 3000$	594/594	219/219	268/594	0/219
9	$\mathbb{Q}(\sqrt{15})$	$q \le 4000$	122/122†	276/276	56/122	0/276
5*	$\mathbb{Q}(\sqrt{3})$	$q \le 3000$	415/415	218/218	197/415	—
7*	$\mathbb{Q}(\sqrt{2})$	$q \le 3000$	415/415	219/219	0/415	—
11*	$\mathbb{Q}(\sqrt{6})$	$q \le 2500$	346/346	180/180	162/346	—
15*	$\mathbb{Q}(\sqrt{3})$	$q \le 2500$	333/333	181/181	165/333	—

†Principal primes. $\mathbb{Q}(\sqrt{15})$ has class number 2; at all 136 non-principal split primes the Frobenius moves every e_p out of the set $\{e_q\}$ (see the text). *Second computation (Lemma VI.43), with new ghosts: every prime ideal $\mathfrak{p}$ of F above a rational prime $q \le 3000$ ($d = 5, 7$) or $q \le 2500$ ($d = 11, 15$) is counted, including those with trivial Frobenius, and the law is required for every generator positive at the stated place. The tested primes meet every class of $\mathrm{Cl}(d\infty_-)$, so the law holds at all primes for these algebraic numbers.

two ghosts, and ϕ describes the action of principal ideals within each member, in accordance with Obs. 8.2 of Ref. 11. Conjecture VI.40 is stated for principal ideals $\varpi\mathcal{O}_F$ for this reason. The class of c_- is principal, generated by $\varpi = -1 - d\delta$, and hence Proposition VI.42 needs only this part of the law.

d. Further dimensions, and a proof of the law for the computed overlaps. The consistency hypothesis of Proposition VI.41 fails in some dimensions, because F may have units positive at ∞_- that are not powers of ϵ_α. Examples are $\eta = -1 - \sqrt{2}$ for $d = 7$, with $\eta^2 = \epsilon_\alpha$, and $\eta = 2 + \sqrt{3}$ for $d = 15$, with $\eta^2 = \epsilon_\alpha$. A weaker hypothesis suffices. Let f_α be the conductor of the order $\mathcal{O}_\alpha$ in $\mathcal{O}_F$. If $\gcd(d, f_\alpha) = 1$, then $\mathcal{O}_\alpha/d = \mathcal{O}_F/d$, and ϕ extends to a ring homomorphism $\mathcal{O}_F \to M_2(\mathbb{Z}_d)$, again denoted ϕ. The proof of Proposition VI.41 shows that the right-hand side of Conjecture VI.40 depends only on the ideal $\varpi\mathcal{O}_F$ as soon as

$$e_{\phi(u)p} = e_p \quad \text{for all } p \neq 0 \text{ and all } u \in \mathcal{O}_F^\times \text{ positive at } \infty_-. \tag{2}$$

In $d = 5$ and $d = 11$, $\mathcal{O}_F^\times = \langle -1, \epsilon_\alpha \rangle$, with $\epsilon_\alpha = 2+\sqrt{3}$ and $5+2\sqrt{6}$, and ϵ_α is totally positive. So the units positive at ∞_- are the powers of ϵ_α, and (2) holds because $\phi(\epsilon_\alpha) = \Lambda^{-1}$. In $d = 7$ and $d = 15$, $\mathcal{O}_F^\times = \langle -1, \eta' \rangle$ with $\eta' = 1 + \sqrt{2}$, resp. $2 + \sqrt{3}$. The units positive at ∞_- are the powers of $\eta = -1-\sqrt{2}$, resp. $\eta = 2+\sqrt{3}$. The stabilizer of (e_p) in $GL_2(\mathbb{Z}_d)$ has order 6 and contains $\phi(\eta)$, which for $d = 7$ has determinant $N(\eta) = -1$. Hence (2) holds in all

four dimensions. In $d = 7$ it holds for only one of the two order-3 elements of the stabilizer in the role of Λ: for the other one, $\phi(\eta)$ does not stabilize (e_p). So in this dimension the consistency requirement alone fixes the choice of Λ.

Lemma VI.43 (one prime per ray class suffices)**.** *Let d be odd, and suppose that $h_F = 1$, that $\gcd(d, f_\alpha) = 1$, that (2) holds, and that M_ε/F is abelian with conductor dividing $\mathfrak{m} = d\infty_-$. If the identity $\left(\frac{M_\varepsilon/F}{\varpi\mathcal{O}_F}\right)e_p = e_{\phi(\varpi)p}$ for all $p \neq 0$ holds for one prime ideal in each class of a generating set of $\mathrm{Cl}(\mathfrak{m})$, then it holds for every $\varpi \in \mathcal{O}_F$ prime to d and positive at ∞_-.*

Proof. Every class of $\mathrm{Cl}(\mathfrak{m})$ contains integral ideals, so it suffices to work with integral ideals $\mathfrak{a}$ prime to d. Choose a generator ϖ of $\mathfrak{a}$ that is positive at ∞_-. This is possible since $h_F = 1$ and -1 is negative at ∞_-. Let $g(\mathfrak{a}) = \phi(\varpi) \in GL_2(\mathbb{Z}_d)$, acting on the labels p. By (2), the class of $g(\mathfrak{a})$ modulo the stabilizer $\mathcal{S}$ of (e_p) does not depend on the choice of ϖ, and g is multiplicative because ϕ is a ring homomorphism. If $\varpi \equiv 1 \pmod{d}$ and $\varpi > 0$ at ∞_-, then $\phi(\varpi) = I$. Hence $\mathfrak{a} \mapsto g(\mathfrak{a})\mathcal{S}$ factors through $\mathrm{Cl}(\mathfrak{m})$; write $g(c)$ for a representative. The Artin map $A(\mathfrak{a}) = \left(\frac{M_\varepsilon/F}{\mathfrak{a}}\right)$ also factors through $\mathrm{Cl}(\mathfrak{m})$, by the conductor hypothesis. Let $S \subset \mathrm{Cl}(\mathfrak{m})$ be the set of classes c with $A(c)e_p = e_{g(c)p}$ for all $p \neq 0$. This condition does not depend on the representative of $g(c)$ modulo $\mathcal{S}$. If $c_1, c_2 \in S$, then $A(c_1c_2)e_p = A(c_1)e_{g(c_2)p} = e_{g(c_1)g(c_2)p}$, and $g(c_1)g(c_2) \in g(c_1c_2)\mathcal{S}$. Hence S is closed under products. Since $\mathrm{Cl}(\mathfrak{m})$ is finite, S is a subgroup, and it contains a generating set by hypothesis. So $S = \mathrm{Cl}(\mathfrak{m})$. □

We applied this to new ghosts in $d = 5, 7, 11, 15$, all with $h_F = 1$, and in each case the tested primes cover *every* ray class. The ghosts were found by a least-squares search restricted to an eigenspace of the Clifford unitary of $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$, and refined by Newton's method to 800, 1500, 6000 and 4000 digits, respectively. The procedure has four steps.

(1) *Exact expressions.* The minimal polynomial of $e_{(0,1)}$ was found by `algdep` and reduced with `polredbest`, giving a generator θ of $K = \mathbb{Q}(e_{(0,1)})$. Each squared overlap e_p was then obtained *exactly*, as the root in K of its own minimal polynomial (by `nfroots`) that agrees with the numerical value. This avoids integer-relation detection in high dimension.

(2) *Exact checks.* The checks $e_pe_{-p} = 1$ for all p and $\sqrt{D_0} \in K$ were done in K, and K was found to contain every e_p. Further, $\mathrm{Gal}(K/F)$ is abelian, inversion κ is an automorphism of K over F, and the conductor of K/F, computed by `rnfconductor`,

is $d\infty_-$. Since $[K : F] = |\mathrm{Cl}(d\infty_-)|$, the field $K = M_\varepsilon$ is the full ray class field modulo $d\infty_-$.

(3) *Frobenius.* For every prime ideal $\mathfrak{p}$ of F above a rational prime $q \le 3000$ (for $d = 5, 7$) or $q \le 2500$ (for $d = 11, 15$), prime to d, to the discriminant of the defining polynomial, and to the denominators of the expressions, the permutation of $\mathcal{E}$ induced by $y \mapsto y^{N\mathfrak{p}}$ modulo a prime $\mathfrak{P} \mid \mathfrak{p}$ of K was compared with $e_p \mapsto e_{\phi(\varpi)p}$, for *every* generator ϖ of $\mathfrak{p}$ that is positive at ∞_-. A few primes at which two distinct squared overlaps have the same residue modulo $\mathfrak{P}$ were skipped: 1, 1 and 19 for $d = 5, 7, 15$.

(4) *Coverage.* The classes of the tested primes in $\mathrm{Cl}(d\infty_-)$ were computed by `bnrisprincipal`.

The law holds at every tested prime, for one of the two order-3 elements of the stabilizer in the role of Λ: $\Lambda = \left(\begin{smallmatrix} 1 & 4 \\ 3 & 3 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 3 & 5 \\ 3 & 3 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 4 & 10 \\ 10 & 6 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 14 \\ 1 & 14 \end{smallmatrix}\right)$ for $d = 5, 7, 11, 15$. The normalization at ∞_+ fails, and so does the other choice of Λ (Table VII). By Lemma VI.43, the law therefore holds at *all* primes for these algebraic numbers. By Proposition VI.42 this gives $c_- = \kappa$ and the place law for the squared overlaps, so that the Galois conjugates of these ghosts above ∞_- are SICs. The choice between Λ and Λ^{-1} does not affect this conclusion, because $\phi(-1 - d\delta) = -I$ for either choice.

Two caveats apply. First, the identification of the numerical overlaps with the algebraic numbers in K rests on agreement to the working precision, as for all exact SIC solutions obtained from numerics. A complete certificate would verify the ghost equations exactly in $K(\varepsilon_p)$, and we have not done so. Second, SICs in $d = 5, 7, 11, 15$ are known, so no new SIC is obtained. The point of the computation is that the reciprocity law, and with it the archimedean relation $c_- = \kappa$, is a theorem for these explicit algebraic numbers. Its proof requires only the abelian structure of M_ε/F and finitely many Frobenius computations.

e. What this changes. Conjecture VI.40 replaces the archimedean statement PL by a statement of reciprocity type, of the form proved for $\zeta(0)$-invariants by Coates–Sinnott, Deligne–Ribet and Arakawa, it is verified exactly, prime by prime, in two SIC dimensions, and for the computed algebraic overlaps it is proved in four (Lemma VI.43). A proof of Conjecture VI.40, or only of its instance at the element $-1 - d\delta$, for infinitely many odd d would prove Problem 1. We have not proved it for the exact Stark solution in any dimension: Lemma VI.43 applies to the algebraic numbers identified from high-precision numerics. For even d the labels live in $\mathbb{Z}^2_{2d}$ and the conductor is $(2d)\infty_-$, and the law needs the corresponding modification; we have not tested it.

H. Prime-power dimensions: the sign at ∞_- from the class group

Proposition VI.30 removes the archimedean input when the whole group $\mathrm{Gal}(M'/F)$ has a unique involution, which happens only on a set of dimensions of density zero. The complex conjugation c_- always lies in a smaller subgroup, however, because every complex conjugation acts trivially on the Hilbert class field of F. In this subsection we show that this smaller subgroup always has a cyclic 2-Sylow subgroup when d is a power of a prime $p \equiv 5 \pmod 6$, or a power 3^k with $k \geq 2$. On this infinite family, under the algebraic part of the Stark picture, $c_- = \kappa$ therefore becomes equivalent to a statement about the action of κ on an extension of F that is unramified everywhere. That statement involves no complex conjugation and no embedding into $\mathbb{C}$.

We use the notation of Sec. VI G: $\tilde{\Pi}$ is a ghost in odd dimension d, $e_p = \varepsilon_p^2$ are its squared overlaps, $M_\varepsilon = \mathbb{Q}(e_p : p \neq 0)$, and κ denotes inversion $e_p \mapsto e_p^{-1} = e_{-p}$. Let H_F be the Hilbert class field of F in the wide sense, that is, the maximal abelian extension of F unramified at all finite and all infinite places, so that $\mathrm{Gal}(H_F/F) \cong \mathrm{Cl}_F$, the (wide) ideal class group. Put $\mathfrak{m} = d\,\mathcal{O}_F\infty_-$. Let $I^{\mathfrak{m}}$ denote the fractional ideals prime to d, $P^{\mathfrak{m}} \subset I^{\mathfrak{m}}$ the principal ones, and $P_{\mathfrak{m},1} \subset P^{\mathfrak{m}}$ those having a generator $\equiv 1 \pmod{{}^\times d}$ that is positive at ∞_-, so that $\mathrm{Cl}(\mathfrak{m}) = I^{\mathfrak{m}}/P_{\mathfrak{m},1}$.

Lemma VI.44 (principal ray classes)**.** *For every odd $d \geq 5$ there is an exact sequence*

$$1 \longrightarrow P^{\mathfrak{m}}/P_{\mathfrak{m},1} \longrightarrow \mathrm{Cl}(\mathfrak{m}) \longrightarrow \mathrm{Cl}_F \longrightarrow 1,$$

and $P^{\mathfrak{m}}/P_{\mathfrak{m},1}$ is isomorphic to a quotient of $(\mathcal{O}_F/d)^\times$.

Proof. The sequence is the standard one: $I^{\mathfrak{m}} \to \mathrm{Cl}_F$ is onto, and its kernel modulo $P_{\mathfrak{m},1}$ is $P^{\mathfrak{m}}/P_{\mathfrak{m},1}$. Sending α to the pair $(\alpha \bmod d,\ \mathrm{sgn}_{\infty_-}\alpha)$ gives the standard isomorphism

$$P^{\mathfrak{m}}/P_{\mathfrak{m},1} \;\cong\; \big((\mathcal{O}_F/d)^\times \times \{\pm 1\}\big)\big/\iota(\mathcal{O}_F^\times),$$

where $\iota(u) = (u \bmod d, \mathrm{sgn}_{\infty_-} u)$. In particular $\iota(-1) = (-1,-1)$. The homomorphism $(x,s) \mapsto sx$ from $(\mathcal{O}_F/d)^\times \times \{\pm 1\}$ onto $(\mathcal{O}_F/d)^\times$ has kernel $\{(1,1),(-1,-1)\} = \langle \iota(-1)\rangle$, because $-1 \not\equiv 1 \pmod d$. Hence the right-hand side is a quotient of $(\mathcal{O}_F/d)^\times$. □

Lemma VI.45 (prime powers)**.** *Let $d = p^k$, where either $p \equiv 5 \pmod 6$ is prime and $k \geq 1$, or $p = 3$ and $k \geq 2$. Then the 2-Sylow subgroup of $(\mathcal{O}_F/d)^\times$, and of each of its quotients, is*

cyclic. More precisely, in the first case p is inert in F and

$$(\mathcal{O}_F/d)^\times \;\cong\; \mathbb{F}_{p^2}^\times \times \big(1+p\mathcal{O}_F\big)/\big(1+p^k\mathcal{O}_F\big),$$

where the second factor has order $p^{2(k-1)}$. In the second case 3 is ramified in F, and $(\mathcal{O}_F/3^k)^\times \cong \mathbb{F}_3^\times \times$ (a group of order 3^{2k-1}).

Proof. With $\Delta = (d-3)(d+1)$ we have $F = \mathbb{Q}(\sqrt{\Delta})$ and $\Delta \equiv -3 \pmod p$ (Lemma VI.36). Since $p \neq 3$, $p \nmid \Delta$, and hence p is unramified in F. It is inert exactly when Δ is a non-square modulo p, that is, when $\left(\frac{-3}{p}\right) = -1$, and by quadratic reciprocity this holds exactly when $p \equiv 2 \pmod 3$. Then $\mathcal{O}_F/p^k$ is a local ring with residue field $\mathbb{F}_{p^2}$, and the Teichmüller splitting gives the displayed decomposition. The first factor is cyclic and the second has odd order. If $d = 3^k$ with $k \geq 2$, then $\Delta = 3\,(3^{k-1}-1)(3^k+1)$, and neither factor in parentheses is divisible by 3. Hence $v_3(\Delta) = 1$, and 3 is ramified in F, with $3\mathcal{O}_F = \mathfrak{p}^2$ and $\mathcal{O}_F/\mathfrak{p} = \mathbb{F}_3$. Then $\mathcal{O}_F/3^k = \mathcal{O}_F/\mathfrak{p}^{2k}$ is a local ring with residue field $\mathbb{F}_3$, and its unit group is $\mathbb{F}_3^\times$ times the group $(1+\mathfrak{p})/(1+\mathfrak{p}^{2k})$ of order 3^{2k-1}. In both cases the 2-Sylow subgroup is cyclic. A quotient of a finite abelian group with cyclic 2-Sylow subgroup again has a cyclic 2-Sylow subgroup. □

Theorem VI.46 (the sign at ∞_- from the class group)**.** *Let $d = p^k$, where either $p \equiv 5 \pmod 6$ is prime and $k \geq 1$, or $p = 3$ and $k \geq 2$. Let $\tilde{\Pi}$ be a ghost in dimension d, that is, a non-Hermitian point of Γ_d. Assume:*

(i) M_ε/F is abelian, with conductor dividing $d\,\infty_-$;

(ii) ∞_- ramifies in M_ε;

(iii) inversion $e_p \mapsto e_p^{-1}$ extends to an automorphism κ of M_ε fixing F.

Then the following are equivalent:

(a) $c_- = \kappa$ on M_ε;

(b) κ acts trivially on $M_\varepsilon \cap H_F$.

If they hold, then $\sigma(\tilde{\Pi})$ is a WH-covariant SIC fiducial for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ above ∞_-, and in particular $\mathrm{Z}(d)$ holds.

Proof. Put $\Gamma = \mathrm{Gal}(M_\varepsilon/F)$ and $\Gamma_0 = \mathrm{Gal}(M_\varepsilon/M_\varepsilon \cap H_F)$. By (i) and class field theory, the Artin map identifies Γ with a quotient of $\mathrm{Cl}(\mathfrak{m})$. Under this identification Γ_0 is the image of $P^{\mathfrak{m}}/P_{\mathfrak{m},1}$, because H_F is the class field of the subgroup $P^{\mathfrak{m}}/P_{\mathfrak{m},1}$ of $\mathrm{Cl}(\mathfrak{m})$ (Lemma VI.44).

By Lemmas VI.44 and VI.45, Γ_0 is a quotient of $(\mathcal{O}_F/d)^\times$, and hence its 2-Sylow subgroup is cyclic. Since Γ_0 is abelian, it contains at most one element of order 2.

Let w be a place of M_ε above ∞_- and c_w the corresponding complex conjugation. It fixes F, and it acts trivially on $M_\varepsilon \cap H_F$ because H_F is unramified at the infinite places. Hence $c_w \in \Gamma_0$. Since Γ is abelian, $c_w = c_-$ does not depend on w (Proposition VI.29(a)), and by (ii) $c_- \neq \mathrm{id}$. So c_- is the unique involution of Γ_0.

By (iii), $\kappa \in \Gamma$ and $\kappa^2 = \mathrm{id}$. Moreover $\kappa \neq \mathrm{id}$. Otherwise $e_p = \pm 1$ for all p, so $|\varepsilon_p| = 1$, and by Lemma VI.1 the ghost $\tilde{\Pi}$ would be Hermitian, hence a SIC and not a ghost. Now (b) says exactly that $\kappa \in \Gamma_0$. If (b) holds, then κ and c_- are involutions of Γ_0 and therefore coincide, which is (a). Conversely, (a) gives $\kappa = c_- \in \Gamma_0$, which is (b).

Assume (a), and let σ lie above ∞_-. For $y \in M_\varepsilon$ we have $\overline{\sigma(y)} = \sigma(c_- y)$, and hence $\overline{\sigma(e_p)} = \sigma(e_p)^{-1}$, so $|\sigma(\varepsilon_p)|^2 = |\sigma(e_p)| = 1$ for all $p \neq 0$. Corollary VI.2 shows that $\sigma(\tilde{\Pi})$ is a SIC projector. □

Corollary VI.47 (odd class-group part)**.** *In Theorem VI.46, if $[M_\varepsilon \cap H_F : F]$ is odd, and in particular if the class number h_F is odd, then (b) holds automatically. Hence (i)–(iii) imply* $\mathrm{Z}(d)$.

Proof. The restriction of κ to $M_\varepsilon \cap H_F$ has order dividing 2 in a group of odd order, so it is trivial. □

The ghost $\tilde{\Pi}$ determines the set $\mathcal{E} = \{e_p : p \neq 0\}$, and an automorphism $\tau \in \Gamma$ either maps $\mathcal{E}$ onto itself or moves $\tilde{\Pi}$ to a different ghost of its Galois multiplet. In $d = 9$ the class group of F has order 2, and the two possibilities are separated exactly by the class group: the automorphisms that preserve $\mathcal{E}$ are those attached to principal ideals (Sec. VI G). We isolate the corresponding statement.

Conjecture VI.48 (multiplet hypothesis, MH)**.** *Every $\tau \in \mathrm{Gal}(M_\varepsilon/F)$ that maps the set $\{e_p : p \neq 0\}$ onto itself acts trivially on $M_\varepsilon \cap H_F$.*

Corollary VI.49 (an infinite family without complex conjugations in the hypotheses)**.** *Let $\mathcal{P} = \{p^k : p \equiv 5 \pmod 6 \text{ prime}, k \geq 1\} \cup \{3^k : k \geq 2\}$. Let $d \in \mathcal{P}$, and suppose that for some ghost $\tilde{\Pi}$ in dimension d hypotheses (i)–(iii) of Theorem VI.46 hold together with Conjecture VI.48. Then* $\mathrm{Z}(d)$ *holds. The set $\mathcal{P}$ is infinite. If these hypotheses hold for infinitely many $d \in \mathcal{P}$, then Problem 1 follows.*

Proof. κ maps e_p to e_{-p} and so maps $\{e_p\}$ onto itself. Hence MH, applied to $\tau = \kappa$, gives (b), and Theorem VI.46 applies. There are infinitely many primes $p \equiv 5 \pmod 6$ by Dirichlet's theorem. □

a. What has and has not changed. Hypotheses (i) and (iii) are the algebraic part of the Stark picture together with R2, exactly as in Proposition VI.30. If M_ε is the full ray class field modulo $d\infty_-$, as in every computed odd dimension, then hypothesis (ii) is condition (C_d) of Proposition VI.35. It holds for all 167 members of $\mathcal{P}$ with $d \le 2000$, and Theorem VI.37 proves it whenever $\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle$. The conductor hypothesis in (i) is essential: if ∞_+ or further primes divided the conductor, the additional sign or unit factors could give Γ_0 a 2-rank larger than 1. Conjecture VI.48 contains no sign condition and no reference to an embedding into $\mathbb{C}$. For $\tau = \kappa$ it is exactly condition (b), so Corollary VI.49 is a reformulation of Theorem VI.46 in terms of a natural structural statement, not an independent reduction. It concerns only the action of $\mathrm{Gal}(M_\varepsilon/F)$ on the finite set of ghosts in the Galois multiplet of $\tilde{\Pi}$, and it is the class-group part of the Shimura-type reciprocity that, in the Stark picture, describes this action. In this sense, on the family $\mathcal{P}$ the archimedean identity $c_- = \kappa$ has been traded for a statement about an everywhere-unramified extension, given the algebraic part of the Stark picture. We stress that neither MH nor the algebraic part of Stark's conjecture is proved here, and that Theorem VI.46 does not prove $\mathrm{Z}(d)$ in any new dimension. Compared with Proposition VI.30, the gain is reach: $\mathcal{P}$ contains 167 dimensions $d \le 2000$ against 24, including half of all primes.

b. Checks. For all 162 members $d = p^k \le 2000$ of $\mathcal{P}$ with $p \equiv 5 \pmod 6$ we confirmed with PARI/GP that p is inert in F and that $|\mathrm{Cl}(d\infty_-)| = 2|\mathrm{Cl}(d)|$. For $d = 3^k \le 2000$ we confirmed that 3 is ramified in F. For the 21 members $d \le 149$ with $p \equiv 5 \pmod 6$, and for $d = 9, 27, 81, 243$, we computed the subgroup $P^{\mathfrak{m}}/P_{\mathfrak{m},1} \subset \mathrm{Cl}(d\infty_-)$ explicitly, as the subgroup generated by the classes of the generators of $(\mathcal{O}_F/\mathfrak{m})^\times$. In every case it has exactly one element of order 2, as Lemmas VI.44 and VI.45 predict. Among the first 21, however, h_F is even in 16 cases, and the 2-Sylow subgroup of $\mathrm{Cl}(d\infty_-)$ is non-cyclic in 15. The class number h_F is odd for exactly 12 members $d \le 2000$, namely $5, 11, 17, 47, 53, 191, 431, 449, 881, 971, 1061, 1681$. These lie in the list of Proposition VI.30, in agreement with Corollary VI.47. Exact computations of M_ε are available for three members of $\mathcal{P}$. In $d = 5$ and $d = 11$, hypotheses (i)–(iii) hold with $h_F = 1$, and $c_- = \kappa$ holds, as

Corollary VI.47 predicts (Table VII). In $d = 9 = 3^2$, where $h_F = 2$, Conjecture VI.48 holds exactly (Sec. VI G), and so does $c_- = \kappa$.

c. Why the restriction to $\mathcal{P}$. If d is divisible by a prime $p \equiv 1 \pmod 6$, which splits in F, or by two distinct odd primes, then $(\mathcal{O}_F/d)^\times$ has 2-rank at least 2, and Γ_0 can contain three involutions. For $d = p \equiv 1 \pmod 6$, for example, these are the images of $(-1,-1)$, $(1,-1)$ and $(-1,1)$ in $\mathbb{F}_p^\times \times \mathbb{F}_p^\times$. Group theory alone then does not single out c_-, and further input is needed. The explicit reciprocity law (Conjecture VI.40) provides it: it predicts that c_- is the image of $(-1,-1)$, that is, $\phi(\varpi) = -I$.

I. Near-group categories: the place law as a unitarity statement

Radchenko and Wheeler prove the algebraicity of the Stark values by realizing every finite quantum dilogarithm as the defining datum of a near-group fusion category, following Izumi[19] and Evans–Gannon[21]. They remark that the real-or-unitary question for finite quantum dilogarithms "is closely related to the distinction between unitary and pseudo-unitary fusion categories" (Ref. 12, §6). This subsection makes that relation precise for the Stark solution E_*. The outcome is that the archimedean input needed for SICs is exactly a unitarity statement: *a SIC exists in dimension d as soon as one pseudo-unitary Galois conjugate of the near-group category of E_* is unitary*, and the place law says that the conjugates that are unitary are precisely the pseudo-unitary ones. By Izumi's classification, the only condition that can fail for a pseudo-unitary conjugate is a single Hermiticity equation.

a. The Izumi fusion ring. Let G be a finite abelian group of order N. The *Izumi fusion ring* R_G has basis $\{[g] : g \in G\} \cup \{[\rho]\}$ and fusion rules

$$[g][h] = [g+h], \qquad [g][\rho] = [\rho][g] = [\rho], \qquad [\rho]^2 = \sum_{g \in G} [g] + N[\rho].$$

A fusion category with Grothendieck ring R_G is a *near-group category of type* (G, N). The invertible objects are the $g \in G$, and ρ is the unique non-invertible simple object. Its Frobenius–Perron dimension is the positive root of $x^2 = N + Nx$,

$$\mathrm{FPdim}(\rho) = \frac{N + \sqrt{N^2 + 4N}}{2} = \frac{N + \sqrt{D}}{2}, \qquad D = N(N+4).$$

A fusion category is *pseudo-unitary* if its global dimension equals its Frobenius–Perron dimension, and *unitary* if it is equivalent to a C^* fusion category.

By Ref. 12 (Thm. 12 and §§5.4–5.5), every $E \in Z(G, \langle\cdot\rangle)$ gives a fusion category $\mathcal{C}_E$ categorifying R_G. It is constructed from endomorphisms of a Leavitt algebra, and no unitarity is assumed. The function E is recovered from $\mathcal{C}_E$ up to relabelling G by automorphisms preserving $\langle\cdot\rangle$. In the spherical structure used there, the invertible objects have dimension 1 and

$$d_\rho(E) = -\frac{\sqrt{N}}{E(0)}.$$

For a Galois conjugate, $\sigma(\mathcal{C}_E) = \mathcal{C}_{\sigma E}$ is the category of the conjugate datum $(\sigma\langle\cdot\rangle, \sigma E)$, with $\sqrt{N}$ replaced by $\sigma(\sqrt{N})$.

Lemma VI.50 (dimensions and pseudo-unitarity)**.** *(a) If $E \in Z$ and $E(0) = (\sqrt{N} + s\sqrt{N+4})/2$ with $s = \pm 1$, then*

$$d_\rho(E) = \frac{N - s\sqrt{D}}{2}.$$

These are the two roots of $x^2 = N + Nx$, and $\mathcal{C}_E$ is pseudo-unitary if and only if $d_\rho(E) > 0$, that is, if and only if $E \in Z_-$.

(b) For the Stark solution and every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$,

$$d_\rho(\sigma E_*) = \frac{N - \sigma(\sqrt{D})}{2}.$$

Hence $\sigma(\mathcal{C}_{E_*})$ is pseudo-unitary if and only if $\sigma(\sqrt{D}) < 0$, that is, if and only if σ lies above ∞_-. In particular $\mathcal{C}_{E_*}$ itself is not pseudo-unitary, and exactly half of its Galois conjugates are.

Proof. (a) Multiplying numerator and denominator by $\sqrt{N} - s\sqrt{N+4}$ gives

$$d_\rho(E) = \frac{-2\sqrt{N}(\sqrt{N} - s\sqrt{N+4})}{N - (N+4)} = \frac{N - s\sqrt{N}\sqrt{N+4}}{2} = \frac{N - s\sqrt{D}}{2}.$$

The global dimension is $\sum_g 1 + d_\rho^2 = N + d_\rho^2$, and the Frobenius–Perron dimension is $N + \mathrm{FPdim}(\rho)^2$. Since d_ρ is one of the two roots $(N \pm \sqrt{D})/2$, whose squares differ, the two dimensions agree exactly when d_ρ is the positive root.

(b) By (a) with $s = +1$, $d_\rho(E_*) = (N - \sqrt{D})/2$ lies in F. The categorical dimension is computed from the data by the same rational expression, and hence $d_\rho(\sigma E_*) = \sigma(d_\rho(E_*)) = (N - \sigma(\sqrt{D}))/2$. This is positive exactly when $\sigma(\sqrt{D}) = -\sqrt{D}$. □

Lemma VI.50(b) refines Lemma VI.18, which records only the sign of $d_\rho(\sigma E_*)$. It also explains the role of $E(0) < 0$ in Sec. VI B: for σ fixing K_0, the conjugate is pseudo-unitary exactly when $\sigma(E_*(0)) < 0$.

b. Izumi's unitarity equations. Izumi classified the C^* near-group categories with multiplicity $m = |G|$ (Ref. 19, Thm. 9.1). In the form of Ref. 19 (Rem. 9.2), they correspond, up to automorphisms of G, to data $(\langle\cdot,\cdot\rangle, a, b, c', d)$, with $a, b : G \to \mathbb{C}$, $c' \in \mathbb{C}$ and $d \in \mathbb{R}$, satisfying, with $n = |G|$:

$$\text{(I1)}\quad c'^3 = \frac{1}{n^2}\sum_{g\in G} a(g), \qquad \text{(I2)}\quad d^2 = dn + n,$$

$$\text{(I3)}\quad R_{c'}b = b, \qquad \text{(I4)}\quad b(0) = -\frac{1}{d},$$

$$\text{(I5)}\quad a(g)b(g)b(-g) = \frac{1}{n} - \frac{\delta_{g,0}}{d}, \qquad \text{(I6)}\quad \text{the cubic (pentagon) equation},$$

$$\text{(I7)}\quad Jb = b, \qquad \text{(I8)}\quad d > 0.$$

These are Izumi's equations (9.6)–(9.13). Here $R_{c'}f(g) = c'a(g)^{-1}\sum_h \langle g,h\rangle f(h)$ is a unitary of period three, and J is the antiunitary of period two

$$(Jf)(g) = \overline{a(g)\, f(-g)}.$$

Izumi notes that (I7) and (I8) "come from the unitarity of the category", that (I1)–(I6) should already give near-group categories in general, and that known solutions have Galois conjugates satisfying all the equations except the last two (Ref. 19, Rem. 9.2). Equations (I1)–(I6) are polynomial with rational coefficients, and hence Galois-stable.

The dictionary between Izumi's data and finite quantum dilogarithms is (Ref. 12, §5.4, footnote 10)

$$a(x) = \langle x\rangle, \qquad b(x) = \frac{E(x)}{\sqrt{N}}, \qquad c = \lambda, \qquad d = d_\rho(E) = -\frac{\sqrt{N}}{E(0)},$$

with Izumi's pairing equal to $\langle x; y\rangle^{-1}$ and $c' = c/\sqrt{n}$. Under it, (I4) is the formula for d_ρ, (I5) is Theorem 5(ii) of Ref. 12 ($E(u)E(-u) = \langle u\rangle^{-1}$ for $u \neq 0$), (I3) is the eigenvalue relation $\widehat{E} = \lambda\langle\cdot\rangle E$ of Theorem 5(iii) there, and (I1), (I2), (I6) are the remaining RW equations. In particular every $E \in Z$ satisfies (I1)–(I6).

Lemma VI.51 (Hermiticity is unitary type)**.** *Let $E \in Z(G, \langle\cdot\rangle)$, and assume that all values of $\langle\cdot\rangle$ have modulus* 1*. Under the dictionary above, Izumi's condition (I7), $Jb = b$, holds if and only if E is of unitary type.*

Proof. $Jb = b$ reads $b(g) = \overline{a(g)}\;\overline{b(-g)}$ for all g. Since $\sqrt{N}$ is real, this is $E(g) = \overline{\langle g\rangle E(-g)}$. Taking complex conjugates and replacing g by $-g$ gives $\overline{E(g)} = \langle -g\rangle E(-g)$ for all g, which is the definition of unitary type. Each step is reversible. □

The placement of the complex conjugations in J was fixed by Izumi's Example 9.3 ($G = \mathbb{Z}/2$, $a(1) = i$, $b(1) = (1-i)/2$, $d = 1+\sqrt{3}$): with $(Jf)(g) = \overline{a(g)f(-g)}$ one has $(Jb)(1) = \bar{i}\cdot\overline{(1-i)/2} = (1-i)/2 = b(1)$, as (I7) requires, while the other placements of the bar fail. The metric values $\langle u\rangle_\gamma$ of the SIC groups, and all their Galois conjugates, are roots of unity, and hence Lemma VI.51 applies to every conjugate of E_*.

Proposition VI.52 (the only obstruction is Hermiticity)**.** *Let $E \in Z(G, \langle\cdot\rangle)$, with all values of $\langle\cdot\rangle$ of modulus 1, and suppose that $\mathcal{C}_E$ is pseudo-unitary. Then the Izumi data of E satisfy (I1)–(I6) and (I8). They satisfy (I7) if and only if E is of unitary type, and in that case $\mathcal{C}_E$ is equivalent to a C^* near-group category, so it is unitary.*

Proof. (I1)–(I6) hold for every $E \in Z$, as explained above. (I8) is $d_\rho(E) > 0$, which is pseudo-unitarity by Lemma VI.50(a). (I7) is Lemma VI.51. If all of (I1)–(I8) hold, Theorem 9.1 of Ref. 19 produces a C^* near-group category with these data. The construction of Ref. 12 (§5.4) uses Izumi's formulas for the endomorphisms, with the Leavitt algebra L_{2N} in place of the Cuntz algebra, and hence gives a category equivalent to it. □

We expect the converse as well: if $\mathcal{C}_E$ is unitary, then E is of unitary type. Izumi's C^* data for $\mathcal{C}_E$ satisfy (I7). By Ref. 12 (§5.5) they determine E up to relabelling G by an automorphism preserving $\langle\cdot\rangle$, and such relabellings preserve unitary type. What remains to be checked is that the normalization of b in Izumi's C^* setting, which uses a vector e with $Je = e$, agrees with the reconstruction formula of Ref. 12 without an extra phase. We have not verified this in detail, and the results below use only the direction proved in Proposition VI.52.

c. SICs from unitary conjugates. Combining Proposition VI.52 with the dictionary of Lemma VI.3 gives the following form of the passage from ghosts to SICs.

Theorem VI.53 (unitarity form of SIC existence)**.** *Let $d \geq 4$, and let $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ satisfy $\sigma(\sqrt{D}) < 0$. Such σ exist, since $\sqrt{D} \notin \mathbb{Q}$, and for each of them:*

(a) $\sigma(\mathcal{C}_{E_*})$ is a pseudo-unitary near-group category of type (G_d, N);

(b) its Izumi data satisfy every unitarity condition except possibly the Hermiticity condition (I7);

(c) if (I7) holds, equivalently if σE_* is of unitary type, then $\sigma(\mathcal{C}_{E_*})$ is unitary, and $\sigma(\tilde{\Pi})$ is a WH-covariant SIC fiducial in dimension d.

Proof. (a) is Lemma VI.50(b), and (b) is Proposition VI.52. For (c), unitarity of $\sigma(\mathcal{C}_{E_*})$ is Proposition VI.52. Unitary type gives $|\sigma(E_*(u))| = 1$ for all $u \neq 0$. By Lemma VI.3, $\sigma(\varepsilon_p) = \sigma(\zeta_p)\,\sigma(E_*(u_p))^{-1}$ with $u_p \neq 0$, and hence $|\sigma(\varepsilon_p)| = 1$ for every $p \not\equiv 0$. Corollary VI.2 gives the SIC. □

Corollary VI.54 (categorical form of the place law)**.** *Suppose $t_u \neq \pm 2$ whenever $2u \neq 0$. Then* PL(u) *holds for all u with $2u \neq 0$ if and only if, for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$:*

- *if $\sigma(\mathcal{C}_{E_*})$ is pseudo-unitary, then σE_* satisfies Izumi's Hermiticity condition (I7), and $\sigma(\mathcal{C}_{E_*})$ is unitary;*
- *if $\sigma(\mathcal{C}_{E_*})$ is not pseudo-unitary, then σE_* is of real type.*

Proof. By Lemma VI.50(b), $\sigma(\mathcal{C}_{E_*})$ is pseudo-unitary exactly when $\sigma(\sqrt{D}) < 0$. Theorem VI.22 says that PL holds if and only if σE_* is of unitary type when $\sigma(\sqrt{D}) < 0$ and of real type when $\sigma(\sqrt{D}) > 0$. Unitary type is (I7) by Lemma VI.51, and it gives unitarity by Proposition VI.52. □

Conjecture VI.55 (pseudo-unitary conjugates are unitary)**.** *For every $d \geq 4$ and every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ such that $\sigma(\mathcal{C}_{E_*})$ is pseudo-unitary, the conjugate σE_* satisfies Izumi's Hermiticity condition (I7); equivalently, σE_* is of unitary type, and $\sigma(\mathcal{C}_{E_*})$ is unitary.*

The relations with the earlier statements are as follows.

- *Conjecture VI.55* $\Rightarrow$ Z(d). One pseudo-unitary conjugate suffices (Theorem VI.53). Hence Conjecture VI.55 for one σ in each of infinitely many dimensions d implies Problem 1.
- *PL* $\Leftrightarrow$ *Conjecture VI.55 + real type below.* By Corollary VI.54, PL is Conjecture VI.55 together with the statement that the conjugates that are not pseudo-unitary are of real type. Thus PL implies Conjecture VI.55.
- *Sign–type correlation.* For σ fixing K_0, pseudo-unitarity is $\sigma(E_*(0)) < 0$ (Lemma VI.50). Conjecture VI.55 is therefore the sign–type correlation of Sec. VI C, extended to all automorphisms, including those that move $\sqrt{N}$.
- *Under RU.* If every conjugate of E_* is of real or unitary type (Conjecture VI.9), Conjecture VI.55 is equivalent to: *no pseudo-unitary Galois conjugate of $\mathcal{C}_{E_*}$ has data of real type.*

For the SIC metric groups, Conjecture VI.55 is therefore of Zauner strength. It is a reformulation of the archimedean input, not a reduction of it to something known. Its value is

that it places the problem in the theory of unitary fusion categories and subfactors, where the available tools differ from those of Stark's conjecture.

d. A norm form of the Hermiticity condition. The Hermiticity condition (I7) can be written as the equality case of an inequality, which is the natural target for a positivity argument.

Proposition VI.56 (norm criterion)**.** *Let E be a finite quantum dilogarithm on a metric group $(G, \langle\cdot\rangle)$ of order N with $E(u)E(-u) = \langle u\rangle^{-1}$ for $u \neq 0$; this holds for every Galois conjugate of E_*. Then*

$$\sum_{u\neq 0} |E(u)|^2 \;\geq\; N-1,$$

with equality if and only if E is of unitary type. In particular, for σ with $\sigma(\sqrt{D}) < 0$, the pseudo-unitary category $\sigma(\mathcal{C}_{E_*})$ is unitary if and only if $\sum_{u\neq 0} |\sigma(E_*(u))|^2 = N-1$.

Proof. The relation holds for E_* because $x_u x_{-u} = 1$ with $x_u = s_u E_*(u)$, $s_{-u} = s_u$ and $s_u^2 = \langle u\rangle$, and it is preserved by Galois conjugation together with the metric. Since $|\langle u\rangle| = 1$, it gives $|E(u)|\,|E(-u)| = 1$. For $u \neq -u$, put $r = |E(u)|^2$; the pair $\{u, -u\}$ contributes $r + r^{-1} \geq 2$, with equality if and only if $r = 1$. For $u = -u \neq 0$ it gives $|E(u)| = 1$. Summing gives the inequality, with equality if and only if $|E(u)| = 1$ for all $u \neq 0$. If $|E(u)| = 1$, then $\overline{E(u)} = E(u)^{-1} = \langle u\rangle E(-u) = \langle -u\rangle E(-u)$, which is unitary type; the converse was noted in Sec. VI. The last statement follows from Theorem VI.53 and Lemma VI.51. □

Proposition VI.56 shows that a pseudo-unitary conjugate fails to be unitary only through an *excess* of the quadratic quantity $\sum_{u\neq 0} |E(u)|^2$ over its minimum. For pseudo-unitary categories, positivity of the categorical structure is automatic, and in the closely related Haagerup–Izumi setting the Cuntz-algebra realization supplies positivity but not Hermiticity (Ref. 21, Thm. 2(c)). A proof of Conjecture VI.55 along these lines would need an identity, for instance a trace or Plancherel identity in the Cuntz-algebra model, that bounds $\sum_{u\neq 0} |E(u)|^2$ above by $N-1$. We have not found such an identity, and none appears in Refs. 19–21.

e. Staying inside one metric group. The sign–type correlation and Theorem VI.14 use an automorphism fixing $K_0 = \mathbb{Q}(\zeta_{4N})$ and moving $\sqrt{N+4}$. The next result says exactly when such an automorphism exists.

Proposition VI.57. *For the SIC group G_d, one has $\sqrt{N+4} \in K_0$ if and only if $d+1$ is a square or twice a square. Otherwise there is $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$ with $\sigma(\sqrt{N+4}) = -\sqrt{N+4}$. Such σ fixes the metric $\langle\cdot\rangle_\gamma$, satisfies $\sigma(\sqrt{D}) < 0$, and maps E_* to a solution $\sigma E_* \in Z_-(G_d)$ of the same system, whose category is pseudo-unitary.*

Proof. Since $N+4 = d^3 - 3d^2 + 4 = (d-2)^2(d+1)$, we have $\sqrt{N+4} = (d-2)\sqrt{d+1}$, and $\sqrt{N+4} \in K_0$ if and only if $\sqrt{m} \in K_0$, where m is the square-free part of $d+1$. A quadratic field lies in $\mathbb{Q}(\zeta_{4N})$ if and only if its discriminant divides $4N = 4d^2(d-3)$. Let p be an odd prime dividing m. Then $p \nmid d$, since $p \mid d+1$, and $p \nmid d-3$, since otherwise $p \mid (d+1) - (d-3) = 4$. Hence $p \nmid 4N$, and $\sqrt{m} \notin K_0$. So $\sqrt{m} \in K_0$ forces $m \in \{1, 2\}$. Conversely, $m = 1$ is trivial, and if $m = 2$ then d is odd, $d-3$ is even, $8 \mid 4N$, and $\sqrt{2} \in \mathbb{Q}(\zeta_8) \subset K_0$. In the remaining cases $\sqrt{N+4} \notin K_0$, and a σ as stated exists. It fixes $\sqrt{N} \in K_0$ and moves $\sqrt{N+4}$, hence moves $\sqrt{D}$, and $\sigma(E_*(0)) = (\sqrt{N} - \sqrt{N+4})/2 < 0$. The last assertion is Lemma VI.50. □

Among $4 \le d \le 100$ the exceptional dimensions are $d = 7$, 8, 15, 17, 24, 31, 35, 48, 49, 63, 71, 80, 97 and 99. For all other d, Conjecture VI.55 can be tested inside the single near-group system of G_d: *the Galois conjugates of E_* that lie in $Z_-(G_d)$ should be of unitary type.* In the exceptional dimensions, any σ with $\sigma(\sqrt{D}) < 0$ must move $\sqrt{N}$ or the roots of unity, and the pseudo-unitary conjugates belong to a Galois-conjugate metric datum; this is the situation described after Table IV in connection with Thornton's theorem.

f. Constraints on a failure. The structure theorems of Sec. VI B restrict how Conjecture VI.55 could fail on the conjugates fixing K_0. Let $L = K_0(\sqrt{N+4})$, $O = \mathrm{Gal}(\overline{\mathbb{Q}}/L) \cdot E_*$, and σ as in Proposition VI.57.

- If the pseudo-unitary conjugate $\sigma(\mathcal{C}_{E_*})$ satisfies (I7), then $h = (\sigma^{-1} c \sigma)c \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ realizes the reflection, $h(E_*) = \rho E_*$ (Theorem VI.7). In categorical terms, $h(\mathcal{C}_{E_*}) \simeq \mathcal{C}_{E_*}$ through an equivalence acting on the invertible objects by $g \mapsto -g$ (Proposition VI.15).
- If the reflection is not realized on O, then none of the pseudo-unitary categories $\mathcal{C}_E$, $E \in \sigma(O)$, satisfies (I7), and the number of them with data of real type is congruent to $|O|$ modulo 2 (Theorem VI.8).

The control experiments of Sec. VI C add a methodological constraint. Square subsystems of the RW equations, and random subspaces with the same symmetries, have non-real and

mixed solutions. Any mechanism forcing (I7), for example a positivity argument from a Cuntz-algebra realization, must therefore use the full overdetermined pentagon system.

g. Relation to unitarity questions for fusion categories. Uniqueness of unitary structures on unitarizable fusion categories is known (Reutter[24]). Existence of a unitary structure on a pseudo-unitary fusion category is open in general, and a proof of Conjecture VI.55 must use the near-group structure. The closest known case is the Haagerup–Izumi programme of Remark VI.16. There the pseudo-unitary conjugates are unitary exactly when an explicit HI matrix is Hermitian (Ref. 21, Thm. 2(c)), which has been verified for odd $N \leq 43$ (Ref. 20). Both settings use double-sine-type special values (Barnes double sine for HI, Faddeev's modular quantum dilogarithm here), and in both the unitarity of a conjugate is decided by the infinite place of a real quadratic field below the embedding. In the SIC setting, Corollary VI.54 identifies this rule with the archimedean rank-one Stark relation over F.

h. Evidence. Every computed case agrees with Conjecture VI.55:

- in all 94 hyperbolic classes with $5 \leq N \leq 30$, every solution in Z_- is of unitary type, so every pseudo-unitary category in these systems satisfies (I7) (Sec. VI B);
- for the SIC group $d = 4$, the same holds for all 16 conjugate metrics (Appendix A 2);
- for the 100 exact trace polynomials (39 classes, $N \leq 18$) and for the SIC dimensions $d = 5, 9, 13$, the place law holds at every embedding (Secs. VI D and VI E), and hence, by Corollary VI.54, every pseudo-unitary conjugate of $\mathcal{C}_{E_*}$ satisfies (I7).

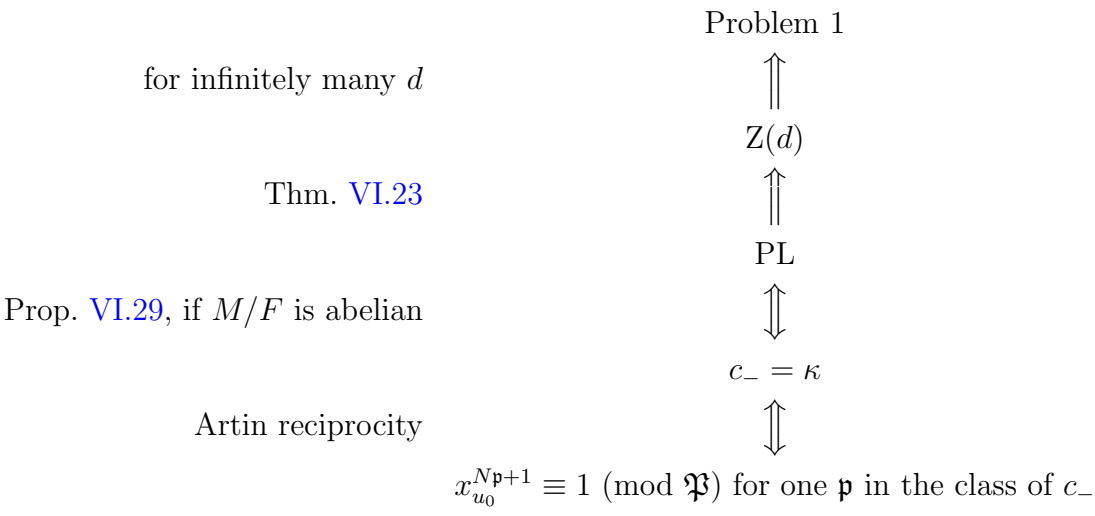
VII. THE REDUCTION IN FULL: WHAT IS PROVED, WHAT IS OBSERVED, AND WHAT REMAINS

This section collects the logical structure of the paper in one place. It records every implication used, the status of each statement, and the computational evidence. It ends with a precise list of what must still be proved to solve Problem 1.

A. Dependency diagram and chain of reductions

The main line of the reduction is shown below. Each double arrow is a proved implication or equivalence, labelled on the left by the result that gives it. The side branches are listed

after the diagram.



Side branches:

- into PL, bypassing abelianness and the congruence: M'/F Galois, $\kappa \in \mathrm{Gal}(M'/F)$, ∞_- ramified in M', and a unique element of order 2 in $\mathrm{Gal}(M'/F)$, for $M' = M$ or the sign-free field M_2 (Prop. VI.30); if M_2 is the full ray class field mod $d'\infty_-$, the last condition means a cyclic 2-Sylow subgroup of $\mathrm{Cl}(d'\infty_-)$; that last condition may instead be replaced by TR, which carries no restriction on the class group but is then equivalent to PL (Prop. VI.32, Cor. VI.33), and the ramification hypothesis is then unconditional for $d = 2m+1$ with m even and m^2-1 squarefree (Prop. VI.35, Thms. VI.37 and VI.38);
- into Z(d): one pseudo-unitary Galois conjugate of the near-group category $\mathcal{C}_{E_*}$ satisfies Izumi's Hermiticity condition (Thm. VI.53); HS + RU (Thm. VI.14); the sign–type correlation (Lemma VI.3, Cor. VI.2); R2 + (G1) plus one unimodular conjugate of one unit (Thm. VI.24);
- out of PL: HS, TR, RU on the orbit of E_*, the sign–type correlation and GR (Sec. VI C); the parity of the unitary count (Cor. VI.27); Conjecture VI.55, that pseudo-unitary conjugates of $\mathcal{C}_{E_*}$ are unitary (Cor. VI.54);
- equivalent form of PL: a conjugate of $\mathcal{C}_{E_*}$ satisfies Izumi's Hermiticity condition exactly when it is pseudo-unitary, and has data of real type otherwise (Cor. VI.54); a pseudo-unitary conjugate is unitary exactly when $\sum_{u\neq 0} |\sigma(E_*(u))|^2 = N-1$ (Prop. VI.56);
- into $c_- = \kappa$ and Z(d), for odd d: the explicit reciprocity law, Conjecture VI.40, at the single element $\varpi = -1 - d\delta$ with $\phi(\varpi) = -I$ (Prop. VI.42).

The archimedean rank-one Stark conjecture implies PL. Its algebraic part implies that M/F is abelian. Computationally, abelianness of M/F and the congruence at the bottom of the main line are confirmed for $d = 4, 5, 9$, and $d = 13$ passes the Frobenius test (Table V). The explicit reciprocity law holds exactly at every prime tested for $d = 5, 9$ (Table VII).

a. The chain of reductions (Steps 0–7). We now spell out the diagram step by step, recording what is proved and what is assumed at each stage.

Throughout, $d \geq 4$ is the SIC dimension. We write E_* for the RW (Stark) special-value solution attached to the AFK admissible tuple of dimension d, with metric group G_d of order $N = d^2(d-3)$ coming from $\gamma = \alpha^3$, $\operatorname{tr}\alpha = d - 1$[11,12]. The parity operator U_P (Section II) corresponds to the reflection $\rho : E \mapsto E(-\,\cdot)$ on finite quantum dilogarithms (Section VI).

Step 0 (the problem). Problem 1 asks for WH-covariant SICs in infinitely many dimensions. It suffices to prove Zauner's existence statement $\mathrm{Z}(d)$ for every d in some infinite set D.

Step 1 (complexification; unconditional). SIC fiducials are the fixed points of $\sigma = c \circ T$ on the complexified variety V_d (Lemma II.2). Ghosts exist in every dimension $d \geq 4$ (Theorem II.4), and hence $V_d(\overline{\mathbb{Q}}) \neq \emptyset$, and V_d has points modulo almost every prime (Corollary II.5).

Step 2 (no geometric shortcut; proved). No element of the extended Clifford group maps a ghost to a SIC (Lemma IV.1). Any passage from ghosts to SICs must therefore be arithmetic.

Step 3 (Galois criterion; proved). $\mathrm{Z}(d)$ holds if and only if some Galois orbit in $V_d(\overline{\mathbb{Q}})$ contains a point Π with $c'(\Pi) = \Pi^T$ for a conjugate c' of complex conjugation (Proposition IV.3, Lemma IV.2). For a ghost, this means some product of two complex conjugations acts on it as a displaced parity (Corollary IV.4). The statements $|S_d| = |\Gamma_d|$ and "ghost $\Rightarrow$ SIC" are each equivalent to $\mathrm{Z}(d)$, given Theorem II.4 (Corollary IV.5). There is also a finite-field version at the primes whose Frobenius is conjugate to c (Theorem IV.6).

Step 4 (two halves; proved). The criterion of Step 3 splits into two parts:

(a) GR: transposition acts on the orbit through *some* Galois element;

(b) that element is a complex conjugation.

By Proposition V.3, GR on the orbit of a ghost is equivalent to the existence of a field automorphism inverting all ghost units $\varepsilon_p = \sqrt{d+1}\,\nu_p$ simultaneously (up to sign).

Step 5 (translation to finite quantum dilogarithms; dictionary). The ghost overlaps

are built from the values $F_\gamma(u)$ (Ref. 13, Prop. 2.1), and $p \mapsto -p$ corresponds to $u \mapsto -u$. GR becomes R2: some $h \in \mathrm{Gal}(\overline{\mathbb{Q}}/L)$ satisfies $h(E_*) = \rho E_*$. The relevant notions match as follows.

- Real type, $\overline{E(u)} = \langle u \rangle E(u)$, is the ghost type. Unitary type, $\overline{E(u)} = \langle u \rangle E(-u)$, which forces $|E| = 1$, is the SIC type.
- The sign of $E(0)$ is opposite to the categorical dimension $d_\rho = -\sqrt{N}/E(0)$.
- Unitary type together with $E(0) < 0$ is exactly Izumi's data for a unitary near-group category (Ref. 19, Thm. 9.1). More precisely, $d_\rho > 0$ is pseudo-unitarity, and Izumi's Hermiticity condition $Jb = b$ is unitary type (Lemmas VI.50 and VI.51).
- In categorical terms, R2 says the category $\mathcal{C}_{E_*}$ is Galois-stable through an equivalence acting by -1 on its invertible objects (Proposition VI.15).

Step 6 (structure theorems; proved).

- Theorem VI.6: if Z_+ is a single reflection pair, R2 can fail only if a non-even real solution with $E(0) < 0$ exists.
- Theorem VI.7: if the $E(0) < 0$ conjugate of E_* is unitary, then R2 holds with an explicit Galois element. This is the dilogarithm form of Z $\Rightarrow$ GR.
- Theorem VI.8: if R2 fails on an orbit O, then $\sigma(O)$ contains no unitary and no even solution, and the number of real solutions in $\sigma(O)$ is congruent to $|O|$ modulo 2.
- Proposition VI.11: TR across all conjugate metric data is equivalent to total reality of the fields $\mathbb{Q}(t_u)$, where $t_u = x_u + x_u^{-1}$.
- Proposition VI.13: HS together with RU produces a unitary-type conjugate of E_*.
- Lemma VI.1, Corollary VI.2, Theorem VI.14: a Galois conjugate of a ghost is a SIC exactly when all its overlap units are unimodular. Hence HS and RU give SICs (via Lemma VI.3).
- Lemma VI.18, Theorems VI.22 and VI.23: over the full Galois group, the type of a conjugate is governed by the sign of its categorical dimension, equivalently by the place of $F = \mathbb{Q}(\sqrt{N(N+4)})$ below it. The place law PL is equivalent to "real above ∞_+, unitary above ∞_-" and gives SICs without RU.
- Corollary VI.27: when $N(x_{u_0}-1)$ is odd, the parity of the number of unimodular conjugates of x_{u_0}, and hence the hypothesis of Corollary VI.26, is determined by $|N(x_{u_0}\pm 1)|$ modulo 4. This is a non-archimedean footprint of the archimedean obstruction.
- Proposition VI.29: if the overlap-unit field M is abelian over F, then PL is equivalent

to $c_- = \kappa$. By Artin reciprocity this is a congruence $x_{u_0}^{N\mathfrak{p}+1} \equiv 1 \pmod{\mathfrak{P}}$ at a single prime in an explicit ray class. This is verified for $d = 4, 5, 9$, and $d = 13$ passes the Frobenius test (Table V). If moreover $\mathrm{Gal}(M/F)$ has a unique involution, PL follows from the algebraic hypotheses alone (Proposition VI.30).

- Proposition VI.32 and Corollary VI.33: if M'/F is Galois, $\kappa \in \mathrm{Gal}(M'/F)$ and ∞_- ramifies in M', then TR alone already gives $c_- = \kappa$, and hence TR, HS and PL are equivalent there, with no condition on $\mathrm{Cl}(d'\infty_-)$.
- Proposition VI.35, Lemma VI.36 and Theorems VI.37, VI.38: ramification at ∞_- is a statement about units; it holds whenever $\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle$, because ε_α reduces to a cube root of unity modulo d, and it is unconditional on a set of dimensions of density $\frac{1}{4}\prod_{p>2}(1 - 2p^{-2}) \approx 0.1613$.
- Theorem VI.24 and Corollaries VI.25, VI.26: under the algebraic conditions (G1)–(G2), or under R2 and (G1), one unimodular conjugate of one generating unit x_{u_0} gives a SIC. A sufficient condition is the integer inequality $P_{t_{u_0}}(2)P_{t_{u_0}}(-2) < 0$.
- Lemma VI.50, Proposition VI.52, Theorem VI.53 and Corollary VI.54: the conjugate $\sigma(\mathcal{C}_{E_*})$ is pseudo-unitary exactly when $\sigma(\sqrt{D}) < 0$. For it, Izumi's unitarity equations reduce to the Hermiticity condition, which is unitary type. If the condition holds for one such σ, a SIC exists, and PL says that it holds for all of them.
- Proposition VI.57: an automorphism fixing K_0 and moving $\sqrt{N+4}$ exists exactly when $d+1$ is neither a square nor twice a square.
- Proposition VI.56: Hermiticity of a pseudo-unitary conjugate is the equality case of $\sum_{u \neq 0} |\sigma(E_*(u))|^2 \geq N - 1$.
- Propositions VI.41 and VI.42: for odd d, the explicit reciprocity law (Conjecture VI.40), in which the Artin symbol of $\varpi\mathcal{O}_F$, $\varpi > 0$ at ∞_-, acts on the squared ghost overlaps by $e_p \mapsto e_{\phi(\varpi)p}$, is well defined, and at $\varpi = -1 - d\delta$ it gives $c_- = \kappa$ and a SIC. It is verified exactly for $d = 5, 9$ (Table VII).

Step 7 (the final decomposition). For the SIC group G_d, with $\sqrt{N+4} \notin K_0$ (which holds unless $d+1$ is a square or twice a square; Proposition VI.57):

$$\boxed{\mathrm{HS}(E_*) \;+\; \mathrm{RU}(G_d) \implies \text{unitary conjugate of } E_* \implies \mathrm{Z}(d)}$$

The first arrow is Proposition VI.13. The second is Theorem VI.14, which rests on the dictionary Lemma VI.3 and on Galois stability of V_d (Lemma VI.1 and Corollary VI.2). In

particular, no transfer of the TCI or of the order-$(2m+1)$ symmetry to conjugate data is needed: the Galois conjugate of the ghost projector already lies in V_d, and only its Hermiticity has to be checked.

The direct route is

$$\text{sign–type correlation for } G_d \ (E(0)<0 \Rightarrow \text{unitary}) \implies \mathrm{Z}(d),$$

because the $E(0)<0$ conjugate σE_* exists whenever $d+1$ is neither a square nor twice a square. Its categorical form, valid for every d, is

$$\boxed{\sigma(\mathcal{C}_{E_*}) \text{ pseudo-unitary and satisfying Izumi's Hermiticity condition, for one } \sigma \implies \mathrm{Z}(d)}$$

(Theorem VI.53). The weaker GR route is

$$\mathrm{TR} + \text{sign statement} + \text{Theorem VI.8} \implies \mathrm{GR}.$$

Section VI D adds two sharper routes:

$$\boxed{\mathrm{PL}(u)\ \forall u \implies \sigma E_* \text{ unitary for every } \sigma \text{ with } \sigma(\sqrt{D})<0 \implies \mathrm{Z}(d)}$$

(Theorems VI.22, VI.23), and

$$\boxed{\mathrm{R2} + (\mathrm{G1}) + \left[\, x_{u_0} \text{ has a unimodular conjugate} \,\right] \implies \mathrm{Z}(d)}$$

(Theorem VI.24, Corollary VI.25). If M/F is abelian, the first route becomes

$$\boxed{M/F \text{ abelian } + \ x_{u_0}^{N\mathfrak{p}+1} \equiv 1 \pmod{\mathfrak{P}} \text{ for one } \mathfrak{p} \text{ in the class of } c_- \implies \mathrm{PL} \implies \mathrm{Z}(d)}$$

(Proposition VI.29). For odd d there is also the reciprocity route

$$\boxed{\left(\tfrac{M_\varepsilon/F}{\varpi\mathcal{O}_F}\right) e_p = e_{\phi(\varpi)p} \text{ at } \varpi = -1 - d\delta \implies c_- = \kappa \text{ on } M_\varepsilon \implies \mathrm{Z}(d)}$$

(Proposition VI.42), in which the archimedean input becomes one instance of a uniform reciprocity law (Conjecture VI.40). Here PL contains HS, RU on the orbit of E_*, and the sign–type correlation, and it implies R2 through Theorem VI.7. The second route separates the problem into algebraic statements, R2 and (G1), and a single archimedean fact about one unit. That fact follows from $P_{t_{u_0}}(2)P_{t_{u_0}}(-2)<0$ (Corollary VI.26).

B. Status of all statements and computational evidence

TABLE VIII: Status of the statements in the reduction, in three blocks: proved, conjectural, and false.

Statement	Status	Evidence or source
Proved		
Ghosts exist for all $d \geq 4$	theorem	Refs. 12 and 13
No Clifford map ghost $\to$ SIC	theorem	Lemma IV.1
Galois criterion, finite-field criterion	theorem	Prop. IV.3, Thm. IV.6
GR $\Leftrightarrow$ unit inversion is Galois	theorem	Prop. V.3
Theorems C, C$'$, orbit version	theorem	Thms. VI.6, VI.7, VI.8
HS + RU $\Rightarrow$ unitary conjugate	theorem	Prop. VI.13
Unit-circle criterion for SICs	theorem	Lemma VI.1, Cor. VI.2
Dictionary: $\varepsilon_p = \zeta_p E_*(u_p)^{-1}$, $u_p \neq 0$	theorem (from Refs. 11 and 13)	Lemma VI.3; only structural facts (i)–(ii) of the following remark are used
HS + RU $\Rightarrow$ SIC	theorem	Thm. VI.14
Sign of d_ρ governs type; $E(0)$ only on $\mathrm{Gal}(\overline{\mathbb{Q}}/K_0)$	theorem	Lemma VI.18
PL $\Leftrightarrow$ real above ∞_+, unitary above ∞_-	theorem	Prop. VI.21, Thm. VI.22
PL $\Rightarrow$ SIC (no RU needed)	theorem	Thm. VI.23
One generating unit suffices (under (G1)–(G2) or R2+(G1))	theorem	Thm. VI.24, Cors. VI.25, VI.26
Parity of unitary count from $\lvert N(x \pm 1)\rvert$ when $N(x-1)$ odd	theorem	Cor. VI.27
That parity equals $[M : \mathbb{Q}]/4$ for every generating unit (under PL)	theorem	Prop. VI.28
If M/F abelian: PL $\Leftrightarrow c_- = \kappa$ $\Leftrightarrow$ one Frobenius congruence	theorem	Prop. VI.29
M'/F Galois, $\kappa \in \mathrm{Gal}(M'/F)$, ∞_- ramified, unique involution $\Rightarrow$ PL, Z(d)	theorem	Prop. VI.30
Same with TR in place of the unique involution; hence TR $\Leftrightarrow$ HS $\Leftrightarrow$ PL	theorem	Prop. VI.32, Cor. VI.33
∞_- ramifies in $\mathrm{Cl}(d'\infty_-)$ $\Leftrightarrow$ no unit $\equiv 1$ (d') is negative at ∞_-	theorem	Prop. VI.35

Statement	Status	Evidence or source
ε_α is a cube root of unity mod d; $\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle \Rightarrow \infty_-$ ramifies, every $d \geq 4$	theorem	Lemma VI.36, Thm. VI.37
That holds unconditionally for $d = 2m+1$, m even, m^2-1 squarefree, density ≈ 0.1613	theorem	Thm. VI.38
$d_\rho(\sigma E_*) = (N - \sigma(\sqrt{D}))/2$; $\sigma(\mathcal{C}_{E_*})$ pseudo-unitary $\Leftrightarrow$ $\sigma(\sqrt{D}) < 0$	theorem	Lemma VI.50
Izumi's Hermiticity condition $Jb = b \Leftrightarrow$ unitary type	theorem	Lemma VI.51 (convention checked on Ref. 19, Ex. 9.3)
Pseudo-unitary $\mathcal{C}_E$: only Hermiticity can fail; Hermitian $\Rightarrow$ unitary	theorem	Prop. VI.52
One pseudo-unitary conjugate of $\mathcal{C}_{E_*}$ Hermitian $\Rightarrow \mathrm{Z}(d)$	theorem	Thm. VI.53
PL $\Leftrightarrow$ (Hermitian $\Leftrightarrow$ pseudo-unitary) for all conjugates of $\mathcal{C}_{E_*}$	theorem	Cor. VI.54
$\sqrt{N+4} \in K_0 \Leftrightarrow d+1$ a square or twice a square	theorem	Prop. VI.57
Pseudo-unitary conjugate unitary $\Leftrightarrow$ $\sum_{u \neq 0} \lvert\sigma E_*(u)\rvert^2 = N-1$	theorem	Prop. VI.56
Reciprocity law well defined; at $\varpi = -1 - d\delta$ it gives $c_- = \kappa$ and $\mathrm{Z}(d)$ (d odd)	theorem	Props. VI.41, VI.42
Reciprocity law: one prime per ray class suffices ($h_F = 1$, M_ε/F abelian of conductor $\mid d\infty_-$)	theorem	Lemma VI.43
$d = p^k$, $p \equiv 5$ (6), or 3^k: under (i)–(iii), $c_- = \kappa \Leftrightarrow \kappa$ trivial on $M_\varepsilon \cap H_F$; automatic if h_F odd	theorem	Thm. VI.46, Cor. VI.47
Same with MH $\Rightarrow \mathrm{Z}(d)$ on the infinite family $\mathcal{P}$	theorem	Cor. VI.49
Conjectural		
Completeness $V_d = S_d \sqcup \Gamma_d$	conjecture	$d = 4, 5, 6$: complete counts; $d = 4, 5$: ghost test
R2 / GR	conjecture	PSLQ certification for $N = 6$–9 and for $d = 4$ ($N = 16$)

Statement	Status	Evidence or source
RU (no mixed type)	conjecture	94 SL_2 classes ($N \leq 30$); 268 cyclic metrics ($N \leq 21$); 0 exceptions
TR	conjecture	same data; exact minimal polynomials for $N = 6, 7, 9$
HS	conjecture	exact minimal polynomials ($N = 6, 7, 9$); $d = 4$: all 16 conjugate metrics, 48/48 split each
Sign statement	conjecture	no non-even real $E(0) < 0$ solution in any computed case
Sign–type correlation	conjecture	holds in every computed case; implies $\mathrm{Z}(d)$
Pseudo-unitary conjugates of $\mathcal{C}_{E_*}$ are unitary (Conj. VI.55)	conjecture	implied by PL; holds in every computed case (94 classes, $d = 4$ over 16 metrics, $d = 5, 9, 13$)
Cyclic 2-Sylow of $\mathrm{Cl}(d'\infty_-)$ for infinitely many d	conjecture	24 dimensions with $d \leq 2000$, thinning out; expected density zero (Sec. VI E)
M/F abelian, ramified at ∞_- only	conjecture	$d = 4, 5$: M is the full ray class field mod $(2d)\infty_-$, resp. $(4d)\infty_-$; conductor $d'\infty_-$ up to signs; $d = 9$ (exact), $d = 13$ (Frobenius test): M_2 is the full ray class field mod $d'\infty_-$
PL (place law, Conj. VI.20)	conjecture	100 exact minimal polynomials, 39 classes ($N \leq 18$, incl. $d = 4$), and $d = 5$ ($N = 50$); no exception
(G1), (G2)	conjecture	exact checks in 18 classes ($N \leq 12$, and $d = 4$): a generating u_0 always exists, and (G2) always holds
Stark (archimedean)	conjecture	implies PL, hence HS, TR and the sign–type correlation
Explicit reciprocity law (Conj. VI.40)	conjecture	exact at every prime tested: $d = 5$, $594 + 219$; $d = 9$, $122 + 276$; proved for the computed overlaps in $d = 5, 7, 11, 15$ (Table VII, Lemma VI.43); all alternative normalizations fail
Multiplet hypothesis (Conj. VI.48)	conjecture	exact in $d = 9$, the only computed dimension with $h_F > 1$; it is vacuous when $h_F = 1$
False		
"$\lvert Z_+\rvert = 2$ always"	false	$\lvert Z_+\rvert \in \{2, 4, 6, 8, 10, 12\}$ occurs

Computation	Result

a. Consolidated computational data. Table IX collects the computations behind the statements in Table VIII.

TABLE IX: Summary of computational evidence.

Computation	Result
All points of V_d (homotopy)	$d=4$: $512=256+256$; $d=5$: $4000=2000+2000$; $d=6$: $6912=3456+3456$
Ghost type of non-Hermitian points	$d=4$: 256/256; $d=5$: 2000/2000 are displaced-parity Hermitian
Non-cyclic abelian Heisenberg SICs	none for $16 \le d \le 36$ (strong for $d \le 20$)
All finite quantum dilogarithms, 94 SL_2 classes, $5 \le N \le 30$	distribution of $\lvert Z_+\rvert$ over classes: 1 (even, $N=5$): 1; 2: 20; 4: 54; 6: 6; 8: 9; 10: 3; 12: 1. In all classes, Z_+ is real and Z_- is unitary; $Z_- = \emptyset$ only when $\sqrt{N+4} \in K_0$
Simplified system (eigenspace + pairing), 268 cyclic metrics	every solution real or unimodular, none mixed; identical to the full system except $N=9$ $(2,2,5)$ and $N=16$ $(3,3,3)$ (16 vs 8 solutions, extras also totally real)
Control A (random subspace, square)	non-real solutions common (e.g. 16 of 64 pure at $N=13$)
Control B (Weil eigenspace, partial pairing)	non-real and mixed solutions appear (e.g. $(16,8,0)$ at $N=17$)
R2 by PSLQ	$N=6,7,8,9$ (cyclic) and $d=4$ ($N=16$): reflection is realized by $\mathrm{Gal}(\overline{\mathbb{Q}}/L)$
Minimal polynomials of t_u	$N=6$: degrees 4, 8; $N=7$: 12; $N=9$: 6, 2. All roots real; exactly half outside $[-2,2]$; all monic (algebraic integers)
$d=4$, all 16 Galois-conjugate metrics	each: $Z_+=4$ real, $Z_-=4$ unitary; t-values: 48 with $\lvert t\rvert>2$, 48 with $\lvert t\rvert<2$, none non-real
Place law, 39 classes $5 \le N \le 18$	100 distinct minimal polynomials (degree ≤ 32): all totally real, $\sqrt{N(N+4)} \in \mathbb{Q}(t_u)$ always, split by the place of F in every case; $\mathbb{Q}(\sqrt{N+4})$, $\mathbb{Q}(\sqrt{N})$ do not predict the split
Single-unit conditions (G1), (G2)	exact checks in 18 classes ($N \le 12$, $d=4$): a generating u_0 exists and (G2) holds in all; the integer criterion certifies a unitary conjugate in 4 classes; $P(2)P(-2)<0$ exactly when m_u is odd (13 of 100)

Computation	Result
$d = 5$ via a U_P-ghost at 400 digits	one trace polynomial of degree 16; PL at all 16 embeddings; $P(0) = 61^2$ odd; $m = 8$, and hence the parity criteria do not apply
Class-field test, $d = 4, 5$	M/F abelian, conductor $(2d)\infty_-$ resp. $(4d)\infty_-$, and $d'\infty_-$ up to signs; Frobenius $= \kappa$ exactly on the class of c_- at all 1305 primes $N\mathfrak{p} \le 5000$
Class-field test, $d = 9, 13$	M_2 is the full ray class field mod $(d)\infty_-$ (exact for $d = 9$; for $d = 13$ by Frobenius orders at 2230 primes $N\mathfrak{p} \le 20000$); M_2^κ has conductor (d); PL sign law on all 36, resp. 96, conjugates; Frobenius $= \kappa$ at all 15 tested primes in the class of c_- $(d = 13)$
Explicit reciprocity law, $d = 5, 9$	exact Frobenius action $e_p \mapsto e_{\phi(\varpi)p}$ at 594/594 split and 219/219 inert primes $(d = 5,\ q \le 3000)$ and at 122/122 principal split and 276/276 inert primes $(d = 9,\ q \le 4000)$; for $d = 9$ all 136 non-principal split primes move the ghost to its Galois partner
Unique-involution census, $4 \le d \le 2000$	2-Sylow of $\mathrm{Cl}(d'\infty_-)$ cyclic with ∞_- ramified for exactly 24 dimensions (15 with $d \le 250$, then $3, 0, 3, 1, 0, 2, 0$ in blocks of 250)
Pseudo-unitary conjugates (Conj. VI.55)	every pseudo-unitary category in the 94 complete systems $(N \le 30)$ satisfies Izumi's Hermiticity condition; by PL, the same for all conjugates of $\mathcal{C}_{E_*}$ in the 100 exact cases and for $d = 5, 9, 13$

b. The first SIC-relevant case. The row of Table IX on the 16 Galois-conjugate metrics of $d = 4$ deserves emphasis. The cyclic examples are only models. The metric group of $d = 4$ is the one that actually produces SICs through the AFK dictionary. For it, every one of the 16 Galois-conjugate metric data satisfies HS, RU and the sign–type correlation *simultaneously*: Z_+ is 4 real solutions, Z_- is 4 unitary solutions, and the traces split 48/48 with none non-real. This is consistent with the known existence of SICs in $d = 4$, and it shows that the decomposition of Step 7 is realized there exactly as predicted.

C. What remains to prove Problem 1

This subsection is self-contained. It recalls the notation needed to state the open problems precisely; it can therefore be read independently of the rest of the paper.

a. The problem. Fix a dimension $d \geq 4$. Let $Xe_j = e_{j+1}$ and $Ze_j = \omega^j e_j$ on $\mathbb{C}^d$, with $\omega = e^{2\pi i/d}$ and indices modulo d. The WH displacement operators are $D_p = \tau^{p_1 p_2} X^{p_1} Z^{p_2}$ for $p = (p_1, p_2) \in \mathbb{Z}_d^2$, where $\tau^2 = \omega$. A *WH-covariant SIC* is a unit vector $\psi \in \mathbb{C}^d$ with

$$|\langle \psi, D_p \psi \rangle|^2 = \frac{1}{d+1} \qquad \text{for all } p \neq 0.$$

Write $\mathrm{Z}(d)$ for the statement that such a ψ exists. Problem 1 asks for SICs in infinitely many dimensions, and hence it suffices to prove $\mathrm{Z}(d)$ for infinitely many d.

b. Ghosts and their overlap units. A *ghost* is a rank-one matrix $\tilde{\Pi}$ with $\tilde{\Pi}^2 = \tilde{\Pi}$, $\operatorname{tr} \tilde{\Pi} = 1$, satisfying the SIC equations $(d+1)\operatorname{tr}(\tilde{\Pi} D_p)\operatorname{tr}(\tilde{\Pi} D_p^\dagger) = 1$ for $p \neq 0$, which is Hermitian only up to parity: $\tilde{\Pi}^\dagger = U_P \tilde{\Pi} U_P$, where $U_P e_j = e_{-j}$. Ghosts with algebraic entries exist for every $d \geq 4$ (Theorem II.4). Their normalized overlaps $\varepsilon_p = \sqrt{d+1}\operatorname{tr}(\tilde{\Pi} D_p)$ are real and satisfy $\varepsilon_{-p} = \varepsilon_p^{-1}$. The basic sufficient condition for a SIC is:

> *if some field automorphism σ of $\overline{\mathbb{Q}}$ satisfies $|\sigma(\varepsilon_p)| = 1$ for all $p \neq 0$, then $\sigma(\tilde{\Pi})$ is a SIC projector* (Corollary VI.2).

c. The arithmetic data. Let $\alpha \in SL_2(\mathbb{Z})$ have trace $d-1$, and put $\gamma = \alpha^3$. Then

$$N = \operatorname{tr} \gamma - 2 = d^2(d-3), \qquad D = N(N+4) = (\operatorname{tr} \gamma)^2 - 4.$$

The field $F = \mathbb{Q}(\sqrt{D})$ is the real quadratic field generated by the fixed point of γ. It has two embeddings into $\mathbb{R}$, the *infinite places* $\infty_+ : \sqrt{D} \mapsto +\sqrt{D}$ and $\infty_- : \sqrt{D} \mapsto -\sqrt{D}$. Every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ restricts to one of them, and we say that σ *lies above* ∞_+ or ∞_- accordingly. Exactly half of all σ lie above each.

On the finite abelian group $G_d = \mathbb{Z}^2/(\gamma - I)\mathbb{Z}^2$ of order N, RW construct an explicit function $E_* : G_d \to \mathbb{C}$, the Stark solution of the finite quantum dilogarithm equations, whose values are special values of the modular quantum dilogarithm and are algebraic. From it we form the *overlap units*

$$x_u = s_u\, E_*(u), \qquad t_u = x_u + x_u^{-1} \qquad (u \in G_d,\ 2u \neq 0),$$

where s_u is a fixed root of unity. These are real algebraic units with $x_{-u} = x_u^{-1}$. By the dictionary of Lemma VI.3, every ghost overlap ε_p equals a root of unity times some $x_u^{\pm 1}$. Thus, $|\sigma(\varepsilon_p)| = 1$ for all p exactly when $|\sigma(x_u)| = 1$ for all u. Let $M = \mathbb{Q}(x_u : u \in G_d)$ be the field generated by the overlap units, and let κ denote inversion, $\kappa(x_u) = x_u^{-1}$. For a prime ideal $\mathfrak{q}$, $N\mathfrak{q}$ denotes its absolute norm, the size of its residue field. For $y \in \overline{\mathbb{Q}}$, $N(y)$ denotes the product of all Galois conjugates of y, a rational number.

d. The open problems. Problem 1 follows if, for infinitely many d, any one of the following routes is carried out.

Route A. The place law (Conjecture VI.20). *For every $u \in G_d$ with $2u \neq 0$ and every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the number $\sigma(x_u)$ is real if σ lies above ∞_+, and has absolute value 1 if σ lies above ∞_-.*

Equivalently, every conjugate $\sigma(t_u)$ is real, with $|\sigma(t_u)| > 2$ above ∞_+ and $|\sigma(t_u)| < 2$ above ∞_-. Equivalently again, the single algebraic number $(t_u^2 - 4)\sqrt{D}$ is *totally positive*: all of its conjugates are positive real numbers. This suffices because any σ above ∞_- makes every $|\sigma(x_u)| = 1$ at once, and hence every $|\sigma(\varepsilon_p)| = 1$ (Theorem VI.23). The place law is a statement about one algebraic number at a time, and it implies all the other conditions studied here (HS, TR, RU and the sign–type correlation).

Route B. The sign–type correlation. *For one automorphism σ that fixes the cyclotomic field $K_0 = \mathbb{Q}(\zeta_{4N})$, which contains $\sqrt{N}$ and all the roots of unity s_u, and sends $\sqrt{N+4}$ to $-\sqrt{N+4}$, one has $|\sigma(x_u)| = 1$ for all u.*

Such a σ exists exactly when $d+1$ is neither a square nor twice a square (Proposition VI.57). It lies above ∞_-, and hence this is the special case of Route A for one particular σ. It suffices by Corollary VI.2 and Lemma VI.3.

Route C. The class-field form (Proposition VI.29). This route has two parts.

(i) *The field M is an abelian Galois extension of F.* Let $\mathfrak{f}$ be the finite part of its conductor.

(ii) *One congruence holds.* Choose $\beta \in \mathcal{O}_F$ with $\beta \equiv 1 \pmod{\mathfrak{f}}$, $\beta > 0$ at ∞_+ and $\beta < 0$ at ∞_-, such that $\mathfrak{p} = (\beta)$ is a prime ideal not dividing $\mathfrak{f}$. For some prime ideal $\mathfrak{P}$ of M above $\mathfrak{p}$ and some u_0 for which x_{u_0} generates M,

$$x_{u_0}^{N\mathfrak{p}+1} \equiv 1 \pmod{\mathfrak{P}}.$$

Part (i) is the algebraic half of Stark's conjecture. Under (i), part (ii) is equivalent to Route A: by Artin reciprocity, the Frobenius element of $\mathfrak{p}$ is the complex conjugation c_- at ∞_-, and (ii) says that c_- acts as inversion. This is the identity $c_- = \kappa$, the classical Stark relation $\varepsilon^{1+c_-} = 1$. For $d = 4$ and $d = 5$ both

parts hold, and M is the ray class field of F of conductor $d'\infty_-$ up to signs, with $d' = d$ for d odd and $d' = 2d$ for d even. For $d = 9$ both parts hold for the sign-free unit field M_2, and $d = 13$ passes the same Frobenius test (Table V). When $\mathrm{Gal}(M_2/F)$ has a unique element of order 2, as for $d = 5, 9$ (and, if M_2 is the full ray class field, for $d = 7, 11, 15, 17, 19, 27, \dots$), part (ii) follows from part (i) together with R2, with no congruence to check (Proposition VI.30). More generally, part (ii) follows from part (i), R2 and TR alone, with no condition on $\mathrm{Cl}(d'\infty_-)$ (Proposition VI.32); the only remaining input is that ∞_- ramify in M_2, which is unconditional whenever $d = 2m + 1$ with m even and $m^2 - 1$ squarefree (Theorem VI.38), a set of density ≈ 0.1613 containing $d = 5, 9, 13$. Read this way, Problem 1 would follow from the algebraic hypotheses (i) and R2 together with TR for the dimensions in that family (Corollary VI.39); since TR is then equivalent to PL, this is a reformulation of Route A on that family, not a reduction of it.

Route D. Norms of $x_u \pm 1$ (Corollaries VI.26, VI.27). Suppose that for some u_0:

(G1) x_{u_0} generates M;

(G2) every field embedding sending $x_{u_0} \mapsto x_{u_0}^{-1}$ sends each $x_v \mapsto x_v^{-1}$ (this follows from R2).

Then a SIC exists as soon as one conjugate of x_{u_0} lies on the unit circle. This holds in particular if

$$N(x_{u_0} - 1) \text{ is odd} \qquad \text{and} \qquad |N(x_{u_0} - 1)| \cdot |N(x_{u_0} + 1)| \equiv 3 \pmod 4.$$

The route therefore needs only the prime factorizations of the ideals $(x_{u_0} \pm 1)$, for example from a Gross–Stark-type formula. However, it detects only the parity of the number of unimodular conjugates. By Proposition VI.28, it can succeed only when $[M : \mathbb{Q}] \equiv 4 \pmod 8$, and it fails for $d = 4$ and $d = 5$.

Route E. Unitarity of a pseudo-unitary near-group category (Sec. VII, Conjecture VI.55). The values $E_*(u)$ define a near-group fusion category $\mathcal{C}_{E_*}$ with simple objects $g \in G_d$ and ρ, and fusion rule $\rho \otimes \rho \cong \bigoplus_g g \oplus N\rho$ (Ref. 12, Thm. 12). The categorical dimension of ρ is $d_\rho = -\sqrt{N}/E_*(0) = (N - \sqrt{D})/2 < 0$, and hence $\mathcal{C}_{E_*}$ is not pseudo-unitary. For σ above ∞_-, the conjugate $\sigma(\mathcal{C}_{E_*})$ has

$d_\rho = (N + \sqrt{D})/2 > 0$ and is pseudo-unitary. Izumi's equations for a C^* near-group category (Ref. 19, Thm. 9.1) then hold for its data, except possibly the single Hermiticity condition

$$\overline{\sigma(E_*(u))} = \sigma\langle -u\rangle\, \sigma(E_*(-u)) \qquad \text{for all } u.$$

Here $\langle\cdot\rangle$ is the quadratic form on G_d of Ref. 12 (Eq. (3)), whose values are roots of unity, and $s_u^2 = \langle u\rangle$. *Show that this condition holds for one σ above ∞_-, that is, that one pseudo-unitary Galois conjugate of $\mathcal{C}_{E_*}$ is unitary.* It gives $|\sigma(x_u)| = 1$ for all $u \neq 0$, and hence a SIC (Theorem VI.53). The place law is the statement that it holds for every σ above ∞_- (Corollary VI.54). This is the near-group analogue of the open Haagerup–Izumi unitarity question (Remark VI.16). Equivalently (Proposition VI.56), show that $\sum_{u\neq 0} |\sigma(E_*(u))|^2 = N - 1$, its smallest possible value.

Route F. An explicit reciprocity law (Sec. VI G, Conjecture VI.40). Let d be odd, and let $\tilde{\Pi}$ be a ghost with Zauner symmetry: the squared overlaps $e_p = \varepsilon_p^2$ satisfy $e_{\Lambda p} = e_p$ for some $\Lambda \in SL_2(\mathbb{Z}_d)$ of order 3 with $\operatorname{tr}\Lambda \equiv d - 1$. Put $\epsilon_\alpha = \frac{1}{2}(d - 1 + \sqrt{(d-3)(d+1)})$, the eigenvalue of α that exceeds 1 at ∞_+, and let $\phi : \mathbb{Z}[\epsilon_\alpha] \to M_2(\mathbb{Z}_d)$ be the ring homomorphism with $\phi(\epsilon_\alpha) = \Lambda^{-1}$. Let $M_\varepsilon = \mathbb{Q}(e_p : p \neq 0)$, and assume it is abelian over F with conductor dividing $(d)\infty_-$, as computed for $d = 5, 9$. *Show that for $\varpi = -1 - d\sqrt{(d-3)(d+1)}$, which is positive at ∞_- and satisfies $\phi(\varpi) = -I$, the Artin symbol of the ideal $\varpi\mathcal{O}_F$ in $\mathrm{Gal}(M_\varepsilon/F)$ sends e_p to $e_{-p} = e_p^{-1}$ for every p.* By Artin reciprocity this Artin symbol is the complex conjugation c_- above ∞_-, and hence every σ above ∞_- gives $|\sigma(\varepsilon_p)| = 1$, and therefore a SIC (Proposition VI.42). The statement is the instance at one element of the conjectured law $\left(\frac{M_\varepsilon/F}{\varpi\mathcal{O}_F}\right)e_p = e_{\phi(\varpi)p}$ for all ϖ prime to d and positive at ∞_-, which holds exactly at every prime tested in $d = 5, 9$. For $\zeta(0)$-invariants the analogous law is a theorem for ideles positive at both infinite places (Coates–Sinnott, Deligne–Ribet; Arakawa[22]). What is needed here is its analogue for the derivative values, including the sign at ∞_-.

Route G. Prime powers and the class group (Sec. VI H, Theorem VI.46). Let $d = p^k$ with $p \equiv 5 \pmod 6$ prime, or $d = 3^k$ with $k \geq 2$, and let H_F be the Hilbert class

field of F. *Show that the field M_ε of squared ghost overlaps is abelian over F with conductor dividing $d\infty_-$ and ramified at ∞_-, that inversion is an automorphism of M_ε, and that inversion acts trivially on $M_\varepsilon \cap H_F$.* The last condition follows from the multiplet hypothesis (Conjecture VI.48): an automorphism that maps the squared overlaps of the ghost onto themselves acts trivially on the class-group part. It is automatic when h_F is odd. None of these statements refers to the infinite places except through ramification at ∞_-. Ramification is condition (C_d), which holds for every $d \le 20000$ and is proved when $\mathcal{O}_F^\times = \langle -1, \varepsilon_\alpha \rangle$. The family $\mathcal{P}$ of such d is infinite.

e. How the routes are related. Routes B–F are not independent of Route A. Route B is the place law for one automorphism fixing K_0. Route C is the place law itself in class-field form, under the algebraic part of Stark. Route D captures one bit of it. Route E is the place law for one automorphism above ∞_-, stated as the unitarity of a pseudo-unitary fusion category; it contains Route B, and unlike Route B it is available in every dimension. Route F is the half of the place law above ∞_-, for the squared ghost overlaps, obtained from one instance of a uniform reciprocity law; it is the route closest in form to proved results. Route G is Route C on the family $\mathcal{P}$, with the congruence replaced by an unramified condition that group theory shows to be equivalent to it. All seven are manifestations of one phenomenon: the archimedean part of the rank-one real-quadratic Stark conjecture for the overlap units. Route E connects it with the open HI unitarity question (Ref. 20, Rem. 3.8), and hence with the methods of subfactor theory and Cuntz-algebra realizations. No step of the present work carries out any route for an infinite family of dimensions. The computations confirm each route in every case examined.

Two technical points also need care in a complete proof.

- The passage from a unitary-type solution to a SIC uses the dictionary Lemma VI.3. Its proof assembles statements of Refs. 11 and 13, and the convention matching should be checked against those papers. The rest of that passage is proved here (Lemma VI.1, Corollary VI.2).
- The complete solution counts rest on numerical homotopy continuation. They are evidence, not proofs. The exact minimal-polynomial, PSLQ and class-field certifications are much more reliable, but they still rest on numerically identified algebraic numbers.

VIII. CONCLUSION

We proved that ghost SICs, which exist unconditionally in every dimension $d \geq 4$, cannot be turned into SICs by any symmetry of the SIC equations, and hence the step from ghosts to SICs is purely arithmetic. We gave Galois and finite-field criteria for that step, and reduced it to Galois inversion of the ghost overlap units. Through the RW finite quantum dilogarithms, we proved structure theorems for the Galois realization of the reflection. We showed that a single place law for the overlap units, real above one infinite place of $F = \mathbb{Q}(\sqrt{N(N+4)})$ and unimodular above the other, implies the existence of a WH-covariant SIC. If the unit field is abelian over F, the place law is equivalent to the identity $c_- = \kappa$, and hence to one Frobenius congruence. Equivalently, in the language of fusion categories, the place law says that a Galois conjugate of the Radchenko–Wheeler near-group category of the Stark solution is unitary exactly when it is pseudo-unitary. For a pseudo-unitary conjugate, Izumi's classification leaves a single Hermiticity condition to be proved, and one such conjugate satisfying it already gives a SIC.

What remains is to prove the place law, or one of its equivalent or sufficient forms (Sec. VII C), for infinitely many d. The place law and the Frobenius congruence hold in every case we computed, including the SIC dimensions $d = 4, 5, 9, 13$. There the (sign-free) overlap-unit field is the ray class field of conductor $d'\infty_-$, as predicted on the AFK side. We have not proved Zauner's conjecture or Problem 1.

For odd d, the identity $c_- = \kappa$ is in turn one instance of an explicit reciprocity law: the Artin symbol of a principal ideal (ϖ), with ϖ positive at ∞_-, acts on the squared ghost overlaps through the matrix of ϖ in the order generated by the Zauner symmetry. This law holds exactly at every prime tested in $d = 5$ and $d = 9$, and its form is that of the reciprocity laws proved for $\zeta(0)$-invariants. It suggests that the missing step is a reciprocity theorem for the derivative values at one sign idele, rather than an isolated archimedean identity.

For d a power of a prime $p \equiv 5 \pmod 6$, or of 3, the archimedean condition can be removed from the hypotheses: under the algebraic part of Stark, $c_- = \kappa$ is equivalent to the triviality of κ on the Hilbert-class-field part of the unit field, and this follows from a sign-free hypothesis on how the class group permutes the ghosts of a Galois multiplet. On this family, the remaining content of Problem 1 is therefore unramified class field theory for the Stark–Shintani units, together with their algebraic Stark property.

The main message is that the SIC existence problem has been reduced to the classical Stark archimedean condition $c_- = \kappa$ for the overlap-unit ray class field, and on an infinite family further to an unramified condition.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Hassan Khodaiemehr: Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available within the article. All code, data and logs are openly available in the GitHub repository of the author at https://github.com/hassankhodaiemehr/ghost-sic-galois, and identical copies are provided as ancillary files of the arXiv submission (directory `anc/`). They contain the homotopy-continuation scripts (Julia, HomotopyContinuation.jl), the complete solution lists for $d = 4, 5$, the parity test, the abelian-Heisenberg search (Python), the finite quantum dilogarithm

solvers, the high-precision PSLQ and minimal-polynomial certificates, the class-field, ray-class and reciprocity-law tests (PARI/GP), and all logs listed in Appendix D 2.

Appendix A: Computational data

This appendix records the computations behind the tables in the main text: methods, tolerances, and the full data. All scripts and logs are available in the GitHub repository https://github.com/hassankhodaiemehr/ghost-sic-galois and as ancillary files of the arXiv submission; the directories and file names are listed in Appendix D 2.

1. Varieties, reduced systems and controls

This subsection gives the full data behind Secs. III and VI C, together with control experiments showing that the reality patterns are not generic.

a. Complexified SIC variety. The complete solution sets and the ghost test for $d = 4, 5, 6$ are as follows.

TABLE X. Complete isolated solution sets of V_d and the displaced-parity ghost test.

d	paths	singular endpoints	$\lvert V_d\rvert$	SICs	non-Hermitian	in $\bigcup_u \Gamma_u$
4	560	48	512	256	256	256 (64 in Γ_{U_P})
5	14 614	9 718	4 000	2 000	2 000	2 000 (80 in Γ_{U_P})
6	218 343	78 192	6 912	3 456	3 456	not tested

b. Reduced system on all cyclic metrics. For each N we used all metrics $\langle x\rangle = \zeta_{2N}^{ax(x+N)}$ with $1 \le a < 2N$ and $\gcd(a, N) = 1$. For the 54 $SL_2(\mathbb{Z})$ classes with $N \le 20$, the reduced and full systems have the same solutions, except $N = 9$ cycle $(2, 2, 5)$ and $N = 16$ cycle $(3, 3, 3)$. Those two have 16 reduced solutions against 8 pentagon solutions, and all are totally real.

c. Control experiments. To check that total reality is a property of the actual finite quantum dilogarithm system, we solved two modified systems of the same shape.

d. Control A. A random ρ- and conjugation-invariant subspace with dimensions (even, odd), chosen such that the pairing system is square.

e. Control B. The true Weil eigenspaces $\lambda = G_0^{1/3}\zeta_3^k$ $(k = 0, 1, 2)$ for the cyclic metric $\langle x\rangle = \zeta_{2N}^{ax(x+N)}$, with a random square subset of the pairing equations. Each entry is

TABLE XI. Conjecture VI.10 on all nondegenerate cyclic metrics, $5 \le N \le 21$.

N	metrics	reduced-system solutions	pentagon solutions	t real	mixed
5	8	40	40	40	0
6	4	16	16	16	0
7	12	48	48	48	0
8	8	32	32	32	0
9	12	48	48	48	0
10	8	32	32	32	0
11	20	160	160	160	0
12	8	32	32	32	0
13	24	192	192	192	0
14	12	72	72	72	0
15	16	128	128	128	0
16	16	64	64	64	0
17	32	256	256	256	0
18	12	96	96	96	0
19	36	144	144	144	0
20	16	128	128	128	0
21	24	240	240	240	0
total	268			all	0

TABLE XII. Control A. Non-real solutions are common once $N \ge 9$.

N	dims	seed	solutions	t real	real or unimodular
7	(2,2)	1	8	8	8
7	(2,2)	2	8	8	8
7	(2,2)	3	8	8	8
9	(3,2)	1	16	16	16
9	(3,2)	2	16	8	8
9	(3,2)	3	16	12	12
11	(3,3)	1	32	12	12
11	(3,3)	2	32	20	20
11	(3,3)	3	32	24	24
13	(4,3)	1	64	16	16
13	(4,3)	2	64	32	32
13	(4,3)	3	64	52	52

solutions / t real / real-or-unimodular. An entry with t real but fewer real-or-unimodular solutions indicates mixed solutions.

TABLE XIII. Control B. Non-real and mixed solutions appear for every N tested.

N	a	seed	$k=0$	$k=1$	$k=2$
11	1	1	8/8/8	8/8/8	4/4/2
11	1	2	8/8/8	8/8/8	4/4/4
11	2	1	8/4/4	4/4/4	8/8/8
11	2	2	8/4/4	4/4/4	8/8/8
11	3	1	8/8/8	8/4/4	4/4/4
11	3	2	8/8/8	8/8/8	4/4/2
13	1	1	8/8/8	16/16/16	8/4/4
13	1	2	8/8/8	16/16/16	8/8/8
13	2	1	16/4/4	8/0/0	8/4/4
13	2	2	16/4/4	8/8/8	8/8/8
13	3	1	8/8/8	16/16/16	8/4/4
13	3	2	8/8/8	16/16/16	8/0/0
17	1	1	16/8/8	32/28/28	32/20/20
17	1	2	16/8/8	32/32/32	32/28/28
17	2	1	16/8/0	32/20/20	32/16/16
17	2	2	16/8/0	32/24/24	32/16/16
17	3	1	32/32/32	32/20/20	16/16/16
17	3	2	32/16/16	32/16/16	16/8/8
19	1	1	32/16/16	32/12/12	64/36/36
19	1	2	32/8/8	32/28/28	64/36/36
19	2	1	32/8/4	64/40/40	32/28/28
19	2	2	32/16/16	64/40/40	32/16/16
19	3	1	32/12/12	64/36/36	32/16/16
19	3	2	32/16/16	64/36/36	32/16/16

2. Arithmetic certificates

This subsection gives the exact and high-precision arithmetic checks behind Secs. VI B–VI D.

a. Reflection realized by Galois (PSLQ). Table XIV summarizes the field-membership tests that certify R2 in the cases computed.

b. Exact minimal polynomials of the traces t_u. For the Stark solution (cyclic metric $a=1$) and $t_u = x_u + x_u^{-1}$:

c. The $d=4$ SIC group over all conjugate metrics. For $\gamma = \alpha^3 = \left(\begin{smallmatrix} 21 & -8 \\ 8 & -3 \end{smallmatrix}\right)$ ($N=16$), we computed for each odd $k \in \{1,3,\ldots,31\}$ the solutions for the metric $\langle\cdot\rangle^k$. The result is the same for every k:

TABLE XIV. Certification of R2. $L = \mathbb{Q}(\zeta_{4N}, \sqrt{N+4})$; digits: 300–400 (cyclic) and 120 ($d = 4$).

Case	test	result
cyclic $N = 6, 7, 8, 9$	$E_*(1) \in L$? $E_*(1) + E_*(-1) \in L$?	no; yes (relation heights $4, 2, 4, 3$)
$d = 4$, $N = 16$	generic θ: $\theta(E) \in L$?	no, for all four $E \in Z_+$
	$\theta(E) + \theta(\rho E) \in L$?	yes
	$\theta(E) + \theta(E') \in L$, $E' \neq \rho E$?	no
	$\sum_{Z_+} \theta \in L$?	yes

TABLE XV. All roots are real. Exactly half lie outside $[-2, 2]$. All polynomials are monic.

N	u	minimal polynomial of t_u	$\lvert t\rvert > 2$	$\lvert t\rvert < 2$
6	1	$t^4 - 4t^3 - 3t^2 + 14t + 1$	2	2
6	2	$t^8 - 16t^6 + 87t^4 - 184t^2 + 121$	4	4
7	1,2,3	$t^{12} - 45t^{10} + 723t^8 - 5424t^6 + 20647t^4 - 38554t^2 + 27889$	6	6
9	1,2	$t^6 - 6t^5 - 18t^4 + 148t^3 - 282t^2 + 207t - 51$	3	3
9	3	$t^2 + 3t - 1$	1	1
9	4	$t^6 + 6t^5 - 18t^4 - 148t^3 - 282t^2 - 207t - 51$	3	3

- Z_+ consists of 4 real solutions;
- Z_- consists of 4 unitary solutions;
- the traces t_u, $u \neq 0$, $2u \neq 0$, over the 8 solutions are 48 with $|t| > 2$, 48 with $|t| < 2$, and 0 non-real.

The remaining coordinates are the three elements with $2u = 0$, where $t = \pm 2$.

d. The place law and the single-unit conditions. The place law and conditions (G1)–(G2) were tested on all classes in which the traces could be identified.

e. Method. For each class we took the real solutions in Z_+ and refined them to 900 digits. For every u with $2u \neq 0$ we found the minimal polynomial P of t_u with PARI/GP `algdep`, testing even degrees up to 40; a polynomial was accepted only if its residual was below 10^{-680} and its height was well within the precision. On each P we then performed four exact computations:

- real-root counting, which tests total reality and the split $|t| \gtrless 2$;
- `polcompositum` with $X^2 - q$ for $q = N(N+4)$, $N+4$ and N, which decides whether $\sqrt{q} \in \mathbb{Q}(t_u)$;
- for each embedding, the sign of $(\sigma(t_u)^2 - 4)\sigma(\sqrt{q})$, normalized such that $\sqrt{q} > 0$ at the embedding of the computed solution;

- the sign of $P(2)P(-2)$.

Polynomials shared by several u, or by several solutions, are counted once. The classes $N = 11$, and a few u for $N = 18$, need degree above 40 at this precision and are omitted.

f. Result. All 100 polynomials are totally real with an exact half split. In every case $\sqrt{N(N+4)} \in \mathbb{Q}(t_u)$, and the split agrees with the place of F at *every* embedding. In contrast, the analogous test with $q = N + 4$ or $q = N$ either fails ($\sqrt{q} \notin \mathbb{Q}(t_u)$) or gives a mixed split. An example is $N = 6$, $u = 2$, where both give 4/4. The sign of $P(2)P(-2)$ is negative for exactly the 13 polynomials with $m_u = \deg P/2$ odd.

TABLE XVI: The place law (Definition VI.19) for the classes in which all or most traces were identified. The columns give the number of distinct minimal polynomials, their degrees, whether PL holds for every one of them, and the number with m_u odd.

N	cycle	$\#P$	degrees of t_u	PL	m_u odd
5	(7)	1	4	yes	0
6	(2, 2, 4)	2	4, 8	yes	0
6	(2, 5)	2	4, 8	yes	0
6	(8)	2	4, 8	yes	0
7	(9)	1	12	yes	0
8	(2, 2, 2, 4)	1	8	yes	0
8	(2, 3, 3)	2	4, 16	yes	0
8	(2, 6)	1	8	yes	0
8	(3, 4)	2	4, 16	yes	0
8	(10)	2	4, 16	yes	0
9	(2, 2, 5)	1	2	yes	1
9	(11)	3	$2, 6^{\times 2}$	yes	3
10	(2, 7)	3	$8^{\times 2}$, 16	yes	0
10	(12)	2	8, 16	yes	0
12	(2, 2, 2, 5)	4	$4^{\times 2}$, 8, 16	yes	0
12	(2, 2, 6)	4	$4^{\times 2}$, 8, 16	yes	0
12	(2, 3, 2, 3)	1	2	yes	1
12	(2, 8)	4	$2^{\times 2}$, $4^{\times 2}$	yes	2
12	(4, 4)	1	2	yes	1
12	(14)	4	$4^{\times 2}$, 8, 16	yes	0
13	(2, 3, 4)	2	$24^{\times 2}$	yes	0
13	(2, 4, 3)	2	$24^{\times 2}$	yes	0
13	(15)	2	$24^{\times 2}$	yes	0
14	(2, 9)	4	6, $12^{\times 2}$, 24	yes	1
14	(3, 6)	4	6, $12^{\times 2}$, 24	yes	1
14	(16)	4	6, $12^{\times 2}$, 24	yes	1
15	(2, 2, 7)	3	8, 16, 32	yes	0

N cycle	#P	degrees of t_u	PL	m_u odd
15 (17)	3	8, 16, 32	yes	0
16 $(2,2,2,6)$	2	2, 8	yes	1
16 $(2,10)$	3	2, 4, 16	yes	1
16 $(3,3,3)$ (SIC, $d=4$)	1	4	yes	0
16 $(4,5)$	4	4, $8^{\times 2}$, 32	yes	0
16 (18)	4	4, $8^{\times 2}$, 32	yes	0
17 $(3,7)$	2	$32^{\times 2}$	yes	0
17 (19)	2	$32^{\times 2}$	yes	0
18 $(2,2,3,4)$	4	8, 16, $24^{\times 2}$	yes	0
18 $(2,2,4,3)$	4	8, 16, $24^{\times 2}$	yes	0
18 $(2,2,8)$	3	$8^{\times 2}$, 16	yes	0
18 $(2,3,5)$	4	8, 16, $24^{\times 2}$	yes	0

g. Single-unit conditions. For the classes in Table XVII we also computed the minimal polynomial P_x of x_{u_0} for each u_0, and checked (G1) and (G2) of Theorem VI.24 exactly. For each v, `polcompositum`$(P_{x_{u_0}}, P_{x_v})$ decides whether $x_v \in \mathbb{Q}(x_{u_0})$, by selecting the factor that vanishes at the computed pair. If it does, `modreverse` gives the polynomial f_v with $x_v = f_v(x_{u_0})$, and we checked the identity $f_v(X)f_v(X^{-1}) \equiv 1 \pmod{P_x}$. That identity is exactly (G2), since an embedding of $\mathbb{Q}(x_{u_0})$ is determined by the image of x_{u_0}. In every class some u_0 generates, and (G2) holds for *every* generating u_0. In four classes a generating u_0 has m_{u_0} odd and $P_{t_{u_0}}(2)P_{t_{u_0}}(-2) < 0$. There, Theorem VI.24 certifies a unitary-type conjugate from the sign of a single integer.

Appendix B: Supplementary table

a. All finite quantum dilogarithms for $SL_2(\mathbb{Z})$ classes, $5 \le N \le 30$. In the table below, $Z_\pm$ are the solutions with $E(0) = (\sqrt{N} \pm \sqrt{N+4})/2$. Each entry gives the number of solutions and their type. "Unit." means unitary type, and "–" means no solutions.

TABLE XVIII: All 94 hyperbolic conjugacy classes with $5 \le N \le 30$.

N cycle $(b_0, \ldots, b_{\ell-1})$	Z_+	Z_-
5 $(3,3)$	1 real+unit., even	4 unit.
5 (7)	4 real	1 real+unit., even
6 $(2,2,4)$	2 real	2 unit.
6 $(2,5)$	2 real	2 unit.
6 (8)	2 real	2 unit.
7 (9)	2 real	2 unit.

TABLE XVII. Exact checks of (G1) and (G2). The columns give the number of u_0 tested (up to $u \mapsto -u$ and repeated polynomials), the number of u_0 satisfying (G1), the degrees of their fields $\mathbb{Q}(x_{u_0})$, whether (G2) holds for all of them, whether $P(0)$ is odd for the generating u_0 (the hypothesis of Corollary VI.27), whether the integer criterion of Corollary VI.26 applies, and whether the 2-adic criterion of Corollary VI.27 applies.

N cycle	u_0	(G1)	$[\mathbb{Q}(x_{u_0}):\mathbb{Q}]$	(G2)	$P(0)$ odd	Cor. VI.26	Cor. VI.27
5 (7)	2	2	8	yes	yes	no	no
6 $(2,2,4)$	2	1	16	yes	yes	no	no
6 $(2,5)$	2	1	16	yes	yes	no	no
6 (8)	2	1	16	yes	yes	no	no
7 (9)	3	3	24	yes	yes	no	no
8 $(2,2,2,4)$	2	2	16	yes	no	no	no
8 $(2,3,3)$	3	2	32	yes	no	no	no
8 $(2,6)$	2	2	16	yes	no	no	no
8 $(3,4)$	3	2	32	yes	no	no	no
8 (10)	3	2	32	yes	no	no	no
9 $(2,2,5)$	3	3	4	yes	yes	yes	yes
9 (11)	4	3	12	yes	yes	yes	yes
10 $(2,7)$	4	2	32	yes	yes	no	no
10 (12)	4	2	32	yes	yes	no	no
12 $(2,3,2,3)$	2	2	4	yes	no	yes	no
12 $(2,8)$	4	2	8	yes	no	no	no
12 $(4,4)$	2	2	4	yes	no	yes	no
16 $(3,3,3)$ $(d=4)$	6	6	8	yes	no	no	no
50 $(4,4,4)$ $(d=5$, ghost$)$	1	1	32	yes	yes	no	no

N cycle $(b_0,\ldots,b_{\ell-1})$	Z_+	Z_-
8 $(2,2,2,4)$	2 real	2 unit.
8 $(2,3,3)$	2 real	2 unit.
8 $(2,6)$	2 real	2 unit.
8 $(3,4)$	2 real	2 unit.
8 (10)	2 real	2 unit.
9 $(2,2,5)$	4 real	4 unit.
9 (11)	2 real	2 unit.
10 $(2,7)$	2 real	2 unit.
10 (12)	2 real	2 unit.
11 $(2,2,3,3)$	4 real	4 unit.
11 $(3,5)$	4 real	4 unit.
11 (13)	4 real	4 unit.
12 $(2,2,2,5)$	4 real	–

N cycle $(b_0, \ldots, b_{\ell-1})$	Z_+	Z_-
12 $(2,2,6)$	4 real	–
12 $(2,3,2,3)$	2 real	2 unit.
12 $(2,8)$	4 real	–
12 $(4,4)$	2 real	2 unit.
12 (14)	4 real	–
13 $(2,3,4)$	4 real	4 unit.
13 $(2,4,3)$	4 real	4 unit.
13 (15)	4 real	4 unit.
14 $(2,9)$	6 real	–
14 $(3,6)$	6 real	–
14 (16)	6 real	–
15 $(2,2,7)$	4 real	4 unit.
15 (17)	4 real	4 unit.
16 $(2,2,2,6)$	4 real	4 unit.
16 $(2,10)$	2 real	2 unit.
16 $(3,3,3)$	4 real	4 unit.
16 $(4,5)$	2 real	2 unit.
16 (18)	2 real	2 unit.
17 $(3,7)$	4 real	4 unit.
17 (19)	4 real	4 unit.
18 $(2,2,3,4)$	4 real	4 unit.
18 $(2,2,4,3)$	4 real	4 unit.
18 $(2,2,8)$	8 real	8 unit.
18 $(2,3,5)$	4 real	4 unit.
18 $(2,5,3)$	4 real	4 unit.
18 $(2,11)$	4 real	4 unit.
18 (20)	4 real	4 unit.
19 (21)	2 real	2 unit.
20 $(2,2,2,7)$	4 real	4 unit.
20 $(2,3,2,4)$	4 real	4 unit.
20 $(2,4,4)$	4 real	4 unit.
20 $(2,12)$	4 real	4 unit.
20 $(3,8)$	4 real	4 unit.
20 $(4,6)$	4 real	4 unit.
20 (22)	4 real	4 unit.
21 $(2,2,9)$	10 real	–
21 $(5,5)$	10 real	–
21 (23)	10 real	–
22 $(2,13)$	2 real	2 unit.
22 (24)	2 real	2 unit.

N cycle $(b_0, \ldots, b_{\ell-1})$	Z_+	Z_-
23 $(2,3,6)$	8 real	8 unit.
23 $(2,6,3)$	8 real	8 unit.
23 $(3,9)$	8 real	8 unit.
23 (25)	8 real	8 unit.
24 $(2,2,2,8)$	4 real	4 unit.
24 $(2,2,10)$	4 real	4 unit.
24 $(2,3,3,3)$	4 real	4 unit.
24 $(2,14)$	4 real	4 unit.
24 $(3,3,4)$	4 real	4 unit.
24 $(4,7)$	4 real	4 unit.
24 (26)	4 real	4 unit.
25 $(2,2,3,5)$	4 real	4 unit.
25 $(2,2,5,3)$	4 real	4 unit.
25 (27)	4 real	4 unit.
26 $(2,15)$	4 real	4 unit.
26 $(3,10)$	4 real	4 unit.
26 $(5,6)$	4 real	4 unit.
26 (28)	4 real	4 unit.
27 $(2,2,11)$	12 real	12 unit.
27 $(2,4,5)$	6 real	6 unit.
27 $(2,5,4)$	6 real	6 unit.
27 (29)	6 real	6 unit.
28 $(2,2,2,9)$	8 real	–
28 $(2,2,4,4)$	4 real	4 unit.
28 $(2,3,2,5)$	4 real	4 unit.
28 $(2,3,7)$	8 real	–
28 $(2,7,3)$	8 real	–
28 $(2,16)$	4 real	4 unit.
28 $(4,8)$	4 real	4 unit.
28 (30)	8 real	–
29 $(3,11)$	4 real	4 unit.
29 (31)	4 real	4 unit.
30 $(2,2,12)$	4 real	4 unit.
30 $(2,17)$	4 real	4 unit.
30 (32)	4 real	4 unit.

Across all 94 classes, no solution is of mixed type, every solution in Z_+ is real, and every solution in Z_- is unitary. The only real solution with $E(0) < 0$ is even ($N = 5$, cycle (7)). The set Z_- is empty only when $\sqrt{N+4} \in K_0$, namely for some classes with

$N = 12, 14, 21, 28$.

Appendix C: Status of the five open problems

We record the status of the five open problems of Horodecki, Rudnicki and Życzkowski[7] as of October 2026.

- **Problem 1 (SICs in an infinite sequence of dimensions):** open unconditionally. There are conditional constructions for all $d > 3$[11], and ghosts exist unconditionally for all $d \geq 4$[12,13]. This paper reduces the remaining gap to Corollary IV.4.
- **Problem 2 (mutually unbiased bases (MUBs) in $d = 6$):** solved[25], not yet peer reviewed. The maximum number of MUBs in $\mathbb{C}^6$ is three, resolving Zauner's dimension-six MUB conjecture. Three MUBs were already known[26]; the new upper bound excludes four bases by a complete certified computation.
- **Problem 3 (36 entangled officers):** solved[27]. It received the Golden KCIK Award in 2021.
- **Problem 4 (negative-partial-transpose (NPT) bound entanglement):** open.
- **Problem 5 (2-copy distillability of $\rho(4, -\frac{1}{2})$):** solved in four independent preprints from July 2026[28–31], not yet peer reviewed. We checked numerically that $\min\langle\psi|(W_\alpha^\Gamma)^{\otimes 2}|\psi\rangle$ over Schmidt-rank-2 vectors, for $W_\alpha = I + \alpha V$ on $\mathbb{C}^4 \otimes \mathbb{C}^4$ (Appendix D 1), equals 0.055, 1.1×10^{-8}, -0.040 and -0.100 for $\alpha = -0.45, -0.50, -0.52, -0.55$. This is consistent with the threshold $\alpha = -\frac{1}{2}$.

Appendix D: Implementation details and related checks

1. Methods and related checks

We list the numerical methods and tolerances, and a check related to Problem 5.

a. Methods and tolerances. The main tools and acceptance criteria were as follows.

b. Homotopy continuation. We used HomotopyContinuation.jl (Julia 1.10) with polyhedral start systems, solving randomized square subsystems of overdetermined systems. Endpoints were kept if the residual of the *full* system was below 10^{-6} (for V_d) or 10^{-7}–10^{-8} (for finite quantum dilogarithms). Duplicates were merged at distance 10^{-6}. Points with $y^T x = 0$, or with a vanishing coordinate in the reduced dilogarithm system, were discarded.

c. Classification. A solution is *real type* if $\max_x |\overline{E(x)} - \langle x\rangle E(x)| < 10^{-6}$, *unitary type* if $\max_x |\overline{E(x)} - \langle x\rangle E(-x)| < 10^{-6}$, and *even* if $\max_x |E(x) - E(-x)| < 10^{-6}$. A solution is *mixed* if it is neither real nor unitary.

d. Eigenspace reduction. For finite quantum dilogarithms we parametrized the λ-eigenspace of $W = \langle\cdot\rangle^{-1}\mathcal{F}$, for each of the three cube roots λ of $\int\langle x\rangle^{-1}dx$. We fixed $E(0) = (\sqrt{N} \pm \sqrt{N+4})/2$ and solved the pairing equations $E(u)E(-u)\langle u\rangle = 1$. The pentagon equation (Ref. 12, Eq. (31)) was then checked on every candidate. Metric groups $G = \mathbb{Z}^2/\mathbb{Z}^2(\gamma - I)$ were built from $\gamma = T^{b_0}S\cdots T^{b_{\ell-1}}S$, with $\langle u\rangle$ and $\langle u; v\rangle$ as in Ref. 12, Eq. (3). The bicharacter identity $\langle u+v\rangle = \langle u\rangle\langle v\rangle\langle u; v\rangle$ was verified numerically to 10^{-15} for every group.

e. High-precision certification. Solutions were refined by Gauss–Newton iteration in `mpmath` at 120–400 digits, reaching residuals 10^{-113}–10^{-398}. Field membership was tested with PSLQ against a $\mathbb{Q}$-basis of $L = \mathbb{Q}(\zeta_{4N}, \sqrt{N+4})$. Minimal polynomials were found with `mpmath.findpoly` at 300 digits, with coefficient bound 10^{12}.

f. Gradient search (Sec. III B). L-BFGS on the frame potential, with an analytic gradient checked against finite differences to 8×10^{-10}.

g. Problem 5 check. On $\mathbb{C}^4 \otimes \mathbb{C}^4$, let I be the identity and V the swap operator, $V(\phi \otimes \chi) = \chi \otimes \phi$. Consider the operator $W_\alpha = I + \alpha V$. This is the Werner state $\rho(4, \alpha)$ before normalization by its trace, which does not affect the sign tests below. Let W_α^Γ be its partial transpose. We minimized $\langle\psi|(W_\alpha^\Gamma)^{\otimes 2}|\psi\rangle$ over unit vectors ψ of Schmidt rank 2 (40 random starts, L-BFGS):

α	-0.45	-0.50	-0.52	-0.55
minimum	0.055	1.1×10^{-8}	-0.040	-0.100

2. Files

All files are in the GitHub repository
https://github.com/hassankhodaiemehr/ghost-sic-galois, and identical copies are in the arXiv ancillary directory `anc/`. Directory names below are relative to the repository root, and each directory has a `README.txt`.

- **Complexified variety** (directory `variety_fqd_highprec/`): `ghost_count.jl`, `ghost_save.jl`, `parity_test.py`, `sols_d4.txt`, `sols_d5.txt`, `s4.log`, `s5.log`,

d6.log.

- **Heisenberg search** (directory variety_fqd_highprec/): hsearch.py, controls.py, big.py, big.log.
- **Finite quantum dilogarithms** (directory variety_fqd_highprec/): metricG.py, checkG.py, fqdW.jl, fqd2.jl, fqdG.jl, gen_gammas.py, scan.jl, scan.log, tscan.jl, tscan_cyc.jl, tcyc.log, control.jl, control.log, scan_d4.jl, d4conj.log.
- **High precision** (directory variety_fqd_highprec/): r2test.py, r2G.py, salem.py.
- **Place law and single-unit checks** (directory place_law_code/): save.jl, refine.py, refx.py, hs2.gp, an.sh, gen.gp, run_gen.sh, logs an_*.log, gen_*.log.
- **Class-field test,** $d = 4, 5$ (directory place_law_code/d5/): ghost_refine.py, gen.gp, an5.gp, xs.gp, cft.gp, cond2.gp, logs a5.log, gen5.log.
- **Ray-class tests,** $d = 9, 13$**, and the unique-involution list** (directory rayclass_code/): ghost and SIC search findghost.py, findsic.py; Newton refinement grefine.gp, refine.gp; refined ghosts g9.gp (2000 digits), g13x.gp and g17x.gp (20500 digits); $d = 9$ analysis d9*.gp, ganal.gp, an.gp; $d = 13$ trace polynomial and conductor d13a--e.gp, d13polys.gp; $d = 13$ Frobenius test d13Rt.gp, d13f.gp; place-law sign checks sl9.gp, sl13.gp; ray class groups rc.gp; 2-Sylow list twosyl.gp ($d \le 150$) and census twosyl2.gp ($d \le 2000$); $d = 17$ attempt d17.gp, d17f.gp; all logs.
- **Coverage figure** (Fig. 1; directory ramification_coverage/): data route_c_coverage_data.csv ($4 \le d \le 2000$; the cyclic 2-Sylow column is the census of Sec. VI E) and plotting script plot_route_c_coverage.py.
- **Ramification at** ∞_-**,** $d \le 20000$ (Sec. VI F; directory ramification_coverage/): place_ramification.py, a self-contained script computing the squarefree part of D, the fundamental unit of F by continued fractions with the index-3 correction, and the condition (C_d), together with the per-dimension output place_ramification_d_le_20000.csv.
- **Explicit reciprocity law,** $d = 5, 9$ (directory reciprocity_code_d5_d9/): $d = 5$ test all5.gp with data g5.gpv, exactness check ver5.gp, inert primes inert5.gp;

$d = 9$ automorphisms `gal9.gp` (output `aut9.gp`), exact overlap expressions `all9h.gp` (output `ex9_all.gp`), Frobenius test `all9i.gp`; logs.

- **Reciprocity law, certified, $d = 5, 7, 11, 15$, and the family $\mathcal{P}$** (directory `recip_v4/`): Zauner-symmetric ghost search `findghost_z.py`; Newton refinement `grefine.gp` with driver `refine_drv.gp`; refined ghost overlaps `eps5.txt`, `eps7.txt`, `eps11.txt`, `eps15.txt` (800, 1500, 6000 and 4000 digits); the test `recip.gp`, with exact expressions saved as `expr*.gp`; the class-field certificate and ray-class coverage `cert.gp`; logs `rec*.log`; the census of $\mathcal{P}$, `fam.gp` (output `family.txt`); and the involution count in $P^{\mathfrak{m}}/P_{\mathfrak{m},1}$, `ker.gp` (log `ker.log`).
- **Problem 5** (directory `variety_fqd_highprec/`): `problem5_check.py`.

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